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**WHAT INFLUENCES THE  
DISSEMINATION OF ONLINE RUMOUR  
MESSAGES IN SOCIAL MEDIA: THE  
ROLE OF MESSAGE, COMMUNICATOR,  
CHANNEL AND RUMOUR FEATURES**

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What Influences the Dissemination of Online Rumour  
Messages in Social Media: The Role of Message,  
Communicator, Channel and Rumour Features

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## TABLE OF CONTENTS

|                                                                     |      |
|---------------------------------------------------------------------|------|
| TABLE OF CONTENTS.....                                              | i    |
| LIST OF TABLES .....                                                | vi   |
| LIST OF FIGURES .....                                               | vii  |
| ACKNOWLEDGEMENT .....                                               | viii |
| LIST OF PUBLICATIONS .....                                          | x    |
| ABSTRACT .....                                                      | xiv  |
| CHAPTER 1 INTRODUCTION.....                                         | 17   |
| 1.1 Research Background and Motivation .....                        | 19   |
| 1.2 Theoretical Gap, Research Questions and Research Objectives.... | 24   |
| 1.3 Overview of Research Design and Methodology.....                | 30   |
| 1.3.1 Scope of Research .....                                       | 30   |
| 1.3.2 Research Methods and Data Analysis .....                      | 32   |
| 1.4 Summary of Results .....                                        | 33   |
| 1.5 Summary of Research Contributions .....                         | 35   |
| 1.5.1 Theoretical Contribution .....                                | 35   |
| 1.5.2 Practical Implication.....                                    | 37   |
| 1.6 Thesis Structure.....                                           | 38   |
| 1.7 Summary of Chapter .....                                        | 41   |
| CHAPTER 2 LITERATURE REVIEW .....                                   | 42   |

|       |                                                               |    |
|-------|---------------------------------------------------------------|----|
| 2.1   | Conceptualisation of Rumour.....                              | 44 |
| 2.1.1 | Defining Rumour.....                                          | 44 |
| 2.1.2 | Interpreting Rumouring.....                                   | 48 |
| 2.1.3 | Distinguishing Rumour from News, Gossips and Legends          | 50 |
| 2.1.4 | Comparing False Rumour with Fake News .....                   | 53 |
| 2.2   | Social Media and Information Dissemination.....               | 54 |
| 2.2.1 | Defining Social Media.....                                    | 54 |
| 2.2.2 | Information Dissemination on Social Media.....                | 55 |
| 2.3   | Conceptualisation of Online Rumour Message Dissemination..... | 58 |
| 2.3.1 | Online Rumour Topic and Online Rumour Message .....           | 58 |
| 2.3.2 | Defining Dissemination of Online Rumour Message .....         | 59 |
| 2.3.3 | Measuring Dissemination of Online Rumour Message .....        | 60 |
| 2.4   | Rumour Theories .....                                         | 62 |
| 2.4.1 | Distortion of Rumour Content.....                             | 62 |
| 2.4.2 | Psychological Motivation of Rumouring .....                   | 63 |
| 2.4.3 | Antecedents of Rumour Dissemination.....                      | 64 |
| 2.4.4 | Rumour in Information Systems Studies.....                    | 67 |
| 2.4.5 | Summary of Rumour Theories and Research Gap .....             | 68 |
| 2.5   | Communication Theories .....                                  | 70 |
| 2.5.1 | Components of Communication.....                              | 71 |
| 2.5.2 | Persuasion Theories.....                                      | 74 |
| 2.6   | Emotion Theories .....                                        | 78 |

|                                                                  |                                                                |     |
|------------------------------------------------------------------|----------------------------------------------------------------|-----|
| 2.6.1                                                            | Typologies of Emotions.....                                    | 79  |
| 2.6.2                                                            | Emotional Content in Computer-Mediated<br>Communications ..... | 80  |
| 2.6.3                                                            | Effects of Emotional Content .....                             | 80  |
| 2.7                                                              | Congruence Theory and Theory of Schemas .....                  | 84  |
| 2.8                                                              | Summary of Theoretical Lens .....                              | 86  |
| CHAPTER 3 THEORETICAL FRAMING AND HYPOTHESES<br>DEVELOPMENT..... |                                                                | 93  |
| 3.1                                                              | Message Features .....                                         | 95  |
| 3.1.1                                                            | Content (Un)ambiguity.....                                     | 96  |
| 3.1.2                                                            | Source (Un)ambiguity .....                                     | 100 |
| 3.1.3                                                            | Emotional Words .....                                          | 102 |
| 3.1.4                                                            | Imagery Words .....                                            | 108 |
| 3.2                                                              | Communicator Features .....                                    | 110 |
| 3.2.1                                                            | Communicator Trustworthiness .....                             | 111 |
| 3.2.2                                                            | Communicator Social Influence.....                             | 112 |
| 3.2.3                                                            | Communicator Likeability .....                                 | 113 |
| 3.3                                                              | Communication Channel Features .....                           | 114 |
| 3.4                                                              | Rumour Features .....                                          | 117 |
| 3.4.1                                                            | Rumour Topic-congruence .....                                  | 118 |
| 3.4.2                                                            | Rumour Message-congruence .....                                | 122 |
| 3.5                                                              | Control Variables .....                                        | 125 |

|           |                                                                                                    |     |
|-----------|----------------------------------------------------------------------------------------------------|-----|
| CHAPTER 4 | RESEARCH METHODOLOGY .....                                                                         | 128 |
| 4.1       | Research Philosophy and Strategy .....                                                             | 130 |
| 4.2       | Research Design .....                                                                              | 132 |
| 4.3       | Research Context and Scope of Research .....                                                       | 135 |
| 4.4       | Method of Data Collection .....                                                                    | 139 |
| 4.4.1     | Identifying Rumour Topics.....                                                                     | 140 |
| 4.4.2     | Identifying Keywords of Rumour Topics.....                                                         | 144 |
| 4.4.3     | Collecting Online Posts .....                                                                      | 144 |
| 4.4.4     | Identifying Online Rumour Messages.....                                                            | 145 |
| 4.4.5     | Collecting Supplementary Data.....                                                                 | 148 |
| 4.5       | Study One: Exploratory Study on Online Rumour Topics and Online Rumour Messages.....               | 149 |
| 4.5.1     | Research Data.....                                                                                 | 150 |
| 4.5.2     | Qualitative Data Analysis .....                                                                    | 150 |
| 4.6       | Study Two: Hypotheses Testing of Message-Level Dissemination Model of Online Rumour Messages ..... | 156 |
| 4.6.1     | Research Data.....                                                                                 | 157 |
| 4.6.2     | <i>Variable Operationalisation</i> .....                                                           | 158 |
| 4.7       | Study Three: Descriptive Analysis on Topic-Level Dissemination Patterns .....                      | 167 |
| 4.8.1     | Aim and Objectives of Study Three .....                                                            | 167 |
| 4.8.2     | Research Data and Research Methods .....                                                           | 167 |

|                                            |                                                            |     |
|--------------------------------------------|------------------------------------------------------------|-----|
| 4.8                                        | Research Ethics .....                                      | 168 |
| CHAPTER 5 DATA ANALYSIS AND FINDINGS ..... |                                                            | 170 |
| 5.1                                        | Findings of the Exploratory Study .....                    | 171 |
| 5.1.1                                      | Thematic Results of Online Rumour Topics .....             | 171 |
| 5.1.2                                      | Thematic Results of Online Rumour Messages .....           | 180 |
| 5.2                                        | Quantitative Analysis of Research Model .....              | 184 |
| 5.3                                        | Patterns Across Different Types of Rumour Topics .....     | 196 |
| CHAPTER 6 CONCLUSION .....                 |                                                            | 208 |
| 6.1                                        | Research Objective Realisation and Research Findings ..... | 209 |
| 6.2                                        | Theoretical Contribution .....                             | 212 |
| 6.3                                        | Practical Implication.....                                 | 215 |
| 6.4                                        | Limitation and Future Research .....                       | 216 |
| Appendix.....                              |                                                            | 239 |

## **LIST OF TABLES**

|                                                                                                                         |     |
|-------------------------------------------------------------------------------------------------------------------------|-----|
| Table 1 Comparison between Online and Offline Rumours.....                                                              | 22  |
| Table 2 Summary of Rumour Definitions from Previous Studies .....                                                       | 45  |
| Table 3 Comparison of Rumour, News, Gossip and Modern Legend .....                                                      | 51  |
| Table 4 Previous Studies on Information Dissemination .....                                                             | 58  |
| Table 5 Overview of Studies.....                                                                                        | 135 |
| Table 6 Labels Used by Websites .....                                                                                   | 142 |
| Table 7 Coding Scheme of Rumour Messages .....                                                                          | 147 |
| Table 8 Operationalisation of Variables.....                                                                            | 159 |
| Table 9 Themes of Online Rumour Topics .....                                                                            | 175 |
| Table 10 Categorisation of Online Rumour Topics .....                                                                   | 179 |
| Table 11 Themes of Rumour Communication Message on Twitter .....                                                        | 181 |
| Table 12 Results of Hypotheses Testing .....                                                                            | 185 |
| Table 13 Summary of Hypotheses Testing .....                                                                            | 189 |
| Table 14 Results of Testing with Binary Measurement of Dissemination<br>.....                                           | 195 |
| Table 15 Comparing Research Framework across Different Types of<br>Rumour Topics (Based on Focal Point of Rumour) ..... | 197 |
| Table 16 Comparing Research Framework across Different Types of<br>Rumour Topics (Based on Object of Rumour) .....      | 202 |
| Table 17 Linguistic Analysis Results of Rumour Topics.....                                                              | 203 |
| Table 18 Measurement of Brand Personality.....                                                                          | 205 |



## LIST OF FIGURES

|                                                                                 |     |
|---------------------------------------------------------------------------------|-----|
| Figure 1 An Overview of Research Questions of this Thesis .....                 | 30  |
| Figure 2 Structure of Chapters.....                                             | 39  |
| Figure 3 Measurement of Dissemination.....                                      | 62  |
| Figure 4 Theoretical Lens .....                                                 | 92  |
| Figure 5 Effects of Message Features (Hypotheses 1 - 4) .....                   | 110 |
| Figure 6 Effects of Communicator Features (Hypothesis 5, 6, 7) .....            | 114 |
| Figure 7 Effects of Communication Channel Features (Hypothesis 8a, 8b)<br>..... | 116 |
| Figure 8 Effects of Rumour Topic-congruence (Hypothesis 9a - 9h).....           | 122 |
| Figure 9 Effects of Rumour Message-congruence (Hypotheses 10a - 10h)<br>.....   | 125 |
| Figure 10 Conceptual Framework .....                                            | 127 |
| Figure 11 Overview of Research Design .....                                     | 134 |
| Figure 12 Processes of Data Collection.....                                     | 140 |
| Figure 13 Process of Qualitative Content Analysis .....                         | 153 |
| Figure 14 Sentiment Analysis Using NLTK API.....                                | 167 |

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## LIST OF PUBLICATIONS

Below is the list of papers that I published during the completion of the thesis:

### <Journal Publications>

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2. Alain Yee-Loong Chong, Ewelina Lacka, **Boying Li**, and Hing Kai Chan (2018) The role of social media in enhancing guanxi and perceived e-commerce institutional mechanisms effectiveness in online marketplace, *Information & Management*.
3. Fangfang Hou, **Boying Li**, Alain Yee-Loong Chong, Natalia Yannopoulou, Martin Liu, and Eugene Ch'ng (2017) Understanding and predicting what influence online product sales? A neural network approach. *Production Planning & Control*, 28 (11-12): 964-975. **(SCI, ABS 3)**
4. **Boying Li**, Eugene Ch'ng, Alain Yee-Loong Chong, and Haijun Bao (2016). Predicting online e-marketplace sales performances: A big data approach. *Computers & Industrial Engineering*, 101: 565-571. **(SCI, EI, ABS 2)**
5. Haijun Bao, **Boying Li**, Jiaying Shen, Fangfang Hou (2016). Repurchase intention in the Chinese e-marketplace: Roles of interactivity, trust and

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6. Alain Yee-Loong Chong, **Boying Li**, Eric W. T. Ngai, Eugene Ch'ng, Filbert Lee (2016). Predicting online product sales via online reviews, sentiments, and promotion strategies: A big data architecture and neural network approach. *International Journal of Operations and Production Management (IJOPM)*, 36 (4): 358-383. **(SSCI, ABS 4)**
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2. **Boying Li** and Martin J. Liu (2018) When service recovery becomes visible: Effects of recovery strategies on electronic word-of-mouth receiver. *2018 International Conference on Information Resources Management (Conf-IRM)*, Ningbo, China, 4-6 June, 2018.

3. Teng Ma, Fangfang Hou, Zhengzhi Guan, **Boying Li**, and Alain Yee-Loong Chong (2018) Importance of community influence and marketer message on brand value co-creation: A study in social commerce. *2018 International Conference on Information Resources Management (Conf-IRM)*, Ningbo, China, 4-6 June, 2018.
4. **Boying Li**, Fangfang Hou, Zhengzhi Guan, Alain Yee-Loong Chong, and Xiaodie Pu (2017) Evaluating online review helpfulness based on Elaboration Likelihood Model: the moderating role of readability. *The 21st Pacific Asia Conference on Information Systems (PACIS 2017)*, Langkawi, Malaysia, 16-20 July, 2017.
5. Fangfang Hou, Alain Yee-Loong Chong, and **Boying Li** (2016). Factors determining a firm's innovativeness: an empirical study of Chinese e-commerce industry. *The 20th Pacific Asia Conference on Information Systems (PACIS 2016)*, Chiayi, Taiwan, 27 June- 1 July, 2016.
6. **Boying Li**, Alain Yee-Loong Chong and Eugene Ch'ng (2015) What triggers sharing in viral marketing? The role of emotion and social feature. *The 19th Pacific Asia Conference on Information Systems (PACIS 2015)*, Singapore, 5-9 July, 2015.
7. **Boying Li**, Martin J. Liu, Alain Yee-Loong Chong, Filbert Lee and Eugene Ch'ng (2015) The Role of Service Recovery in Negative Word of Mouth Communication. *The 9th International Conference on Operations*

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8. **Boying Li**, Ke Li, Meijiang Li, Wen Li, Youran Liu (2015) How does computer-mediated communication influence e-commerce repurchase intention? The role of perceived effectiveness of e-commerce institutional mechanism. *The 9th International Conference on Operations and Supply Chain Management, Ningbo, P.R. China, 12-15 July, 2015.*
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## **ABSTRACT**

Social media enables efficient and easy exchange of information. However, this not only enhances the sharing of valid information but also facilitates the dissemination of rumours which may have harmful impacts on individuals, companies and the society. To manage the impacts of rumours, it is important to understand the drivers and patterns of online rumour message dissemination.

Considering the uniqueness of social media as information exchange platforms, this thesis aims to understand the themes and traits of online rumour topics and messages. Moreover, traditional rumour theories often study individual motivation of rumouring; however, in social media, messages are mostly broadcasted and can be accessed by the public, diminishing the influences of individual recipients' attributes. This thesis thus seeks to develop a message-level framework to delineate how message features, communicator features and channel features influence the dissemination of online rumour messages and how rumour features moderate those effects. Specifically, this thesis considers online rumour messages as persuasive messages. Combining rumour theories, persuasion theories, emotion theories and congruence theories, this thesis aims to investigate the roles of cognitive appeals (i.e. content unambiguity and source unambiguity), affective appeals (i.e. message pleasantness, activation and imagery), communicator attributes (i.e. communicator trustworthiness, proximity, social influence and likeability), communication channels (i.e. hashtag and @) and rumour features (i.e. rumour target reputation and rumour valence) in influencing online rumour message



dissemination in social media. Furthermore, the patterns of online rumour message dissemination may vary across rumour topics. To deepen the understanding of rumour message dissemination, this thesis also aims to summarise topic-level dissemination patterns of online rumour messages.

Based on the research aims and objectives, this thesis takes a multi-method research design. First, an exploratory qualitative content analysis was carried out to understand the types of online rumour topics and the themes of online rumour messages. The findings were used to support the development of research framework. Second, the research framework and corresponding hypotheses were tested quantitatively using regression. Third, linguistic analyses were applied to describe how rumour messages vary across different types of rumour topics, and the effects of research framework were compared.

Qualitative exploration finds that an online rumour topic may target a business's brand, its offerings, its people or its operations and may focus on what happens, why or with what impacts. Results of regression analysis reveal that expression certainty, emotional activation, community communication and communicator attributes are positively associated with the number of retransmissions (i.e. the number of retweets in this research context) of online rumour messages, whereas the source explicitness, source accessibility, emotional pleasantness, emotional imagery and direct communication are negatively associated with the number of retransmissions. In addition, both topic-congruence (i.e. congruence between rumour target reputation and rumour topic valence) and message-congruence (i.e. congruence between rumour target

reputation and rumour message valence) play important moderating roles. Results also show that for different types of rumour topics, the effects of cognitive and affective appeals on online rumour dissemination are different.

This thesis contributes to the existing literature in five aspects. First, it applies rumour theory to the context of social media and updates rumour theories by introducing expression certainty, use of evidence, source explicitness and source accessibility as new dimensions of information unambiguity. Second, this thesis investigates the effects of emotional pleasantness, activation and imagery on rumour dissemination, bringing new insights to rumour theories from emotion theories. Third, built on congruence theories and persuasion theories, this thesis investigates how topic-congruence and message-congruence moderates the effects of cognitive appeals and affective appeals on rumour dissemination. By doing so, this thesis extends rumour theory by combining the rumour context features and rumour message features. Moreover, this thesis identifies different types of online rumour topics and illustrates that online rumour messages are disseminated differently across rumour topics, enriching the understanding of online rumour dissemination on social media. Furthermore, while traditionally researchers have found it difficult to identify rumours and trace their disseminations, this thesis uses social media data to track rumours and their disseminations. Using social media data, this thesis tests the proposed research framework at rumour message level and explores the patterns of aggregate rumour message dissemination at rumour topic level. Practically, the findings of this thesis can be used to identify viral rumour messages so that practitioners can monitor and control rumours efficiently and effectively.

## **CHAPTER 1 INTRODUCTION**

---

This thesis focuses on understanding the dissemination of online rumour messages in social media. With the ubiquity of the Internet, it is increasingly prevalent for rumours to get spread online. The dissemination of online rumour messages can have negative and severe impacts on organisations and individuals and has thus captured the attention of both academics and practitioners. To manage the impacts of online rumouring, it is important to understand the factors that influence the dissemination of online rumour messages. However, traditional rumour theories may not be sufficient to explain such phenomenon considering the uniqueness of social media. Moreover, social media also offers new opportunities for research that can hardly be achieved in offline rumour context, such as the tracking of rumour messages. Therefore, what drives the dissemination of online rumour messages in the context of social media still requires theoretical understanding and empirical investigation.

This thesis aims to first explore the themes of rumour topics and online rumour messages that are propagated online. It enhances the overall understanding towards rumours topics and messages that disseminate online and

lays the foundation for the investigation of dissemination. This thesis then aims to develop a theoretical framework to understand the factors that influence the dissemination of online rumour messages on the rumour message level. Such framework extends the rumour research to the social media context and is empirically examined using real social media data. Further, this thesis aims to describe the general patterns of the dissemination of online rumour messages on the rumour topic level. Aggregating all the messages of each rumour topic, the descriptive patterns supplement the message-level dissemination framework and add insights towards the dissemination of online rumour messages in social media.

This opening chapter provides an overall introduction and overview to the thesis. Specifically, this chapter is organised as follows. Section one introduces the general research background and expounds the importance of this research topic. Section two elaborates the research gap and specifies the research questions and objectives. Section three explains the scope of the research and the research methodology. The results and findings are summarised in section four. After this, section five highlights the key theoretical contributions and practical implications. Following section five, section six outlines the structure of the thesis. Finally, a summary of the chapter is presented.

## **1.1 Research Background and Motivation**

Social media is becoming one of the mainstream channels for communication and information dissemination (Ozturk et al. 2015). With social media, information can be created, consumed and exchanged more easily and quickly than ever before. However, the use of social media not only smooths the sharing of valid information, it also facilitates the dissemination of rumours (Oh et al. 2013; Shin et al. 2018).

Referring to previous literature, rumours are unverified statements, propositions or beliefs that are concerned with certain circumstances, objects or people and are interesting to the social networks that they are passed along (Difonzo and Bordia 2007; Huang et al. 2015; Lee et al. 2015; Oh et al. 2013). Although sociologists often consider rumour-mongering as a collective social behaviour that assists people to interpret uncertain situations, evidences from research in information systems, marketing, organizational behaviours and finance disciplines have shown that the wide dissemination of rumours, especially the false ones, can have harmful impacts on individuals, organisations and the society (Difonzo et al. 1994; Difonzo and Bordia 2007; Shin et al. 2018). For individuals, rumours can cause misunderstanding to the situation, create stress, and mislead the people who intend to help those in need (Difonzo et al. 1994). For organisations and companies, rumours have been found to have negative influences on productivity and brand reputation, causing drops in sales, profit and stock price (Ahern and Sosyura 2015; Bordia and Difonzo 2004; Difonzo et al. 1994). For governments and the society, rumours may disturb

police investigation, create chaos and panic and seduce stigma and impulsive behaviours (Huang et al. 2015).

Rumours can be created, disseminated and consumed easily on social media platforms. The dissemination of rumour in the online social media has some characteristics that differentiate it from the dissemination in the traditional offline context. In offline rumourmongering, the sender and receiver must meet face-to-face to spread the rumour. However, in social media, without the constraints of physical network, distance and time, online rumour messages can reach out to an unprecedented scale within a very short period of time. CNN reported that misinformation online “spread faster than ever” (Leopold 2012). A rumour claiming an Italian restaurant chain finances Trump’s re-election bid was retweeted over 250,000 times within 48 hours (Daniels 2019). Additionally, messages in social media are persistent, and they are available to others and can be consumed at any time unless being deliberately removed. Due to the persistency and lack of physical constraint, the dissemination of online rumour messages in social media does not require the co-presence of sender and receiver. Also, because of these unique characteristics, the dissemination of online rumour messages in social media also tends to be easier and faster than in offline context, thus may cause stronger financial and social harms than offline rumours may cause (Ahern and Sosyura 2015; Bordia and Difonzo 2004; Difonzo et al. 1994; Shin et al. 2018). For example, in 2018 a rumour about the accounting practices of an Indian company named Infibeam Avenues Ltd was disseminated quickly on WhatsApp and caused the company to lose around 71% of its market value within a very short period of time (Patwa, 2018). Moreover, social media

supports a wide range of message receivers and is scalable (Berthon et al. 2012). In the offline context, rumour messages are passed along to certain intended receivers, and the number of receivers is often limited due to time and space constraints. In social media, however, rumour messages can be posted to oneself as an intrapersonal communication, shared with specific others as an interpersonal interaction, broadcasted to the user network as a sharing to the public. Regardless of who the intended receivers are, the messages can reach wide range of receivers, including the unintended ones. Furthermore, social media lowers the requirement for comprehension. In offline context, because a rumour needs to be passed along in sender's own words, comprehending the rumour is a precondition of spreading the rumour. A person needs to understand a rumour before communicating it to others. Whereas in social media, the messages can be directly forwarded or easily copied and pasted without fully comprehending the message. In terms of the delivery of rumour messages, social media in general is less competent at showing non-verbal cues such as gesture than face-to-face communication. Instead, social media enriches textual content with pictures, videos, weblinks and networks (Berthon et al. 2012). Moreover, the reputation or social standing of users, which is often implicit in the offline world, can be explicit in the social media via functions such as number of followers and likes (Kietzmann et al. 2011). Also, while receivers of a rumour mainly get information from a sender and can hardly refer back to the source in the offline context, the source of rumour can be visible and traceable in social media via the sender. Social media also enables the tracking of rumours that is hardly achievable in face-to-face context. Traditionally rumours are thought to be fleeting phenomena that are difficult to trace, and the most common methods

of studying rumouring is through experiments (Bordia and Difonzo 2004). With social media, it is possible to capture the lifecycle of a rumour. This allows the tracking of rumours and rumour messages in the real world, or putting it more accurately, in the real online world. The comparison between online and offline rumour dissemination is summarised in Table 1.

**Table 1 Comparison between Online and Offline Rumours**

|                                                            | Rumour<br>Dissemination in<br>Offline Context | Rumour<br>Dissemination in<br>Social Media |
|------------------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| Physical Constraint                                        | High                                          | Low                                        |
| Persistency                                                | Low                                           | High                                       |
| Co-presence of communicator<br>and receiver simultaneously | Required                                      | Not Required                               |
| Reach                                                      | Narrowly targeted<br>receivers                | Wide range of<br>receivers                 |
| Comprehending being a<br>precondition                      | Yes                                           | No                                         |
| Non-verbal cues                                            | Strong                                        | Weak                                       |
| Verbal cues                                                | Strong                                        | Stronger                                   |
| Pictures, videos, weblinks and<br>networks                 | Weak                                          | Strong                                     |
| Reputation of sender                                       | Implicit                                      | Explicit                                   |
| Visibility of source                                       | Low                                           | High                                       |
| Tracking of rumour                                         | Very Difficult                                | Feasible                                   |



Because of the above differences between rumour dissemination in social media and in offline context, to manage the tough challenges posted by online rumour messages, it is important to understand how those messages disseminate in the context of social media. This study seeks to understand the drivers and patterns of online rumour message dissemination, so that theoretical understanding on dissemination can be advanced to help control rumour dissemination, develop rumour combating strategies and manage the potential harmful consequences.

While previous research mainly studied rumours in the research contexts of political campaigns, natural disasters and social crises, business-related rumours, the rumours that are about brands, companies, their offerings or operations, are prevalent in social media yet are often neglected in prior research. Among business-related rumours, the false ones are usually beyond the rumour target's expectation and it is almost impossible for companies to prepare an upfront handling mechanism prior to their disseminations. The control of such false business-related rumours thus heavily relies on understanding of the dissemination of rumour messages. This makes this kind of rumour a representative and important context to study the dissemination of online rumour messages in social media.

## **1.2 Theoretical Gap, Research Questions and Research Objectives**

Traditional rumour research mainly studies the dissemination of rumour in the offline context. Because offline rumour dissemination heavily relies on word-of-mouth and communications between individuals, previous works pay strong attention to the individual recipients, emphasizing how individual motivations and perceptions shape rumour sharing behaviours (Allport and Postman 1945; Bordia and Difonzo 2004; Kimmel and Audrain-Pontevia 2010). In social media, the importance of individual characteristics tends to be weakened because interpersonal relationships are less significant in broadcasting communications in social media. Instead, the message feature tends to play a more important role and persuasive appeals are important for understanding the roles of message features (Das et al. 2015; Rosselli et al. 1995). Recent studies in information systems (IS) start to investigate rumour phenomenon in social media from message perspective (Mondal et al. 2018; Oh et al. 2013; Zhao et al. 2016). IS scholars are paying increasing attentions to online rumours. Oh et al. (2013) applied rumour theory to evaluate the quality of citizen-generated information in microblogs, and Nourbakhsh et al. (2016) studied semantic characteristics of rumours online. These studies used message-level variables and social media data for analysis, providing guidelines and supports for this thesis.

Although increasing attention is paid to studying rumours in social media from the message perspective (Mondal et al. 2018; Oh et al. 2013; Z. Zhao et al. 2016), limited attention has been paid to the structural and informational themes

of those online rumours and rumour messages. Considering that social media is a unique platform for rumour dissemination, and the messages spread on social media are often unstructured, prior to the investigation of rumour message dissemination, it is important to explore the themes and characteristics of rumours and rumour messages that are spread online in social media. Therefore, the first research question is:

***Research Question 1: What are the themes of online rumours and online rumour messages that are disseminated in social media?***

This question aims to develop a thematic understanding on the rumour topics that are dispersed in social media. Traditional rumour theories have identified different types of rumour and various motivations for rumouring. Via thematic analysis, this thesis attempts to extend the understanding of traditional rumour theories in the context of social media and focuses on exploring the typology and traits of online rumour topics and rumour messages. Therefore, ***the first research objective is to explore the typology of online rumour topics and the themes rumour messages with thematic analysis.*** When exploring the typology of online rumour topics, the unit of analysis is every rumour topic, whereas when understanding the themes of online rumour messages, the unit of analysis is each piece of online rumour message. This exploratory study facilitates the investigation of online rumour message dissemination in research question 2 and 3.

The dissemination of rumour in social media entails messages being

transmitted among individuals. Such transmission can be considered as the communication process from a communicator to receivers. Therefore, communication model is used to frame the components of factors that influence the dissemination of online rumour messages in social media. Traditional rumour theories usually emphasise on individual's motivations of rumouring behaviours, and often employ experiments to study how rumours emerge and what makes individuals create, believe and distribute rumours (Bordia and Difonzo 2004; Bordia and DiFonzo 2002). In social media, online rumour messages are often broadcasted and can be accessed by a group of receivers or the public. The effects of individual recipients' attributes on rumour diffusion are thus diminished in such context. Instead, the message content and sender characteristics tend to play a more important role. Therefore, investigating the factors that influence the dissemination of online rumour messages from the message and sender perspectives can provide helpful insights to existing rumour research. Based on this, the second research question this thesis is:

***Research Question 2: What cause certain online rumour messages get spread more frequently than others in social media?***

This question aims to understand the dissemination of online rumour messages on the rumour message level. The dissemination of online rumour messages entails messages to get forwarded and spread out. With each piece of online rumour message as a unit of analysis, this study proposes and examines a theoretical framework of the dissemination of online rumour messages. Specifically, rumour theory and communication theory are used as the

overarching theoretical background to identify the factors that influence the dissemination of online rumour messages in social media, and persuasion theory, emotion theory and congruence theory are used to specify how the variables influence the number of retransmissions of online rumour messages in social media.

Existing rumour theories have recognised the importance of ambiguity, belief and anxiety in the spreading of rumour. However, content ambiguity was found to play an insignificant role in one of the recent rumour studies in social media context (Oh et al. 2013). Given the inconsistent findings from prior research, further investigation is necessary to understand the role of ambiguity. Thus, ***the second research objective is to investigate how ambiguity of online rumour message influences its number of retransmissions.***

Moreover, existing research has almost exclusively focused on the role of anxiety. Rumours, however, can evoke other emotions such as fear, anger and hope (Allport and Postman 1946; Rosnow and Foster 2005; Zhu and Smith 2016). Given that different emotions may influence people's beliefs, attitudes and behaviours differently, there is a need to incorporate the effects of different emotions on rumouring. Therefore, ***the third research objective is to investigate how emotions expressed in online rumour message influences its number of retransmissions.***

In addition, the credibility and social influence of rumour communicators are hard to capture in the offline context and thus are often omitted in rumour

related research. Social media, on the contrary, displays the social standing and potential influence of each user (Gunarathne et al. 2018; Kietzmann et al. 2011). Moreover, in social media, the communicator's social network may determine the breadth of message dissemination. Because of the above, understanding the influence of communicator's attributes are important especially for rumour message dissemination in social media. Moreover, social media also provides various interactive functions that allow rumour communicators to communicate with different targets. Therefore, ***the fourth research objective is to investigate how rumour communicator's attributes and communication channels influence the number of retransmissions of online rumour messages.***

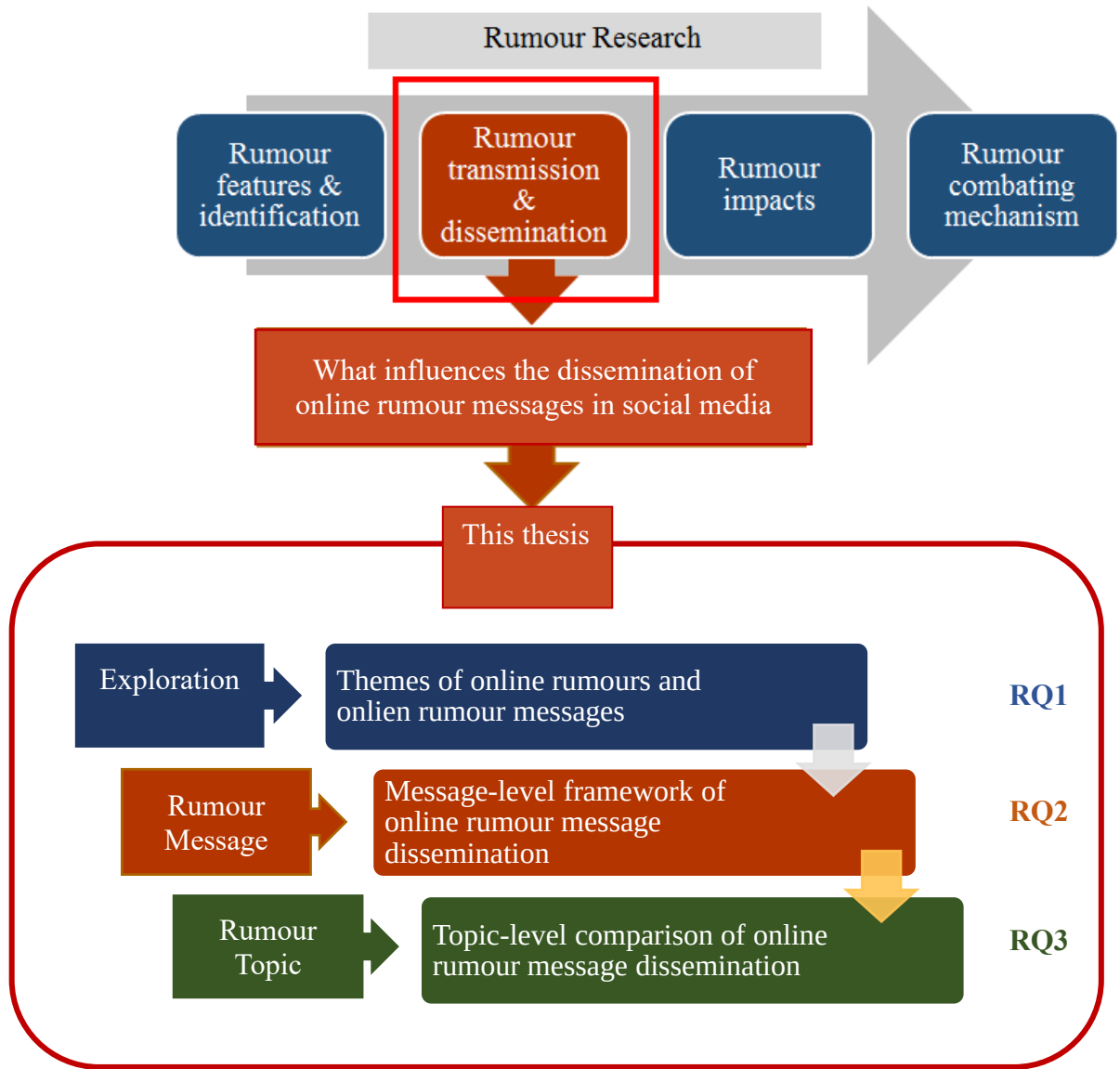
Furthermore, belief is commonly accepted as an important antecedent of rumour dissemination based on traditional rumour theories. However, there are other variables that can lead to rumour dissemination without the mediating of belief. Prior attitudes on rumour target is one of such variables. However, existing literature rarely investigates the attributes of rumour target. Therefore, ***the fifth research objective is to investigate how rumour target's attributes influence the number of retransmissions of online rumour messages.***

Apart from the variance in number of retransmissions among rumour messages, the dissemination pattern of rumour messages may vary across rumour topics. Thus, the third research question of this thesis is:

***Research Question 3: How are rumour messages and rumour message disseminations different between different types of online rumour topics?***

This question aims to understand the dissemination of online rumour messages on the rumour topic level. This study compares the online rumour message dissemination model under different types of online rumour topics. Moreover, with all rumour messages of each rumour topic as the unit of analysis, this study describes the general characteristics of rumour dissemination in social media and add topic-level insights to supplement the message-level dissemination framework in question 2. Accordingly, ***the sixth and final research objective is to summarise the rumour message dissemination pattern at the topic-level.***

Although it can also be important to identify rumours from non-rumours, to study the impacts of rumours and to investigate the strategies to combat rumours, this thesis focuses on investigating what influences the dissemination of online rumour messages in social media. The three research questions of this thesis are summarised in Figure 1.



Note: RQ1=research question 1; RQ2=research question 2; RQ3=research question 3

**Figure 1 An Overview of Research Questions of this Thesis**

## 1.3 Overview of Research Design and Methodology

### 1.3.1 Scope of Research

To address the research questions proposed, this thesis focuses on business-related false rumours and selects Twitter as the social media platform. Business-



related false rumours that are related to specific brands, companies, their products, services and/or business operations and that turned out to be false, are targeted for the following reasons. First, business-related rumours are prevalent in social media and can cause harmful impacts, yet limited attention has been paid to them. The spreading of business related rumours on social media is not uncommon and has led to harmful consequences for companies such as drop in stock market price and damage to reputation (Boland 2017; Bordia et al. 2005). However, business related rumours are usually studied in the offline context, focusing on the effects of workplace rumours on employee performance (Difonzo et al. 1994). Most previous studies on rumours in social media focuses on natural disasters, social crises, and social events (Liu et al. 2014; Mondal et al. 2018; Oh et al. 2013; Shin et al. 2018; L. Zhao et al. 2016), yet business-related rumours are often overlooked. Second, false rumours, compared to true rumours, are harder to predict and understanding the dissemination is especially important for rumour control. Unlike rumours that turned out to be true, business crises or word-of-mouth complaints which are based on events or incidents that actually happened, false rumours are usually beyond the rumour target's expectation. Because of this, it is almost impossible to think of an upfront handling mechanism prior to its occurrence. The control of such false rumours thus heavily relies on understanding the dissemination of rumour messages. Understanding the dissemination allows the rumour target to track and handle such rumours and minimize the harms they may cause. Therefore, business-related false rumour serves as an important context for investigating the research questions.

Twitter is chosen because it is one of the most popular and representative social media platforms and has been widely accepted by researchers who study rumours in social media (Gunarathne et al. 2018; Liu et al. 2014; Oh et al. 2013). Twitter was launched in 2006 and has around 330 million monthly active users in 2018 (Aslam 2018). By the end of 2017, approximately 100 million users on Twitter are daily active users and about 500 million tweets are posted each day (Aslam 2018). Moreover, it has been reported that a large number of false rumours were disseminated on Twitter (Usui et al. 2015). Given the popularity of Twitter usage, the prevalence of false rumours on Twitter, as well as the publicity of Twitter data, previous studies have considered Twitter as an appropriate research context for studying online rumours (Oh et al. 2013). Therefore, conducting research in the context of Twitter enables the addressing of listed research questions.

### **1.3.2 Research Methods and Data Analysis**

A multi-method design is used to address the research questions regarding the dissemination of online rumour messages. Both qualitative and quantitative research methods with inductive and deductive reasoning are employed to analyse social media data collected from Twitter. To address the first research question, inductive qualitative content analysis is applied to analyse the themes and features of online rumours and rumour messages disseminated in social media. At this stage, this thesis explores the online rumour phenomenon and enhances the understanding towards such phenomenon. The results of this stage are used to support the development of research model and hypotheses, which are concerned with the second research question. The research model and

hypotheses are tested using regression analysis with social media data operationalised via manual coding and sentiment analysis. To answer the third research question, linguistic analyses are applied to describe the patterns of online message for different types of online rumour topics, and the framework of online rumour message dissemination is compared across types of online rumour topics as well.

## **1.4 Summary of Results**

Through qualitative analysis, this thesis finds that online rumour topics can be mainly categorised into brand-related rumour, offerings-related rumour, people-related rumour and operations-related rumour based on the object of rumour, and into description-oriented rumour, reason-oriented rumour and impact-oriented rumour based on the focal point of rumour. Online rumour messages are found to consist of three general themes, namely content unambiguity, source unambiguity and affective appeal. Specifically, content unambiguity includes expression uncertainty and use of evidence, source unambiguity consists of source explicitness and source accessibility, and affective appeal comprises emotional words and imagery words. Based on persuasion theory, content unambiguity and source unambiguity can be considered as the cognitive appeals of online rumour messages. Both cognitive appeals and affective appeals can influence message persuasiveness and message receivers' behaviours. The findings of the themes of online rumour messages are used to support the development of a message-level research framework of online rumour message dissemination.

Using regression to examine the proposed research framework, the results show that the communicator features (i.e., communicator trustworthiness, communicator social influence and communicator likeability) are all positively associated with the number of retransmissions, in this case, the number of retweets of online rumour messages. Regarding the effects of communication channel, community communication is positively associated with the number of retweets, whereas the direct communication is negatively associated with the number of retweets. In terms of how cognitive and affective appeals of the message influence the number of retweets, the results suggest that expression certainty and emotional activation are positively associated with the number of retweets. However, the source explicitness, source accessibility, emotional pleasantness and emotional imagery are negatively associated with the number of retweets. Also, the use of evidence, is found to have no significant influence on the number of retweets.

For the effects of rumour features, the results reveal that both topic-congruence (i.e. the congruence between target reputation and rumour topic valence) and message-congruence (i.e. the congruence between target reputation and rumour message valence) are negatively associated with the number of retweets. Apart from the direct effects, both types of congruence moderate the effects of message features on the number of retweets. Specifically, topic-congruence significantly moderates the effects of affective appeals on number of retweets. It also significantly moderates the effect of source accessibility on number of retweets. However, it is not a significant moderator for the effects of other cognitive appeals on number of retweets. In terms of message-congruence,

it significantly moderates the effects of use of evidence, source accessibility, emotional pleasantness and emotional imagery on number of retweets. However, message-congruence does not moderate the effects of expression certainty, source explicitness and emotional activation on number of retweets.

Through comparing the online rumour message dissemination at the topic-level, this thesis finds that for different types of rumour topics, the effects of variables on driving online rumour dissemination are different. Further exploration also reveals that the number of rumour messages and duration of survival vary across rumour topics. Moreover, dissemination of rumour messages tends to have a tipping point where the count of rumour messages accelerated dramatically and then faded quickly. For most of the rumour topics, only one tipping point is observed, whereas for a few rumour topics, there are several bursts.

## **1.5 Summary of Research Contributions**

### **1.5.1 Theoretical Contribution**

This thesis contributes to the literature in the following ways. First, this thesis extends traditional rumour theory to the context of social media by introducing new dimensions to the key concepts of rumour theory according to the new context. Expression certainty and use of evidence are introduced as new dimensions of content unambiguity, whereas source explicitness and source accessibility are introduced as new dimensions of source unambiguity. Through examining the associations between these new variables and online rumour

dissemination, this thesis clarifies how information unambiguity influences rumour dissemination in the new context.

Second, this thesis contributes to rumour theory by investigating the effects of emotional content on the dissemination of online rumour messages. Combining rumour theory with emotion theory, this thesis uses pleasantness, activation and imagery to measure emotional valence, level of arousal and message vividness which are the three general aspects of emotional content. By doing so, this thesis contributes to rumour theory by generalising the effects of emotions on influencing rumour dissemination.

Third, this thesis combines congruence theory and persuasion theory to provide new insights to rumour theory. The congruence between rumour context features and rumour message features is considered. Specifically, by taking topic-congruence and message-congruence as moderators, this thesis clarifies how the effects of cognitive appeals and affective appeals on rumour dissemination differentiate.

Moreover, this thesis enriches rumour theory by outlining how rumour disseminates differently across rumour topics. A typology of online rumour topics is proposed, classifying online rumour topics based on their objects and focal points. Referring to the types of online rumour topics, this thesis compares how the research framework of online rumour message dissemination works differently for different types of online rumour topics.

Methodologically, this thesis uses social media data to track online rumour messages and to study online rumour message dissemination. By capturing social media data, this thesis is able to trace rumours that were traditionally untraceable. Using social media data, this thesis is able to capture the actual rumour message, communicator attributes and dissemination behaviours. By doing so, this thesis investigates the actual rumour dissemination in social media and contributes to extant rumour research.

### **1.5.2 Practical Implication**

This thesis has several practical implications. This thesis can guide practitioners regarding what features of online rumour messages they should pay attention to so that practitioners can monitor and control rumours efficiently and effectively. In general, practitioners can set alert to the online rumour messages that are certain in expression but does not have a clear source or a link to the source. Practitioners should also pay special attention to the online rumour messages that communicate negative emotions with high level of arousal, such as anger and hatred. Because these online rumour messages are likely to be disseminated, practitioners should put such rumour messages in priority when controlling online rumours. If the practitioners have a track of their companies' reputations, they should be aware that whether their reputations are congruent with the rumour topic valence and each rumour message valence and pay different levels of attention to different types of rumours. In addition, social media users tend to have different rumouring motivations, and there are different types of rumour topics. Therefore, a company's response to online rumour messages should be tailored to meet the specific motivation according to the types of rumour topic.

To sum up, understanding message-level factors that influence online rumour message dissemination, the themes of online rumour topic and the topic-level dissemination patterns of online rumour messages can guide practitioners to develop appropriate coping strategies and to minimize the damage to rumours.

## **1.6 Thesis Structure**

This thesis focuses on understanding the dissemination of online rumour messages in social media and seeks to investigate this phenomenon from rumour message level (Research Question 2). However, to support and supplement the message-level framework, qualitative exploration on rumour topic and rumour message is conducted (Research Question 1). Moreover, the framework is compared across rumour topics to explore message dissemination pattern at rumour-topic level (Research Question 3). Specifically, the thesis is structured with 6 chapters (see Figure 2).



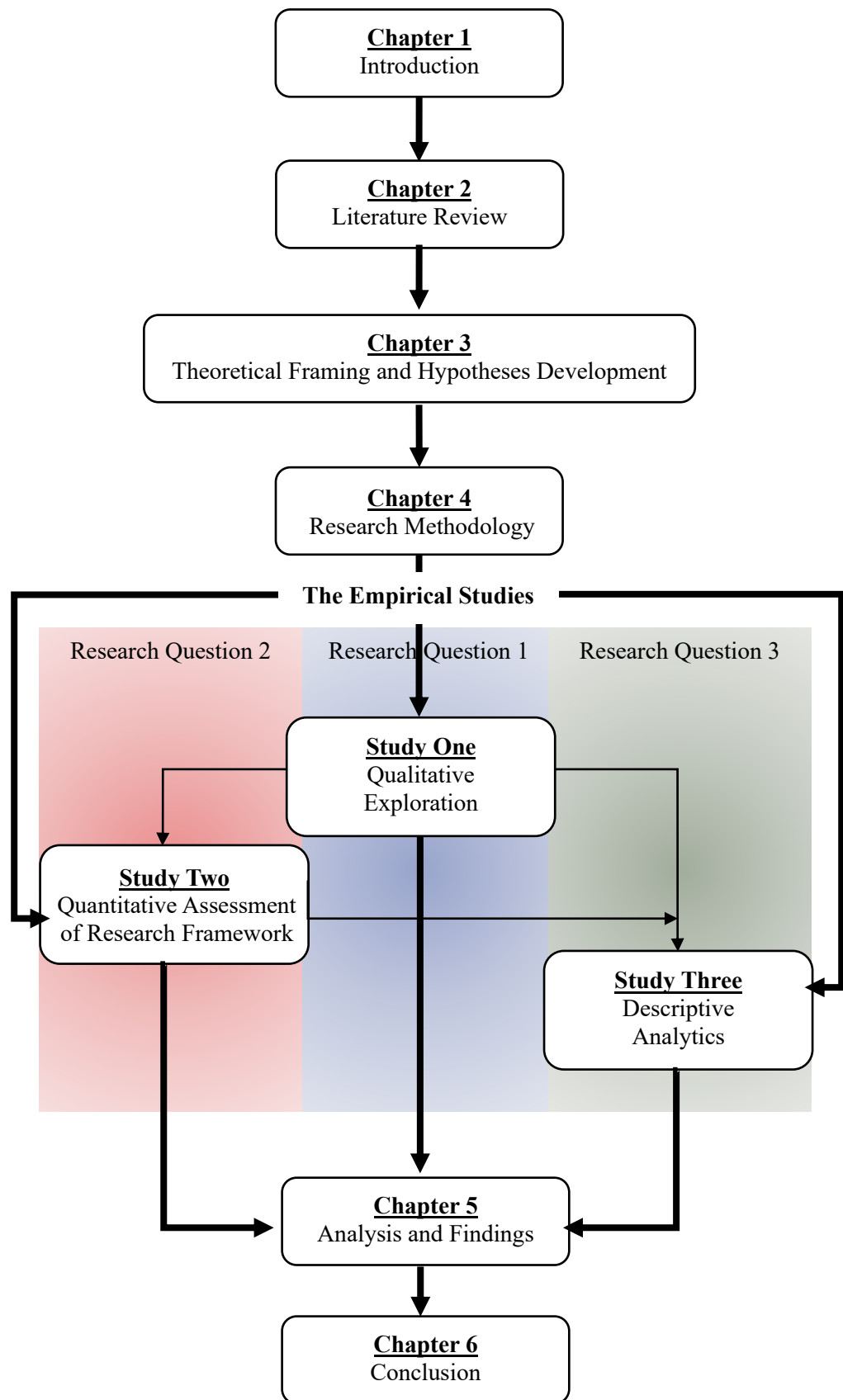


Figure 2 Structure of Chapters

*Chapter 1 Introduction* provides an overview to the research background and motivation and summarises the thesis. First, research background and motivation are presented. Second, theoretical gap and research questions are highlighted. Then, the overview of research methodology is presented, following with a summary of research findings. In addition, the main theoretical contribution and practical implications are summarised. After this, the structure of the thesis is presented, following with the summary of the chapter. *Chapter 2 Literature Review* defines key concepts of this thesis and provides a general overview of relevant literature and theories. In this chapter, the concept of rumour is first defined and distinguished from other similar constructs. The definition of social media is then clarified. Afterwards, the rumour theory, communication theory, emotion theory and congruence theory are reviewed to provide the theoretical foundation for the research. Based on extant theories, *Chapter 3 Theoretical Framing and Hypotheses Development* outlines the research framework and builds up the hypotheses. In *Chapter 4 Research Methodology*, the philosophical foundation of the research strategy is described first based on which the research design and methods are selected. Then, the research design is presented, providing an overall picture of the three studies conducted in this thesis, namely the qualitative exploration (study 1), the quantitative assessment of research framework at the message level (study 2) and the comparison of research framework at the topic level (study 3). After that, the context and scope of research, the process of data collection, and the main research methods of data analysis are described. Following this, the details of each study are explained respectively. A discussion on the research ethics is also included before the summary to the chapter. *Chapter 5 Data Analysis and*

*Findings* reports the results and findings of the studies and discusses the findings referring to existing literature. Finally, *Chapter 6 Conclusion* summarises the entire thesis. It first recaps the main research aims, design and findings. Second, it highlights the theoretical and practical contributions this thesis has. Finally, it concludes the thesis with a discussion on research limitations and future research directions.

## **1.7 Summary of Chapter**

This chapter introduces the research background and motivation of this thesis, explains the theoretical gaps and research questions, summarises the research design and data analysis, presents the theoretical contributions and practical implications and outlines the entire thesis. By doing so, this chapter offers a broad overview of the entire thesis and lays the foundation for the rest of the chapters. The next chapter will conceptualise the key concepts and review relevant theories and literature concerning online rumour message dissemination.

## **CHAPTER 2 LITERATURE REVIEW**

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The dissemination of online rumour messages in social media is the core of this thesis. While people tend to hold intuitive perceptions on what rumour, online rumour message and social media mean in daily life, the definitions on these key concepts vary across previous studies. To clearly define and conceptualise the key concepts of the thesis, this chapter first reviews previous literature on rumour, rumour dissemination and social media.

Specifically, the first section defines rumour and online rumour messages. A comprehensive definition of rumour is provided with a brief overview of different classifications of rumour. Meanwhile, the context choice of this thesis is explained, justifying why this thesis focuses on false business-related rumours on Twitter. Then, diverse interpretations of rumouring behaviours are presented. Afterwards, rumours are distinguished from similar concepts including news, gossips and legends. In particular, because false rumour is often confused with fake news, the similarity and difference between false rumour and fake news are discussed. Following this, the meanings of online rumour and online rumour messages are clarified. The second section defines the context (i.e. social media)

and reviews how online information dissemination is operationalised. Section three, based on previous definitions and existing literature, conceptualises the dissemination of online rumour messages.

Following the conceptualisation of key concepts, section four to section seven identifies the core theories and reviews relevant studies. Specifically, section four summarises rumour theories, especially the works that are related to rumour dissemination. Section five reviews relevant works in communication theories, including Lasswell's Model of Communication and the Elaboration Likelihood Model. Emotion theories and related works are reviewed in section six. The final section reviews the congruence theory from the schematic information processing perspective.

## **2.1 Conceptualisation of Rumour**

### **2.1.1 Defining Rumour**

The definition of rumour varies across existing literature. Summarising previous studies (Table 2), this thesis defines a rumour as an unverified statement, proposition or belief that refers to certain circumstance, object or person and is topically relevant to the social networks that it is passed along (Difonzo and Bordia 2007; Huang et al. 2015; Lee et al. 2015; Oh et al. 2013). Such definition conceptualises a rumour as consisting of four elements, namely unverified hypothesis or belief, referent of statement, topical relevance, and communication.

**Table 2 Summary of Rumour Definitions from Previous Studies**

| <b>Definition of Rumour</b>                                                                                                                                                                                                | <b>Study</b>                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| An unconfirmed message passed from one person to another in face-to-face interaction that refers to an object, person, or situation rather than an idea or theory.                                                         | Buckner (1965, p. 55)                                                |
| Unverified proposition or belief that bears topical relevance for persons actively involved in its dissemination.                                                                                                          | Bordia and Difonzo (2004, p. 33)<br>Rosnow and Kimmel (2000, p. 122) |
| Any report, statement, or story that one may have heard for which there is no immediate evidence to verify its truth.                                                                                                      | Rosnow et al. (1988, p. 32)                                          |
| A belief that diffuses, usually rapidly, among members of a particular population. The belief is often unverified, and thus it can be factually wrong.                                                                     | Bhavnani et al. (2009, p. 878)                                       |
| Messages circulating within a social network that carry instrumentally relevant information to help people manage uncertainty or threats when verified information is absent                                               | Zhu and Smith (2016, p. 1345)                                        |
| Unverified information, usually of local or current interest, intended primarily for belief... propositions or allegations colored by various shades of doubt, because they are not accompanied by corroborative evidence. | Difonzo (1994, p. 50)                                                |
| Unverified and instrumentally relevant information statements in circulation that arise in contexts of ambiguity, danger, or potential threat and that function to help people make sense and manage risk.                 | Brooks et al. (2013, p. 5)<br>DiFonzo and Bordia (2007, p. 13)       |

Rumours are *unverified hypothesis or belief*. They emerge when no valid and reliable explanations are available (Bordia and Difonzo 2004). Receivers of rumours are not always exposed to the verifications. Sometimes even when official announcements are made by the mass media, government or corporate management, such announcements are not necessarily received by the rumour receivers, and even if they do, some receivers may doubt the credibility of verifications. In fact, the truth and plausibility in one's mind can be entirely different from those in another's mind. Nevertheless, rumour messages still get spread. Therefore, this study does not use individual subjective perception to measure rumour; instead, this study follows the code scheme developed by Oh et al. (2013) to define rumour, and treats information as rumour if it arises in situations with no credible and commonly accepted substantiation. Notably, the non-substantiation does not mean that rumour is false information. It is always possible that unverified statements are true. However, because of lack of authentic knowledge, such true information may be interpreted and spread as a rumour (Mondal et al. 2018).

To be a rumour, such unverified hypothesis or belief needs to have a *specific referent* or a *target*, which might be a person, a group of people, an object or an event. Moreover, the hypothesis or belief is usually *related to topical issues* that people are interested in or concerned about (Bordia and Difonzo 2004). Traditionally, some studies refer to such interest as "local and current interest" (Difonzo et al. 1994, p. 50); however, because of the Internet, physical contact is no longer a requirement for transmitting rumours. Therefore, the interest towards rumour topic is not necessarily local. In addition, another very important



element of rumour is the *communication* of the hypothesis or belief within the social networks from one person to another (Buckner 1965). Having the proposition in one's mind without communicating it to others would not make the proposition a rumour. Communication is a crucial mechanism of rumour emergence and transmission (Bhavnani et al. 2009).

Rumours can be categorised from different perspectives. Rumour, by nature, is unverified information and thus can turn out to be true or false (Allport and Postman 1947). Based on the credibility of information carried, rumours can be categorised into true rumours and false rumours. Based on the valence of consequences, rumours can be divided into positive rumours and negative rumours. Furthermore, based on the emotions they evoke, rumours can be categorised into dread rumours, wedge-driver rumours and pipedream rumours (Knapp 1944). Dread rumours are mainly about fearful and disappointed consequences and usually evoke fear and anxiety, wedge-driver rumours are mostly driven by hatred and aggression and tend to divide groups and cause stigma, and wish rumour or pipe-dream rumour are those that raise hopes and represent wishful thinking (Allport and Postman 1946; Knapp 1944).

Given that false rumours are usually beyond the rumour target's expectation and are hard to be identified and controlled, these rumours tend to be the most threatening ones. Moreover, although the context of rumour has not been used to categorise rumours, there are indeed certain circumstances where rumours are likely to become viral. Rumouring is especially common during crises, natural disasters and social events or when targeting politicians and businesses (Liu et

al. 2014; Oh et al. 2013; Weeks and Garrett 2014). As pointed out by Kamins et al. (1997) and Bordia et al. (2005), many companies, such as Procter and Gamble (P&G) and McDonald's, have been the targets of rumours, including false rumours, and such rumours may cause serious negative consequences. While recent studies have studied crises/disaster-related rumours and politics-related rumours in the online environment, limited attention is paid to the business-related rumours on the Internet (Garrett 2011; Liu et al. 2014; Mondal et al. 2018; Shin et al. 2018; L. Zhao et al. 2016). Business-related rumours can cause severe impacts such as decrease in employee performances, drops of stock market price, decrease in sales and reputation damage (Difonzo and Bordia 2002; DiFonzo and Bordia 2000). Therefore, this thesis focuses on the business-related false rumours. Understanding the dissemination of such rumour messages on social media allows companies to track and handle the unexpected false rumours and minimize the harms they may cause.

### **2.1.2 Interpreting Rumouring**

The emergence and transmission of rumour can be interpreted from different perspectives. Existing research often conceptualises rumouring as the collective sense-making, the group anxiety releasing, or the dissemination of information.

Sociologists and social psychologists perceive rumouring as a collective reaction to information ambiguity and scarcity in uncertain situations (Huang et al. 2015; Oh et al. 2013; Shibutani 1966). From this perspective, rumouring is a process of information convergence (L. Zhao et al. 2016). Sometimes people fail to understand what has happened or what is going on in a situation. To meet

information needs and control social tension, people will seek and exchange information. When valid information is not available from mainstream media, people turn to their informal social networks or other sources and collaborate with each other to understand the situation. Based on improvised information, people try to portray their own versions of truth, which might then be consumed and processed by others (Huang et al. 2015; Oh et al. 2013). Thus, rumours are created and transmitted to understand the social world and take effective action (DiFonzo et al. 2013). Through collective and collaborative community efforts, rumouring can lead to a unified understanding towards the uncertain situation, which is likely to be the correct one. From this perspective, rumours are products of collective efforts to solve information ambiguity, cope with situational uncertainty and release emotional tension (Shibutani 1966). Such collective sense-making is similar to several other phenomena in social psychology, such as attitude polarization, collective information processing, group problem-solving and social contagion (DiFonzo et al. 2013).

Notably, Allport and Postman (1946) pointed out that rumour is not merely a substitute for reliable information. When important issues occur, unless people are certain that they know the facts, even with plenty of news reports, they may still go beyond simply accepting the story told by others (even when the source is credible) and create their own fantasies.

Apart from functioning as helping people make sense and manage risk, rumour also functionalises to relieve and to rationalise (Allport & Postman 1946). Rumours allow people to relieve certain emotional urge and reduce

anxiety, and at the same time give people the chance to justify his/her feelings to himself/herself or to other people.

Rumouring can also be seen as the dissemination of information and distribution of messages. This perspective is mostly adopted by researchers who seek to understand the spreading of rumour and thus is the most relevant to this research. Some studies from this perspective focuses on modelling the propagation or diffusion of rumours mathematically (Z. Zhao et al. 2016), whereas the rest of the research from this perspective usually consider rumouring as the process of communication and transmission of messages (Kamins et al. 1997; Liu et al. 2014). This thesis is based on the latter standpoint and conceptualises the dissemination of online rumour messages as a communication process.

### **2.1.3 Distinguishing Rumour from News, Gossips and Legends**

In everyday life, people tend to mix up rumours with similar concepts such as gossips (Michelson and Mouly 2002). Scholars who study rumour have recognised this issue. To clarify the conceptualisation of rumour, scholars have identified a few similar concepts, mainly news, gossips and legends, and distinguish rumour from them (DiFonzo et al. 2013; Mullen 1972; Rosnow and Foster 2005). Summarising from previous literature, Table 3 compares the traits of rumour, news, gossip and modern legends.

**Table 3 Comparison of Rumour, News, Gossip and Modern Legend**

|                                        | <b>Rumour</b>          | <b>News</b> | <b>Gossip</b>                            | <b>Modern legend</b>                     |
|----------------------------------------|------------------------|-------------|------------------------------------------|------------------------------------------|
| <b>Information contained</b>           | Unverified             | Verified    | Unverified                               | Unverified                               |
| <b>Topic of relevance</b>              | Relevant               | Relevant    | Relevant or irrelevant                   | Irrelevant                               |
| <b>Aim of communication</b>            | To fill knowledge gaps | --          | To entertain or change social structures | To entertain or communicate moral themes |
| <b>Level of importance</b>             | Important              | Important   | Not important                            | Not important                            |
| <b>Role of belief in dissemination</b> | Significant            | --          | Not significant                          | Not significant                          |

Similar to news, rumours are usually about important issues with collective concerns (Difonzo et al. 1994). However, different from news which are substantiated, rumours are unverified (Bordia and Difonzo 2004; Difonzo et al. 1994). Despite ambiguity being often associated with rumours, unambiguity is a key factor of news in which the events or stories need to be clearly meaningful and interpretable (Perry 2002).

Rumours, gossips and legends all share the traits of non-substantiation. Nevertheless, rumours mostly concern topical issues of significant importance to the public or certain groups of people, and such topic can be about people, events, or objects, whereas gossips are usually idle talk regarding people that aims for entertaining, maintaining interpersonal relationships or isolating someone from the social group (Bordia et al. 2005; Difonzo and Bordia 2007).

Compared to legends and folklore, rumour usually involve non-traditional materials and does not need to have a narrative element (Mullen 1972). In fact, legends can be considered as the conventionalized accounts of rumours. While there is no particular point during transmission when a rumour becomes traditional, legends are rumours that has become part of the verbal heritage of a people.

However, modern legends is also not entirely traditional. Modern legends, also known as urban legends, are usually narratives with horrible, humorous or unusual events that contain modern or contemporary themes and moral implications and that are usually circulated orally (Bordia and Difonzo 2004; Mullen 1972). Modern legends, similar to gossips, tend to have no major significance or importance (Difonzo et al. 1994); rumours, however, has a sense of urgency and are intended to solve information scarcity and fill in knowledge gaps in situations where reliable information is not available or people lack the cognitive ability to understand the situation clearly (Bordia and DiFonzo 2002; Difonzo et al. 1994; Rosnow 1988).

Moreover, rumours, gossips and modern legends influences transmission processes differently at both individual and network level. When rumours are primarily motivated by fact-finding, belief in the rumour will play a leading role in deciding whether to transmit it. In contrast, gossip is most often spread without regard to the transmitter's belief in it. Nor is belief important in spreading urban legends; they are often told primarily for entertainment or education motives.

#### **2.1.4 Comparing False Rumour with Fake News**

As discussed, the focus of this thesis is false rumour. False rumour is often mixed-up with fake news by practitioners and existing literature. Some studies distinguish false rumour from fake news, whereas other studies treat these two concepts interchangeably (Shin et al. 2018). This thesis acknowledges the similarities between false rumour and fake news yet considers false rumour distinguishable. While the previous sub-section has distinguished rumour from news, this sub-section compares rumour with fake news to identify their similarities and differences. Through such comparison, this thesis clarifies the nature of false rumour.

Both false rumour and fake news carry false information which is the main attribute of both. Notably, false information can be classified into misinformation and disinformation based on the intent of the creator or the source (Kumar and Shah 2018). Misinformation is created without the intent to mislead or deceive the readers, whereas disinformation is created with the intent to mislead or deceive. Without the intent to lie, misinformation is often caused by “misinterpretation, distortion of an original piece of true information... lack of understanding, attention or even cognitive biases” (Kumar and Shah 2018, p. 4). Therefore, misinformation is often generated and spread by the creator unintentionally. In contrast, disinformation is created and first communicated with the intent to mislead, aiming to sway public opinions or manipulate online traffic. Fake news is a form of disinformation (Kumar and Shah 2018). However, while some false rumours are generated deliberately, most of them are created unwittingly. False rumour is therefore a specific type of false information and

may carry misinformation or disinformation.

Although the creators may have different intents of creating and initially distributing fake news and false rumours, the receivers of both fake news and false information may re-transmit the information without being aware of its falseness. Rumour's contribution to collective sense-making often creates and spread false information unintentionally. However, rumours are also acknowledged for being generated and dispersed as a propaganda campaign with the intention, even malicious intention, to influence others (Difonzo and Bordia 2011).

## **2.2 Social Media and Information Dissemination**

### **2.2.1 Defining Social Media**

Social media is pervasive nowadays (Ngai et al. 2015). With the quick development and rich diversity of social media, researchers and practitioners may be confused about the definition of social media. The development of the Internet and the emergence of a series of popular platforms such as MySpace, Facebook and Twitter shaped the modern definition of 'social media' (Kietzmann et al. 2011; Witkemper et al. 2012). More recently, social media has been considered to consist of two key components: the rich and two-way interactions enabled by the technology (i.e. "social") and the rich and timely information exchanged via the technology (i.e. "media") (Gustafson et al. 2019).

Social media can be defined as the web or mobile based platforms on which



users can create, co-create and exchange user-generated content or link to other creators of content highly interactively (Hollebeek et al. 2014; Kaplan and Haenlein 2010; Laroche et al. 2013; Majchrzak et al. 2013). Usually in social media individuals can establish a profile available to the public or within certain restricted network where they can connect with a group of other users (Ellison 2008). Social media content includes text, pictures, videos, links and networks, and the use of social media facilitates the creation and sharing of information (Stieglitz and Dang-Xuan 2013). People may post and spread information on social media for different purposes. For example, Java et al. (2003) examined the characteristics of Tweets and identified several key themes of Twitter's usage, including daily chatter, conversations, information/weblink sharing and news spreading.

### **2.2.2 Information Dissemination on Social Media**

The diffusion of messages within a social network in face-to-face manner has long been studied (Katz et al. 1973). A great amount of recent research attention has been paid to the diffusion or dissemination of information on social media. Studies from the area of physical and computational science focuses on building mathematical or epidemiological model of information propagation (Petrovic et al. 2011; Sun et al. 2009; Z. Zhao et al. 2016). Research from the sphere of marketing studies the sharing of eWOM and viral marketing (Stieglitz and Dang-Xuan 2013). Research from organisational behaviour or information systems associate information dissemination with knowledge sharing and message dissemination. The information dissemination has been studied under different social media platforms, including microblogging websites such as

Twitter (Stieglitz and Dang-Xuan 2013), social networking sites (SNS) such as Facebook (Luarn et al. 2014), blogs (Wattal et al. 2010), picture or video sharing services such as Instagram and YouTube (Liu-Thompkins and Rogerson 2012; Qiu et al. 2015).

Previous studies investigated the effects of various factors, including content features and user characteristics, in influencing the dissemination of information in social media. Information is likely to be believed and shared if it is well written, long, and is well-referenced (Kumar and Shah 2018). Some examples of these previous studies are listed in

Table 4. Previous research studied the purpose of information dissemination and found that people mainly retransmit online posts to entertain others, to comment on specific post, to save the information for future access, etc. The quantity of retweets, speed of retweeting and the structure of retweet network have been used as indicators of message dissemination, and many factors have been studied as drivers of message dissemination (Yang and Counts 2009). For example, Suh et al. (2010) investigate the effects of URL, hashtag, number of followers, etc. in influencing retweet quantity. Nagarajan et al. (2010) found that the structure of information dissemination network varies depending on the characteristics of message. If a message calls for action, engages crowdsourcing or entails collective group identity making tweet, its dissemination network tends to be sparser than the retweet network of a message that emphasises information sharing. Previous studies found that the quantity of sharing is the most commonly used measure of information diffusion in previous studies (Stieglitz and Dang-Xuan 2013).

**Table 4 Previous Studies on Information Dissemination**

| <b>Study</b>                   | <b>Dependent variable</b>                         | <b>Independent variable</b>                                           | <b>Platform</b> |
|--------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|-----------------|
| Sutton et al. (2014)           | Information transmission<br>(quantity of retweet) | Tweet content, style, and network properties                          | Twitter         |
| Petrovic et al. (2011)         | Decision to retweet                               | Social features about the author and tweet features about the content | Twitter         |
| Zeng et al. (2016)             | Rate of rumour transmission (speed of retweet)    | Figures about authors, message content and external context           | Twitter         |
| Stieglitz and Dang-Xuan (2013) | Information diffusion (quantity of retweet)       | Sentiment of content                                                  | Twitter         |
| Susarla et al. (2012)          | Rate of diffusion                                 | Channel's social influence                                            | YouTube         |
| Bampo et al. (2008)            | Viral campaign performance (reach and speed)      | Social structure of digital networks                                  | --              |
| Berger and Milkman (2012)      | Decision to share                                 | Sentiment of content                                                  | --              |

## 2.3 Conceptualisation of Online Rumour Message Dissemination

### 2.3.1 Online Rumour Topic and Online Rumour Message

Online rumour in this study refers to a rumour that is created and/or transmitted

via the Internet. Specifically, this study focuses on online rumours that are transmitted through social media. In social media, online rumours are spread via online posts. Such posts that carry online rumours are defined as online rumour messages.

### **2.3.2 Defining Dissemination of Online Rumour Message**

The spreading of rumour, which is often used to describe the general rumouring phenomenon, consists of three stages: the genesis of rumour, the diffusion of rumour and the control of rumour (Liu et al. 2014). The genesis of rumour is about how and why rumour occurs. The sociological studies identify two conditions that are most commonly agreed for generating rumours, namely the importance of the rumour topic or target and the lack of valid evidence in the situation (Iyer and Debevec 1991). The control of rumour is about rumour combating strategies and the natural decline of rumour. Denial is one of the most commonly studied strategies to control rumour (Bordia et al. 2005). The diffusion of rumour is about the dissemination and transmission of rumour which this thesis is interested in. The diffusion of rumour has been the focus of many previous studies. Although some scholars content that the interpersonal dynamics of rumour transmission can be similar to the dynamics of sharing novel information, news or new products (De Fleub 1962), considering the uniqueness of social media, dissemination of online rumour messages in social media may have different dynamics. Sociologists tend to commonly agree that for rumours to diffuse, there needs to be cognitive unclarity, issue importance and common relevance to the social network (De Fleub 1962). It is acknowledged that the diffusion, transmission, retransmission, propagation and

dissemination of rumour all emphasise the flow of rumour; however, considering that dissemination tends to have a sense of broadcasting which is often the nature of social media communication, this thesis adopts the term dissemination to label the diffusion of online rumour messages in social media.

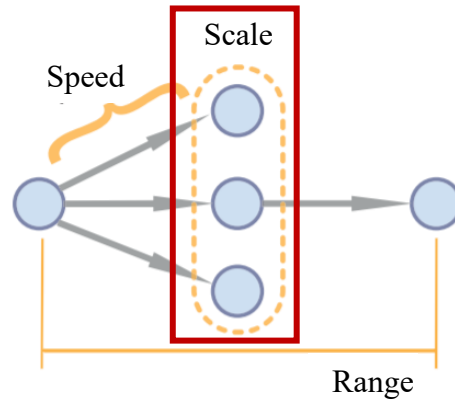
Marketing perspective tends to study rumour message dissemination under the rubric of word of mouth (WOM) because they both entail multiple actors and tend to influence the opinions of receivers (Hornik et al. 2015). In the online context, the dissemination of online rumour message has thus been considered as the communication of electronic WOM (eWOM). Rumour studies from information systems perspective heavily focuses on online rumour messages. From this perspective, the dissemination of rumour message is often assimilated with online information dissemination or online communication. Moreover, the rumour message tends to be considered as having persuasive power on the receivers. Therefore, apart from the rumour theory, communication theory including persuasion theory, emotion theory and information processing theory are important theoretical foundations for understanding the dissemination of online rumour messages.

### **2.3.3 Measuring Dissemination of Online Rumour Message**

Although many studies have investigated offline rumour spreading using experiments, limited research has paid attention to quantify the measurement of online rumour messages. Since the dissemination of online rumour message can be considered as a form of information diffusion, previous studies on information propagation and diffusion provide references for measuring the

dissemination in social media.

In Petrovic et al. (2013)'s research on information propagation on Twitter, whether a tweet will be reposted or not are used to label the diffusion. Apart from the binary retweeting measurements, quantity and speed of retweeting and reposting have also been used to measure online information diffusion (Stieglitz and Dang-Xuan, 2013; Hoang and Lim, 2011). Yang and Counts (2010) model information diffusion in social media with three dimensions, namely speed, scale and range (see Figure 3). Speed relates to whether the information is reposted and when the first diffusion instance takes place. Scale refers to the number of affected instances at the first degree, that is, the number of people who repost the same topic as the first-degree child nodes in the network. Range is about how far the diffusion chain can continue in depth, achieved by tracing a topic from an initial node to its antecedent nodes. Among these measurements, scale is most commonly used in previous studies (Yang and Counts 2009). A few studies that investigate online rumours on Twitter also adopt the number of retweets as the key research output considering. Therefore, this thesis applies the scale dimension to measure the dissemination of online rumour messages since it is the most commonly used one. Furthermore, social media enables the tracking of digital traces, which is hardly available in offline rumouring.



Source: Yang and Counts (2009)

**Figure 3 Measurement of Dissemination**

## 2.4 Rumour Theories

Traditional rumour theories have two dominant streams, one that models the diffusion of rumours using mathematical, physical and epidemiological theories, the other studies rumouring motivations, behaviours and patterns using psychological and sociological theories (L. Zhao et al. 2016). This thesis is built upon the latter perspective.

### 2.4.1 Distortion of Rumour Content

Sociological perspective often studies the content distortion of rumour messages during dissemination (Allport and Postman 1947; De Fleub 1962). The changes in content as rumour spread mainly can be summarised as having three characteristics - levelling, sharpening and assimilation. They are often referred to as the three-part pattern of distortion. Levelling means when rumour spread, the message is likely to become shorter and more concise. Sharpening means



that, as a rumour passes along, limited selective details stay. Assimilation means the distortion of rumour message tends to shift towards certain directions, and such directions are usually in line with the interests and attitudes of the social network (De Fleub 1962). Based on previous research, a rumour often ends up being increasingly condensed during the transmission.

#### **2.4.2 Psychological Motivation of Rumouring**

Individual's psychological motivational states of rumouring have been outlined to include fact-finding, self-enhancement and relationship enhancement (Bordia et al. 2005; Difonzo and Bordia 2011). Fact-finding refers to one's motivational state that seeks facts and tries to be correct. In the context of rumour, people check the credibility of sources and proposing and evaluating possible explanations, which may lead to the spreading of rumours. Self-enhancement is one's desire to boost the sense of self. This includes improving one's social status, increasing his or her sense of uniqueness and achieving personal value. To enhance the sense of self, people are likely to be involved in rumouring behaviours. Relationship enhancement means an individual is motivated to strengthen his or her relationship with others in the social network. This is a strong motivation for rumouring because communicators of rumours sometimes spread rumour messages to maintain and further develop the relationship with the recipients and to make the recipients think better of the communicator (Bordia et al. 2005).

### **2.4.3 Antecedents of Rumour Dissemination**

Apart from framing rumouring with the motivational frame, rumour theories have identified four general influencers of rumour transmission: uncertainty, importance, anxiety and belief (Difonzo and Bordia 2002). These four predictors are used to describe recipient's features. That is, whether a receiver would pass the message along is influenced by the uncertainty, anxiety, importance and belief of the receiver. Thus, they are usually measured using survey or manipulated in experiments to study whether the uncertainty, anxiety, importance and belief of a recipient would pass the rumour along.

Some studies emphasise the effects of importance and uncertainty or ambiguity on driving the spread of rumour (Allport and Postman 1946, 1947; Oh et al. 2013; Rosnow 1991). On individual level, uncertainty refers to a psychological state of doubt and cognitive unclarity (Difonzo and Bordia 2011). A seminal rumour model proposed by Allport and Postman (1947) highlights the roles of topic importance and ambiguity. For a rumour to occur and to be disseminated, both the sender and the recipient need to consider the topic of story to be important and the truthfulness of story to be ambiguous. Topic importance provides the incentive for individuals to spread the story to others. Meanwhile, ambiguity stimulates subjective elaboration and interpretation which drive the story to be passed along.

Considering that the importance of rumour is hard to capture, Anthony (1973) brought in anxiety to measure importance. Different from Anthony (1973), Rosnow (1991) used personal involvement to measure the importance

of rumour and treated anxiety as a separated influencing factor. Different levels of anxiety have been considered for their roles in influencing rumour transmission, including rumour-specific anxiety (the anxiety about the content and consequence of the rumour), situational anxiety (the anxiety about the environment where rumour is spread) and trait anxiety (the anxiety as a personality trait) (Pezzo and Beckstead 2006; Thatcher and Perrewe 2002). Notably, while the uncertainty is concerned with the cognitive dimension of transmitting decision, anxiety deals with the emotional dimension (Difonzo and Bordia 2011).

Belief in the rumour is one of the commonly agreed factors that drives the dissemination of rumours (Jaeger et al. 1980; Rosnow et al. 1988). Previous rumour studies have spent great efforts in identifying factors that makes people believe in a rumour which is believed to further drive the dissemination of rumour (DiFonzo and Bordia 2007; Rosnow 1991).

Difonzo et al. (1994) studied rumours in organisational context and summarised three stages of rumour: its generation, evaluation and dissemination. Among the three stages, evaluation is closely related to the belief of rumour. Difonzo et al. (1994) argued that rumours arise from uncertainty and anxiety. When situations are unclear, rumours offer explanations and interpretations that help check the truth and ease discomforts to enhance the sense of control and alleviate the feeling of anxiety. Evaluation means that people tend to transmit rumours that they believe is true, and thus people who spread a rumour are usually credulous and perceiving rumour to be fit with their cognitions that are

produced based on available information, capability and stereotypes. This is because recipients recognise that spreading false rumours would cost on the transmitter. Giving false hope or triggering false alarm are likely to cause disappointment and fear and anxiety on others, which in turn damages the reputation of the transmitter. In terms of the dissemination stage, repetition enhances belief. And as a rumour being disseminated, it is continuously refined by the transmitters based on their own interpretation and belief, in turn causing a rumour to evolve and becomes more plausible as it transmits. Based on the three stages, Difonzo et al. (1994) proposed strategies and guidelines to prevent and neutralise rumours.

A number of factors may lead to belief, such as the credibility of source, repetition of message and belief of other people (Bordia et al. 2005; Wang et al. 2018). The effect of belief on driving dissemination decision of rumour may be moderated by the involvement with the rumour and perceived anxiety (Kimmel and Keefer 1991; Wang et al. 2018). With high level of involvement, the effect of belief is alleviated. That is, an individual who is highly involved in a rumour may spread the rumour message regardless of his or her belief (Rosnow et al. 1988). However, if the level of involvement is low, belief may not be enough to trigger transmission.

Prior research has paid huge attention to the antecedents of rumour dissemination. Given the general importance of belief and involvement in shaping rumour dissemination, persuasion theories are commonly used theories to explain the dissemination of rumours. However, there still lacks research to

investigate the antecedents of rumour dissemination in the context of social media where both message and rumour features are traceable.

#### **2.4.4 Rumour in Information Systems Studies**

In recent years, information systems scholars are paying more attentions to rumours and rumour theories, especially in the context of Internet and social media. Oh et al. (2013) applied rumour theory to evaluate the quality of citizen-generated information about social crises in micro-blogging service, and studied what factors make messages more likely to be rumours. They found that source ambiguity is the most significant factor that distinguish rumours from non-rumours, followed by personal involvement and anxiety. DiFonzo et al. (2013) referred to Dynamic Social Impact Theory to understand online rumour dissemination. They found that people tend to pay more attention to a rumour and are more likely to believe the rumour if they receive the rumour from someone in their own social network cluster. Lee et al. (2015) applied the innovation diffusion theory to the rumour context and studied the effects of a list of tweet features on the degree of message diffusion. In their study, they used message type (rumour related or non-rumour related) to measure the “*innovation itself*”, reaction time between event time and first posting time to represent the “*time*”, hashtag usage to measure “*communication channel*”, and number of followers to represent the “*social system*”. The results showed negative relationship between reaction time and degree of diffusion and revealed positive effects of number of followers and use of hashtags on message diffusion. However, the message type was not a significant influencer, indicating that

rumour and non-rumour related messages diffused to the same extent. Nourbakhsh et al. (2016) studied semantic characteristics of rumour. Liu et al. (2014) develops a rumour retransmission model based on rumour theory and elaboration likelihood model. Wang et al. (2018) studies the herding effects on online rumour dissemination behaviour and the group polarization tendency in rumour dissemination networks. These studies used message-level variables and social media data for analysis, providing guidelines and theoretical supports for the development of a message-level dissemination model.

#### **2.4.5 Summary of Rumour Theories and Research Gap**

Traditional rumour studies mainly describe what rumour is, why rumour is created and what stimulates people to create and transmit rumours (Allport and Postman 1947; Jaeger et al. 1980). Prior research has focused on the receiver characteristics when studying the dissemination of rumours (Kimmel and Keefer 1991; Rosnow et al. 1988). Rumour theories highlight the importance of receiver's belief, perceived importance, perceived uncertainty and level of anxiety in rumourmongering (Allport and Postman 1947; Difonzo et al. 1994; Difonzo and Bordia 2002; Kimmel and Audrain-Pontevia 2010). However, comparably limited attention has been paid to the rumour message itself. Although recently IS scholars start to take the message perspective to understand rumour and rumour dissemination, few studies have systematically investigated how the communicational components of online rumour message shape the dissemination. This thesis intends to refer to models of communication components to develop a systematic model of rumour message dissemination in social media.

Since belief is a key driver of rumourmongering, for a rumour to disseminate on social media, its persuasive appeals can be important. Prior research has applied and validated rumour theory in the context of social media, yet few studies have taken the persuasive appeals perspective and the findings tend to be controversial (Oh et al. 2013; Zhao et al. 2016). Rumour theories have recognized the importance of ambiguity in rumouring; however, in the context of social media information ambiguity has been found to have inconsistent effects on dissemination (Oh et al. 2013). To delineate the effects of information (un)ambiguity, this thesis seeks to examine the effects of cognitive appeals on rumour message dissemination in social media. Moreover, previous studies have almost exclusively focused on the role of anxiety in understanding rumour; however, rumours can communicate other emotions which may influence people's beliefs, attitudes and behaviours differently such as fear, anger and hope (Allport and Postman 1946; Zhu and Smith 2016). To understand the roles of emotions, this thesis intends to investigate the effects of affective appeals on rumour message dissemination online. Therefore, persuasion theories and emotion theories are incorporated to study the effects of cognitive appeals and affective appeals on rumour message dissemination in social media.

In addition, while recent studies in IS had started to study the features of rumour messages, limited attention has been paid to the features of rumour topics (Oh et al. 2013). This study seeks to incorporate rumour topic features with rumour message features to examine their interaction effects on rumour message dissemination in social media. Rumour theory posits that belief and importance are important drivers of rumourmongering. As the perceived

trustworthiness and importance of a new piece of information can be shaped by schemas and information congruence, this thesis combines rumour theory with congruence theory to delineate the effects of topic-congruence (i.e. the congruence between rumour target's reputation and rumour topic valence) and message-congruence (i.e. the congruence between rumour message valence and rumour topic valence) on rumour message dissemination in social media.

The following sub-sections review literature related to components of communication, persuasion theories, emotion theories and congruence theories.

## **2.5 Communication Theories**

Communication is one of the key components of rumour and its dissemination. Rumour can be viewed as a process of social interaction (Mullen 1972). According to Shibutani (1966), when the communicative activity ceases, the rumour no longer exists. Therefore, communication theories can play an important role in understanding the dissemination of online rumour messages.

While information theory model generally considers communication as a transmission where people pursue power, the balance models reflect communication as a way for people to flee anxiety. This is similar to the rumour theories where some scholars focus on the dissemination of rumours while others consider rumouring as a collective way of sense making and dealing with anxiety. While traditional rumouring behaviours usually are through interpersonal communication, with the Internet and social media, an online rumour message may not necessarily be intended for one person and is often



broadcasted. Thus, the dissemination of online rumour messages in social media is characterised by both interpersonal and mass communication features.

Communication can be conceptualised from the transmission view or the ritualistic view. Communication can be defined as the process whereby “messages are transmitted and distributed in space for the control of distance and people” (Carey, 1989, p.15). However, because in this research, the key concern is the dissemination and distribution of rumour messages, information theory is referred to as the guideline of this research.

### **2.5.1 Components of Communication**

Communication theorists refer to several models to understand the components of communication. Shannon and Weaver (1949)’s model of communication is commonly considered as the origin of communication theories. According to Shannon and Weaver, communication can be understood as a linear process consisting of source, message, transmitter, signal, channel, noise, receiver and destination. Due to the simplicity of Shannon and Weaver’s model, many derivatives were developed based on it. A commonly acknowledged verbal version of Shannon and Weaver’s model is the Lasswell’s model (1948) which focuses on mass communication. Lasswell’s model explains the mass communication process as *who (sender), says what (message), in which channel (medium), to whom (receiver), with what effect (feedback)*. Similar to Shannon and Weaver’s model, Lasswell’s model considers communication as the linear process of information transmission. As rumour dissemination in nature entails the transmission of information, Lasswell’s model of communication is used in

this study to classify factors that influence dissemination of online rumour messages.

Attributes about message source are one of the most popular aspects in communication studies (Roskos-Ewoldsen and Fazio 1992). There are in general two most well-known model of source features. One is source-credibility model, which includes source expertise and source credibility. The other model is source-attractiveness model, which contends that the effectiveness of the message is influenced by source's likeability (Ohanian 1990). There are also other categorisations of source features such as source's credibility, attractiveness and power. The effects of source variables are also found to be moderated by other factors such as vividness and involvement. The source variables have been found to provide a significant role in messages.

Message features are important parts of communication as well. Specifically, message attribute refers to the manner where communications are presented. Message features in communication studies can be categorised into three dimensions: the content, the structure and the style (Dillard and Shen 2013). Content is related to the choice of topic, theme or story being told in the content of the message. This is essentially what the message is about. (Dillard and Shen 2013). Apart from the choice of topics and themes, supporting evidence and sidedness are another two dimensions of message content. In general, there are four types of evidence: statistical evidence, testimonial evidence, anecdotal evidence and analogical evidence (Dillard and Shen 2013). Statistical evidence refers mainly to the statistics used to support a claim, including the frequencies

and percentages. Testimonial evidence uses one's personal experience, personal opinion or an eye-witness interpretation as evidence. Anecdotal evidence involves having one's observations of the world to support the claim. Analogical evidence, which often is used for novel topics, is about using analogies as the evidence of a claim, comparing one idea or one situation to another. Apart from the content of a persuasive message, the structure of the message is also important. Previous research has found that the number of arguments, the order of the arguments, and whether opposite ideas are recognised are important structure-related variables that shape effectiveness of communication. Along with the content and structure, message style is also another important component. Message style concerns the choice of word, figure of speech and other aspects of language use. The process of persuasion and the effectiveness of persuasive message may vary depending on different features of messages.

Communication channel refers to the method an information or message is delivered to the audience. In communication studies, there are mainly two types of communication channels, namely the personal channels and non-personal channels.

Communicator, message and channel are three controllable aspects in communication processes. In this thesis, the dissemination of online rumour message is considered as a communication process and is examined using variables from these three aspects of communication.

### **2.5.2 Persuasion Theories**

All communication in nature can be argued to be persuasive (Stiff and Mongeau 2016). In psychology, the outcomes of persuasion are usually considered as attitude change and specific behavioural intentions stimulated by attitude change. Rumour messages, because they arise under situations of uncertainty, often help understand the situation and reduce the uncertainty, either intentionally or unintentionally. As a result, these rumour messages usually shape or change the responses of other people. As pointed out by Difonzo et al. (1994), rumours are intended primarily for belief. Even though some rumour messages are not intended to be persuasive, they may still have influence on others' beliefs, attitudes and behaviours. Moreover, the adoption and dissemination of rumour messages tend to increase when the messages are perceived as truthful (Goode & Ben-Yehuda, 1994). The precondition of dissemination of rumour message may be the persuasiveness of message (Iyer and Debevec 1991). Therefore, the dissemination of rumour messages may be influenced by the persuasiveness of the message. In this regard, persuasion theory is strongly useful in understanding rumour message dissemination.

Belief in the information is a key factor for individuals to disseminate a rumour message. Mullen (1972) contented that a rumour must be at minimum partially accepted before it is passed on. Thus, a rumour message that is persuasive tend to shape receivers' belief, stimulating the acceptance and further the dissemination of the message. Therefore, persuasion theory is an important theory that can be used to understand dissemination of online rumour messages. O'Keefe (1990) argued that persuasion should not be accidental or coercive;

instead, it is inherently communicational.

Elaboration Likelihood Model (ELM) is one of the most well acknowledged theories in persuasion, especially in the online context. According to ELM, persuasion can be considered as a cognitive process. Receivers of persuasive messages use different mental cognitive efforts to process the message and come up with the decision to accept or reject them. ELM is a dual process theory of persuasion. Initially proposed by Petty and Cacioppo (1981), ELM contends that a person processes information either via the central route or via the peripheral route. The choice of the route depends on whether the person is motivated and able to elaborate the arguments. A person tends to go through the cognitively effortful central route if he or she has strong incentive and/or if he or she has the cognitive capability to scrutinise the detailed arguments when processing the information. Alternatively, if a person is not motivated or able to process the information, he or she will go through the peripheral route and relies on simple cues such as source attributes more than argument quality (Petty et al. 1997). From the message perspective, messages that are likely to be processed through the central route usually have a wealth of information, rational arguments, and evidence to support certain opinions or conclusions. Such messages tend to have longer-term impact than the messages that are processed through peripheral routes.

ELM has been used to study rumour dissemination. In the context of rumour dissemination, both central cues and peripheral cues exist. Strong arguments can be important as people tend to seek facts and accurate information.

Peripheral cues such as source credibility are crucial as well considering people usually refer to peripheral cues when there lacks verified truth.

Cognitive dissonance theory (CDT) suggests that persuasion is not about one person adding new beliefs to others, but rather the persuasion occurs intrapersonally when there is inconsistency between one's existing belief and the newly presented information. When such inconsistency occurs, individuals experience dissonance (Festinger, 1957). Because people tend to feel uncomfortable holding inconsistent beliefs, they are likely to make attempt to minimize the discomfort, and this often serves as a motivation to elaborate the arguments of the message or even seek additional information to reach a new belief. Congruence effect can play significant roles in influencing the information processing procedure and the effectiveness of persuasive messages. To further understand how congruence may work, congruence theories are reviewed in the following section.

Previous studies on online message persuasiveness have identified key persuasive variables including source credibility, source expertise, communicator attractiveness, argument strength, receiver's prior belief, receiver's prior attitude and level of involvement (Andrews and Shimp 1990; Kim and Bock 2011). Receivers tend to believe the information if it is from a source with high level of credibility and expertise. The physical attractiveness of the communicator may also influence the persuasiveness of messages. Receivers are also likely to believe the information if the argument of message is convincing and valid. Confirmation and disconfirmation of prior belief also

influence the credibility and persuasiveness of information. The existence of a prior attitude on the object or event also has an impact on the perceived persuasiveness of received information. The level of interest in the object or event is also shown to have influence on message persuasiveness.

Previous research on persuasion has identified different persuasive appeals and investigated their persuasive power. A persuasive appeal may be cognitive and/or affective, having an effect on the thoughts and/or feelings. Cognitive and affective appeals of persuasive communication are of great importance in terms of understanding the persuasive effects of messages. Psychological theorists distinguish cognitive persuasive appeals from affective persuasive appeals and recognise that the distinction between cognitive persuasive appeals and affective persuasive appeals is meaningful. Cognitive persuasive appeals, also known as intellectual, logical or rational appeals, refer to the appeals that involve logical arguments and provide facts. Affective persuasive appeals, also known as emotional appeals, are the appeals that use emotionally charged languages to generate or influence emotions and feelings (Rosselli 1995). Emotional appeals serve to create salient emotions or trigger the retrieval of emotional experiences related to the object of interest. Persuasion studies have investigated the processes that cognitive and affective appeals shape persuasiveness and compared the influences of different appeals on cognitive elaboration, attitude changes, memories and behaviours. Although many studies of persuasion did not find differences between the impacts of affective appeals and the impacts of cognitive appeals, many other studies did find the impacts of different appeals to be different. Persuasion theorists proposed that persuasive appeals are most

effective when the appeal matches the underlying nature of the attitude. Prior studies that investigate the effects of persuasive appeals are mainly about purposive persuasive communication such as advertisements and persuasive speeches, and many of such studies consider persuasive appeals as dichotomous. However, persuasion studies find that persuasive communications are not exclusively cognitive or affective; instead, the same message may contain both kinds of appeals, and limited research seeks to understand how those two appeals work when a message contains both. This thesis thus considers the roles of cognitive appeals and affective appeals together in influencing the behavioural intentions of receivers of non-purposively persuasive messages, specifically, how cognitive and affective appeals affect the dissemination of online rumour messages in social media.

## **2.6 Emotion Theories**

As commonly agreed in rumour theories, anxiety- a specific emotion category- plays a key role in influencing rumour dissemination (Anthony 1973; Oh et al. 2013). On this basis, it is plausible that rumour dissemination is associated with emotions. Therefore, to gain a good theoretical understanding on emotions and its effects on the dissemination of online rumour messages, theories and existing literature about emotions, especially the emotions in computer-mediated communications, are reviewed.

Emotions are the internal mental states resulted from evaluative responses to events, agents, or objects (Dillard and Shen 2012). In general, emotions are



short-lived and vary in intensity. However, the detailed definition of emotions differs across theories. While some theories define emotions with the emphasises on the physiological characteristics including neural processes and facial expression, other theories focus on the psychological features such as appraisals, action tendencies and motivations (Dillard and Shen 2012).

### **2.6.1 Typologies of Emotions**

Emotion theories can be categorised into different classes (Yin et al. 2014), most well-known streams are the dimensional theories of emotions and the cognitive appraisal theories of emotions. Dimensional theories treat emotions as varying along “a limited number of fundamental, abstract dimensions” (Yin et al. 2014, p. 541). There is no agreed definition regarding the scales and dimensions of emotions; however, valence, arousal and power are of the most commonly used dimensions by previous studies (Yin et al. 2014). Valence, sometimes also called the pleasantness or evaluation, is a bipolar scale that evaluates the affective state from positivity to negativity or from pleasant to unpleasant (Li et al. 2015; Russel et al. 1989). Level of arousal, or activation, refers to the state of mobilization that measures expression intensity (Heinrichs and Koob, 2004), where high arousal (or activation) is a state of activity, and low arousal (or deactivation) is a state of relaxation (Berger and Milkman, 2012, pp. 193). Circumplex model of Russell (1980) and two-dimensional (valence-arousal) model are most popular in dimensional theories (Yin et al. 2014).

The other approach to study emotions is with the cognitive appraisal theories. The appraisal theories of emotions argue that “emotional reactions to

an event are a result of personal interpretations (appraisals) of the event itself and the situational environment” (Yin et al. 2014, p. 542). Based on appraisal theories, an emotion is elicited by certain cognitive appraisals, and other situations that match the appraisal pattern can evoke that kind of emotion. Previous studies have identified various appraisal dimensions, and the most well-known dimensions are: pleasantness, certainty and control (Yin et al. 2014).

### **2.6.2 Emotional Content in Computer-Mediated Communications**

Emotional states and feelings are often expressed in online messages people posted in social media. In computer-mediated communication (CMC), emotions are represented via emotional words, nonverbal paralinguistic cues such as emoticons, and emotional framing (Stieglitz and Dang-Xuan 2013). Such emotions in CMC are often also referred to as sentiments. Emotions of senders can be transmitted in CMC and detected by receivers (Stieglitz and Dang-Xuan 2013). Previous studies also found that emotions of an online message can be diffused to its replies (Dang-Xuan and Stieglitz 2012).

### **2.6.3 Effects of Emotional Content**

Emotions are influential in shaping people’s judgements and decisions. Emotional content has effective persuasive power (Stieglitz and Dang-Xuan, 2013). The valence of WOM is found to influence message persuasiveness. Some studies found that messages with positive emotions are more persuasive than negative emotions, whereas others suggest that message that communicates negative emotions is more attention grabbing and more persuasive (Wang et al.

2015).

In online communications, both verbal and non-verbal content can be used as cues for receivers to understand the sender's emotions. The verbal cues include emotional words and lexical or syntactical linguistic emotion encodings, while the non-verbal cues include the emoticons or images (Stieglitz and Dang-Xuan 2013). Studies have shown that emotional online communication content can influence the sense of community and the amount of feedbacks (Joyce and Kraut, 2006; Huffaker 2010). Moreover, the sentiment in online communication may also be contagious and influence the sentiments of the subsequent replies and comments (Huffaker 2010).

Emotional content can also influence the dissemination of information (Stieglitz and Dang-Xuan 2013). Emotional words, sentences or framing of messages in written format can elicit emotional effects that stimulate cognitive and affective processes (Bayer et al. 2012). In terms of the cognitive process, the emotional stimuli in written online messages attract receivers' attentions, increase the cognitive involvement of receivers, and drive the receivers to employ responding behaviours such as sharing the information (Heath, 1996).

Apart from the increase in cognitive involvement, emotional stimuli in messages may also evoke high physiological level of arousal, which tends to stimulate behaviours including information sharing (Berger and Milkman 2012). When someone experiences high level of arousal, the person is in a state of mobilization, whereas if someone experiences low level of arousal, he or she

feels a sense of relaxation and deactivation (Berger 2011). Because of this, the message that contains high-arousal emotional content is more likely to be shared and disseminated than those that contains low-arousal emotional content.

In addition, emotional content may also stimulate the emotional contagion. Emotional contagion is the transmission of emotional states via simple exposure (Hatfield et al. 1994). As emotional contagion theory suggests, one person's emotional state can be changed as it exposes to another emotional person (Pugh 2001). Traditionally, such contagion of emotional states was argued to only occur in face-to-face interactions. When communicating face-to-face, communicator's emotional state reflected by facial expression, voice and gesture can cause the observer to develop similar emotional state (Hatfield & Cacioppo, 1994). More recent research suggests that emotional contagion can also occur in computer-mediated communications and such contagion has been observed (Hatfield and Cacioppo 1994; Kramer et al. 2014; Hancock et al. 2008). With emotional contagion, the receivers of emotional content from another online user tend to be emotionally affected by the message and thus arises certain emotions and develops intention to share.

In sum, the emotional content can influence information dissemination in three mechanisms: firstly, the emotional stimuli may directly evoke the receiver's cognitive process, and the receivers may share the message after cognitively process the information; secondly, the emotional stimuli may directly cause arousal-related effects which stimulate sharing behaviours without going through detailed cognitive processes; thirdly, the emotions

expressed by the sender may be transferred to the receivers, and thus indirectly influence the receivers' sharing decisions (Heath 1996). In addition, Emotional appeal may not only have direct influence on the dissemination of messages. Instead, emotions may attract attention and induce high level of cognitive involvement so that it may moderate the effect of persuasive factors on message dissemination (Kissler et al. 2007; Stieglitz and Dang-Xuan 2013).

Prior studies have confirmed that emotions play important roles in influencing communication and information dissemination. According to Boyd (2010), people consume content that stimulates their mind and senses. Such content is not always the most informative content, but it is the emotional content that triggers a reaction. Emotional online communication may evoke emotional responses of receivers, and thus stimulates their attention. Such process emotional transference can be seen as an extension to conventional emotional contagion. The study by Berger and Milkman (2012) indicated that the sharing behaviour via e-mail is positively correlated to having emotions expressed. Also, they found the content that elicits high arousal emotions (e.g. anger) is spread more than that triggers low-arousal emotions (e.g. sadness). Many studies have investigated the influence of sentiment in social media. Huffaker (2010) studied the effects of emotion in the context of discussion form. He found that affective language tends to stimulate more feedback than non-affective language. Duan-Xuan and Stieglitz (2012) found that sentiment tends also to diffuse on social media via comments and replies to original posting. In addition, a sentiment can be classified as positive and negative or into more dimensions such as happy, angry, and sad. While many information studies and sentiment analysis have

used the positive-negative approach to study the sentiment of postings, only a few studies adopted multiple-dimensional emotion measures (e.g. Bollen, et al., 2011).

In rumour theories, emotions, especially anxiety, are perceived as important factors of rumour transmission. While increasing attention has been paid to the situational anxiety, rumour-specific anxiety and trait anxiety (Pezzo and Beckstead 2006), the influence of other types of emotions, such as fear, anger, disgust, and wish relief and hope, and other measurement of emotions, such as the general emotional valence and level of arousal, have rarely been studied. With the extended considerations and measures of emotions, it is possible to study how various dimensions of emotions and the congruences between different levels of emotions influence the dissemination of online rumour messages.

## **2.7 Congruence Theory and Theory of Schemas**

The influence of congruence on information processing has been widely studied in psychology, social psychology and marketing literature (Fleck and Quester 2007). Heckler and Childers (1992) conceptualise the congruence as having two dimensions, namely relevance and the expectancy. Relevance describes the extent to which the stimulus is relevant to and favours the theme being communicated, whereas expectancy defines the extent to which a piece of new information fits in a pre-determined schema. The effects of expectancy congruence can be of particular importance for the dissemination of online

rumour messages. The processing of information may be affected by the perceiver's expectation about the referring target (Stangor and McMillan 1992). Similarly, online rumour messages communicate rumour topic whose target may already be perceived by the public in certain ways. The judgement on the rumour message and decision on dissemination may thus be influenced by how the newly received rumour message falls into the existing schema about the rumour target. Apart from cognitive information, emotions may serve as a form of information and the emotion congruence may also influence the processing of information (Niedenthal and Setterlund 1994). Therefore, theories that explain the congruence effects are reviewed in this thesis to understand what role congruence plays in the dissemination of online rumour messages.

Social expectations can be signified as schemas or schemata from schematic information-processing perspective, and the effects of expectancy dimension of congruence can be explained using schema theory (Rifon et al. 2004; Stangor and McMillan 1992). According to schema theory, schemas can be understood as “organised interconnected knowledge structures that develop through past experiences and guide the processing and interpretation of new information” (Mor and Inbar 2009, p. 392). The new information may be congruent or incongruent with the schema. The schema-congruent information, sometimes also refers to expectancy-congruent information or simply congruent information, means the newly received information is consistent with prior expectancy developed from pre-existing schema; conversely, schema-incongruent information, or expectancy-incongruent information or incongruent information, refers to the new information that contradicts the expectancy based

on pre-existing schema (Schmidt and Hitchon 1999).

Schema theories tend to agree that congruent information is processed differently from incongruent information. However, previous studies have contradicting findings on how congruence influence information processing. While some studies argue that congruent material can grab attention, increase cognitive elaboration, and enhance memory (Mor and Inbar 2009; Stangor and Duan 1991), others suggest that schema-incongruent information is more likely to stimulate cognitive evaluation and increase the number of elaborations than schema-congruent information (Um and Kim 2016). The rationale behind the former perspective is that, compared to schema-incongruent items, it is easier for schema-congruent materials to be assimilated with the schematic knowledge. However, according to the latter perspective, when schema-incongruent information can hardly be ignored or assimilated, the information drives perceivers to pay more attention to the information and spend more time and efforts in processing the information elaborately (Schmidt and Hitchon 1999).

A schema can be about the perceiver oneself (i.e. self-schema), about other targets (i.e. social-schema), and about the perceiver's relationship with others (i.e. relational-schema). In this thesis, the schema mainly refers to the social-schema that is associated with the company that a rumour targets (Schmidt and Hitchon 1999).

## **2.8 Summary of Theoretical Lens**

Traditional rumour theories studying rumour dissemination entail two dominant



streams: one models the dissemination using mathematical, physical and epidemiological theories, and the other understands the motivations of dissemination using psychological and sociological theories from individual perspective (L. Zhao et al. 2016). In recent years, IS scholars are paying attention to online rumour dissemination. An increasing number of IS studies adopt the psychological and sociological theories to study rumour dissemination behaviour from the message perspective (Mondal et al. 2018; Oh et al. 2013). Such studies provide support for understanding rumour dissemination from the message perspective. The dissemination of online rumour messages can be considered as a communication process of persuasive messages. To systematically investigate online rumour message dissemination from message perspective, Lasswell's model of communication can serve as an overarching framework to group the variables that influences the dissemination of online rumour messages. Because in social media the dissemination of rumour messages no longer requires the co-presence of rumour sender and rumour receivers, in this thesis, message features, communicator features, communication channel features, and context (rumour) features are considered to be the key components of rumour message communication.

In addition, the persuasion theories, emotion theories and congruence theory can supplement traditional rumour theories and give new directions to the choice of variables that may influence the dissemination of online rumour messages. Rumour theories have identified belief, ambiguity, anxiety and importance as key factors that influence rumour dissemination (Allport and Postman 1947; Difonzo and Bordia 2002). Although rumours in nature entail uncertainty,

rumour theories suggest that belief is crucial for rumour to disseminate (Dubois et al. 2011; Wang et al. 2018). Considering that rumour messages tend to shape or change people's attitudes about the companies, they can be considered as persuasive communications whose dissemination in social media highly relies on persuasive appeals (Stieglitz and Dang-Xuan 2013). Persuasion theorists suggest that persuasive appeals entail cognitive and affective components (Okazaki et al. 2013). Cognitive appeals involve facts and rational arguments and reflect the quality of message content, whereas affective appeals use emotional languages to create emotions and feelings. Cognitive appeals shape responses by message-based cognitive elaborations, and affective appeals are likely to influence persuasion through creating emotions which influence the valence and amount of cognitive responding. Rumour theories also acknowledge that the dissemination of rumour messages may be influenced by cognitive and affective factors. A seminal rumour model proposed by Allport and Postman (1947) highlights that for the birth and transmission of rumour, the truthfulness of the story must be masked with some level of ambiguity, and the theme of the story must be important to both message sender and recipient. The ambiguity is associated with the cognitive aspect of rumour message persuasive appeal. Considering that the importance of rumour is hard to capture, Anthony (1973) brought in the anxiety to reflect importance, which forms the affective aspect of rumour message persuasive appeal.

Previous rumour research mainly focuses on investigating the roles of cognitive appeals. Information ambiguity has been considered as an important factor that stimulates subjective elaboration and interpretation which drive

rumour generation and dissemination (Allport and Postman 1947; Oh et al. 2013). Oh et al. (2013) further conceptualizes information ambiguity as consisting of two dimensions, namely content ambiguity and source ambiguity. Existing research has applied and validated rumour theory in the context of social media. However, there are contradictory findings on the effects of information ambiguity, especially the content ambiguity (Oh et al. 2013; Shin et al. 2018). The controversial may exist because cognitive appeals are multidimensional and there needs further conceptualization to content ambiguity and source ambiguity. Therefore, this study seeks to conceptualize the sub-dimensions of content ambiguity and source ambiguity and to delineate their roles in influencing rumour message dissemination in social media.

Scholars are also paying increasing attention to understand the effects of affective appeals on rumour dissemination. Traditional rumour theories have recognized anxiety as a crucial factor in stimulating rumour dissemination (Bordia et al. 2005; Kimmel and Audrain-Pontevia 2010). Anxiety has been studied at different levels, including personal chronic anxiety, rumour-specific anxiety caused by the rumour and its potential consequences, and situational anxiety that is related to the situation where the rumour occurs (Kimmel and Audrain-Pontevia 2010). Research suggests that rumours that evoke feeling of anxiety, such as rumours about food poisoning, tend to have high level of dissemination (Kimmel and Keefer 1991). However, rumours can evoke other emotions such as anger, fear and hope. Because rumour theories illustrate that rumour messages can have different emotions and the emotions can be measured on different levels, measuring the influence of just one specific type of emotion

may not be sufficient. In stead, it is important to capture the more general effects of emotions. Considering that information transmission studies in other contexts (e.g. word-of-mouth) have differentiated the influences of various affective appeals on the transmitting behaviour, it will be interesting to extend the analysis of emotions and other affective appeals in understanding the dissemination of rumour messages in social media. In this thesis, the most commonly used measurements of emotions- the valence and level of arousal of emotions – are used to measure the emotions in a general approach.

Cognitive appeals and affective appeals are not dichotomous and message receivers do not perceive message appeals as exclusively cognitive or affective. Rumour messages in social media may have cognitive appeals, affective appeals or both. Therefore, cognitive appeals and affective appeals are important for understanding how message features influence dissemination of rumour messages in social media.

Social schema of the rumour target captures the effects of expectancy congruence based on the Theory of Schemas. Specifically, this thesis is interested in two types of congruence: topic-congruence and message-congruence. Here topic-congruence is defined as the congruence between the rumour target's reputation and the rumour topic's valence, and the message-congruence refers to the congruence between the rumour target's reputation and the rumour message's valence. The existing reputation of the rumour target serves as the schema that represents the expectancy based on existing knowledge, whereas the valence of the rumour topic and the valence of the rumour message

serve as the new information. According to the Schemas Theory, both types of congruence can influence the processing of the information (i.e. the online rumour message) and further affect relevant behaviours.

The theoretical lens is summarised in Figure 4. Combining Lasswell's model of communication, persuasion theories, emotion theories and congruence theory, the research model is developed.

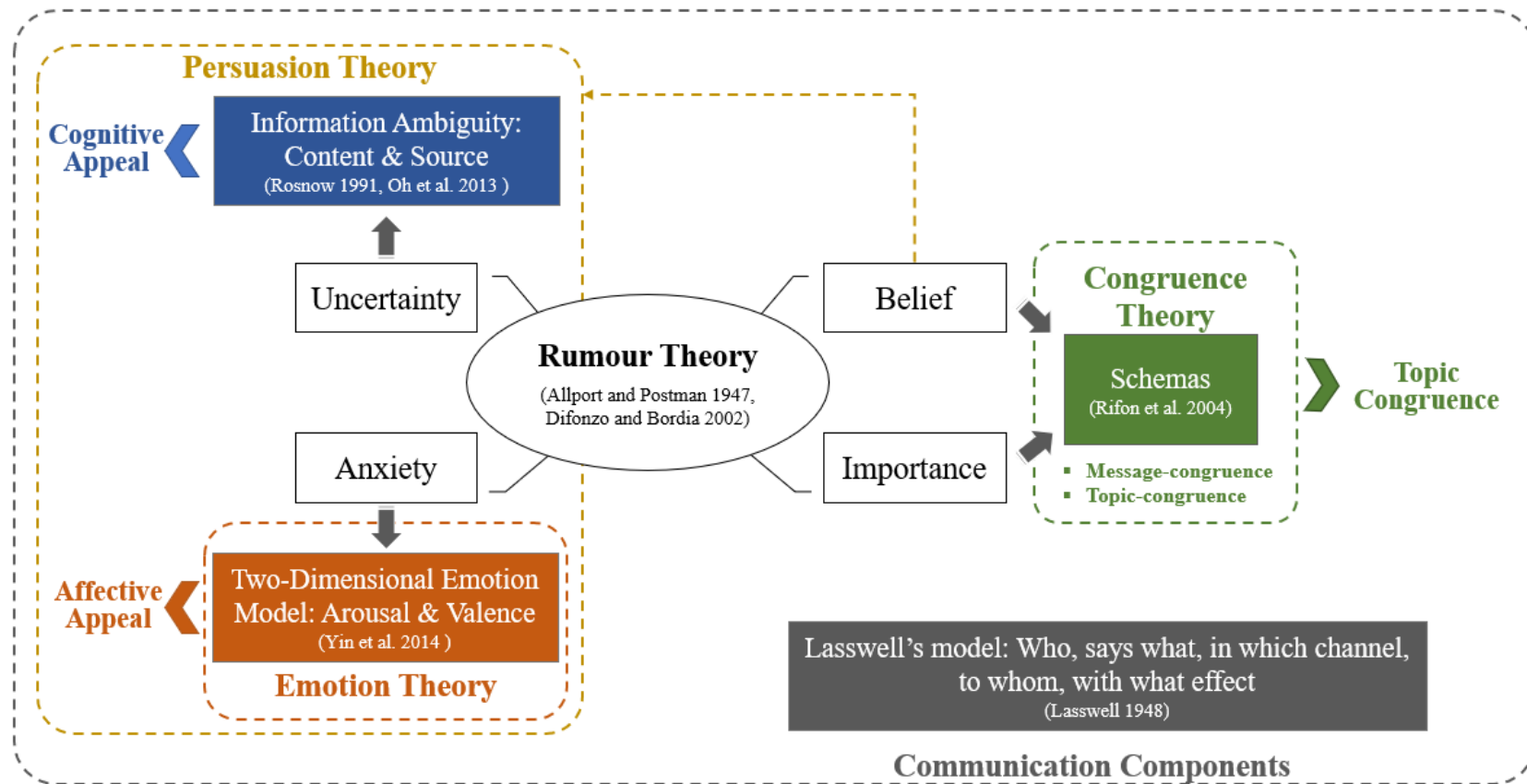


Figure 4 Theoretical Lens

## **CHAPTER 3 THEORETICAL FRAMING AND HYPOTHESES DEVELOPMENT**

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Given the uniqueness of online rumour message dissemination in social media, previous rumour theory alone may be unable to describe its mechanism. Since there lacks a well-developed theory to explain the phenomenon, a preliminary conceptual framework is first developed to support the qualitative exploration based on which a comprehensive research model and corresponding hypotheses will be developed. Incorporating the preliminary conceptual framework with the findings of the exploratory qualitative study, this chapter expands the original theoretical understanding on the dissemination of online rumour messages and further refines the research model and develops research hypotheses. Specifically, the message features are extended and refined to better fit the online rumour message dissemination context.

As the dissemination of online rumour message in social media strongly entails communication, this thesis refers to communication process to structure the factors that influence the dissemination. Main components of

communication process can be decomposed into transmitter (communicator), message, channel, receiver (recipient) and context. However, because the social media context diminishes the influences of recipients, factors from recipient's perspective will not be considered. Therefore, this thesis identifies factors that influence the dissemination from communicator, message, channel and context perspectives.

With communication components defining the overall structure of the theoretical framework, rumour theories are used as the pillar theory that lays the core theoretical foundation for this thesis. However, given that rumour theory is unable to explain the whole picture in the context of online social media, persuasion theories, emotion theories and congruence theories are used to extend the rumour theory and supplement the understanding towards the phenomenon.

This chapter is organised as the following. First, this chapter identifies the key message features that influence the dissemination of online rumour messages and proposes the hypotheses accordingly. Second, variables related to communicator characteristics and their potential relationship with online rumour message dissemination are proposed. Third, the associations between channel features and rumour dissemination are hypothesised. Following this, the effects of rumour valence and referent (i.e., the rumour target) reputation on the dissemination of online rumour messages are proposed in section 4. Finally, some control variables are identified and included in the research model.



### **3.1 Message Features**

As mentioned in Chapter 2, online rumour messages can be considered as persuasive communications as they tend to shape or change people's attitudes about certain object or event. Message features such as the content and tone have been found to have effect on the persuasiveness of communication messages (Iyer and Debevec 1991). Content characteristics of messages, such as argument quality, can influence the persuasive effects cognitively. Apart from content features, past research also has clearly documented the importance of the tone in influencing the receivers' interpretation of a message content and the level of persuasiveness in face-to-face context (Iyer and Debevec 1991). Similarly, the "tone" of written communication may also play similar role in shaping persuasiveness. To understand the impacts of content and "tone" of online rumour messages, cognitive and affective persuasive appeals are considered. Persuasive theorists suggest that persuasive communication entails cognitive and affective components (Okazaki et al. 2013). Cognitive appeals involve facts and rational arguments and reflect the quality of message content, whereas affective appeals use emotional languages to create emotions and feelings. Cognitive appeals and affective appeals are not dichotomous and message receivers do not perceive message appeals as exclusively cognitive or affective. An online rumour message can contain cognitive appeals, affective appeals or both. Traditional psychological studies of persuasion mainly focus on understanding the effects of cognitive appeals (Rosselli et al. 1995). More recently, increasing attention has been paid to studying the impacts of affective appeals.

Both cognitive and affective appeals shape responses by message-based cognitive elaborations. Cognitive response theory suggests that thoughts and feelings as cognitive responses have influence on the formation and the change of attitudes (Okazaki et al. 2013). Based on this theory, a persuasion message can induce different types of cognitive responses with different appeals. Affective appeals are also likely to influence the process of persuasion through creating emotions which influence the valence and amount of cognitive responding.

While online rumour messages are not like advertisement which are created to shape people's attitudes purposively, in the context of uncertainty, online rumour messages did serve as a persuasion message that form or alter the opinions and attitudes of people, and the message are related to inducing thoughts or feelings or both. In addition, rumour theory also suggests that rumouring is driven by two dimensions of motives: the affective dimension and the cognitive dimension (Oh et al. 2013). Therefore, cognitive appeals and affective appeals are considered to understand how message features influence people's behavioural responses towards online rumour messages. Specifically, content unambiguity and source unambiguity are studied as cognitive appeals, whereas emotion valence and level of arousal are considered as affective appeals.

### **3.1.1 Content (Un)ambiguity**

Content ambiguity refers to “the level of interpretative ambiguity contained in the information” (Oh et al. 2013, p. 410). Individuals experience interpretative ambiguity if they recognize that two or more valid competing interpretations

could be made of the same message (Bonitatibus 1996). The existence of more than one plausible interpretations may cause cognitive dissonance and people may take efforts to resolve the dissonance. From this perspective, ambiguous content is likely to drive individuals to engage in information seeking and sharing, and thus stimulate rumour dissemination. However, from the persuasion perspective, ambiguous content lacks clarity and are less understandable, resulting in low argument quality. Compared with ambiguous content, unambiguous content are stronger arguments, which have stronger persuasive power, and people who receive the rumour message are more likely to accept the message and further spread it out.

In Oh et al. (2013)'s coding scheme, they treat content ambiguity as the expression of distrust, ambiguity and/or confusion regarding message content. However, apart from the expression of uncertainty, content ambiguity may also occur if there lacks informative content. Therefore, in this study, we extend Oh et al. (2013)'s definition and conceptualises content (un)ambiguity as consisting of two dimensions: expression certainty and use of evidence. When the level of expression certainty is low and when evidence is absent, the level of content ambiguity of the message is high. While expression certainty largely matches Oh et al.'s (2013) original defining perspective of content ambiguity, use of evidence and message elaborateness extend it by highlighting the effects of evidence on providing more information and reducing confusion.

#### ***3.1.1.1 Expression certainty***

Expression certainty is defined as the level of confidence or conviction

expressed in a message. This concept is derived from attitude certainty, a subjective feeling of confidence about one's own evaluation or attitude. Karmarkar and Tormala (2010) extended attitude certainty to the context of interpersonal communication and studied the influence of "certainty expressed by a message source... regarding his or her own opinion or recommendation" on others' attitudes and behaviours (p.1033).

People tend to shape and share their opinions and attitudes in social contexts such as rumour-mongering (Karmarkar and Tormala 2010). Rumour theory suggests that, after receiving a rumour message, a person is likely to interpret and evaluate the information before further passing it on, and belief is one important factor for making transmitting decision. Apart from forming belief, people also tend to form secondary cognition about their beliefs, reflecting how convictional they feel about their beliefs (Dubois et al. 2011). With social media, because of the retweeting and sharing function, it is easier, than in face-to face manner, to pass on rumours without strong understanding and interpretation. Therefore, levels of certainty in rumour communication are likely to vary, and the sense of certainty may shift from primary cognition of the rumour (such as belief and other attitudes) to the rumour itself. For example, there are posts with external URLs and asking "is this (rumour) true or false?". Expression certainty in this study measures the general level of certainty expressed in a message, including both the source's certainty feeling on the rumour and on the belief of rumour. In this case, high expression certainty indicates high level of confidence, which lead to strong persuasiveness naturally (Karmarkar and Tormala 2010), and strong persuasive communication tend to

be accepted and further shared more by its receivers. Finding from Kamins et al.'s (1997) study is consistent with this argument, such that information labelled as rumour was transmitted less than information with no rumour label. Therefore, this study hypothesises that:

*H1a: Rumour message with high expression certainty is disseminated more than rumour message with low expression certainty.*

#### **3.1.1.2 Use of evidence**

Evidence provides relevant information about the uncertain situation and thus is likely to enhance the clarity of message. In persuasion studies, evidence is perceived as important to support a claim. Evidence can be categorized into four types, namely the statistical evidence which uses statistics and numeric values, the testimonial evidence that uses eye-witness or personal experiences, the anecdotal evidence that uses one's subjective observation and interpretation of the world, and the analogical evidence which uses analogies to compare one situation or idea to another (Dillard and Shen 2012). Among the four types of evidence, statistical and testimonial evidences are the most widely used and most relevant ones.

Interestingly, although rumours are unverified, the presentation of rumours sometimes involve the use of evidence. Rumours might be generated by those who witnessed an anxious situation such as a disaster or crisis (Liu et al. 2014). Moreover, it is also common to include numerical values in the communication of rumours. For example, a rumour related to Ford mentions "...Ford donated

1,000,000 to BLM (black-lives-matter) campaign...”. Previous research has compared the persuasive effects of statistical and testimonial evidence, and a meta-analysis shows that statistical evidence tends to be more persuasive than testimonial evidence, though both are more persuasive than having no evidence (Dillard and Shen 2012). The possible reasons behind this may be that: compared to no evidence, arguments with evidence are likely to be perceived as stronger arguments; while statistical evidence tend to make an argument more explicit and easier to interpret than testimonial evidence can (Gold et al. 2008). Considering that a message with higher persuasive power tend to be accepted and distributed more than others, in this study, it is hypothesised that:

*H1b: Rumour message with evidence is disseminated more than rumour message without evidence.*

### **3.1.2 Source (Un)ambiguity**

In communication studies, the source, sender and communicator of messages are often used interchangeably. However, in the context of rumour transmission, a communicator of a rumour message is not necessarily the source of the rumour. With mentioning the source of rumour, the communicator is likely to be perceived as a mere transmitter of the rumour by recipients. Therefore, in this study, the communicator is used refers to the sender of the message, while the source is used to describe any source of rumour mentioned.

In Oh et al.'s (2013) study, source ambiguity measures whether the rumour message contains “an external sources (such as name of media or links to

external media, video, picture, etc.)” or/and expresses “distrust and/or ambiguity about the source”(p. 425). The current study adopts Oh et al.’s definition, and further explicates the source to be the source of rumour, and distinguish the (un)ambiguity in terms of the explicitness of a source and the direct accessibility towards the source. While Oh et al. (2013) found source ambiguity the most important variable causing rumours, it would help deepen the understanding of rumouring by studying whether source (un)ambiguity also influences the dissemination of rumour messages, and which component of source (un)ambiguity are more important.

#### ***3.1.2.1 Source explicitness***

Source explicitness refers to whether the rumour message mentions a clear and explicit rumour source. Intuitively, a message with an external source may increase the perceived credibility of the message as having external information support. Nourbakhsh et al. (2016) also found that most rumours on Twitter come from other social media services and platforms or from websites and blogs. However, little empirical study has examined on how source explicitness influences rumour dissemination. This study intuitively proposes the effects of source explicitness on rumour message dissemination, and intends to test the hypothesis empirically.

*H2a: Rumour message with explicit rumour source is disseminated more than rumour message with no explicit source.*

### 3.1.2.2 Source accessibility

Source accessibility concerns whether the rumour message contains a link that can directly lead access to the rumour source, such as providing a website hyperlink to an external source or mentioning (i.e. using @) the source. While studies on retweeting behaviour have proved that the use of URLs is positively associated with retweet rate (Suh et al. 2010), URLs are found to link to news, blogs, videos and other types of sources. In the context of rumour, messages with clear and verified reference to solid news are not counted as rumour, therefore, the effects of access to rumour source on message dissemination remain an interesting issue. The inclusion of hyperlink or mentions or other possible direct access to the source of rumour functions as a form of reference or citation to the source of rumour (Weller et al. 2011). Therefore, direct link illustrates the existence of the source. Moreover, because the source of rumour may contain additional information for evaluation, the inclusion of link to source potentially increase the amount of information provided, which in turn facilitates information processing and persuasion. Therefore, it is hypothesized that:

*H2b: Rumour message with direct link to rumour source is disseminated more than rumour message with no direct link.*

### 3.1.3 Emotional Words

Apart from cognitive appeals (i.e., content unambiguity and source unambiguity) affective appeals of the message are also important as it influences the impacts of persuasive messages from affective perspective. Emotions can be the motivations of rumouring, and rumours and rumour communications can also



evoke emotional responses. As Zhu and Smith (2016) suggested, this thesis distinguishes situational emotions (i.e. the emotions induced by the situations, e.g. referent reputation) from affective appeals of rumour messages, which are emotions communicated by specific online rumour messages. Moreover, in offline environment, it is more difficult for a receiver to immediately spread out the message- because it is likely that no further receiver is available at that time. However, in online environment, the receiver can spread the message easily without the requirement of physical appearance of the audience. Therefore, emotional content is likely to play an important role in the message, stimulating the spreading behaviour.

Traditionally, the rumours are usually diffused via word of mouth in face-to-face interactions (Allport and Postman, 1947; Buckner 1965). In this era of computer-mediated communication, the dissemination mechanism of rumours is different from traditional face-to-face manner. In an offline environment, people tend to interpret the message before being able to transmit it. Moreover, when passing along messages through face-to-face interactions, people tend to modify the rumours every time when they pass it, because it is difficult for humans to repeat the exact information details, and in conversations there are likely to be language redundancies. However, via computer-mediated communication such as microblogging, people can directly quote and retweet others' messages. Thus, transmitting rumours in social media may require fewer personal cognitive efforts. Moreover, spreading rumour in offline context requires the physical appearance of audiences; however, in social media, receivers can spread rumour messages anytime they want with no restriction of

time and space. As a result, emotional appeal of the message content may play an important role in stimulating the sharing behaviours.

Previous research has measured anxiety in different levels. Anxiety has been measured on individual level as an individual dispositional trait, on situation level to describe the situation where rumour occurs, and on rumour level to capture the consequences caused by the rumour. Personal anxiety is found to have positive effect on rumour dissemination consistently; however, prior studies have different findings regarding the influences of situational and rumour-specific anxieties on rumour dissemination. In our study, because we are not going to track individual recipient, the personal chronic anxiety is not taken into consideration in this study. Rather, we focused more on the anxiety and other emotions evoked by the situation and by the rumour message. As mentioned before, the relationship between emotion and rumour are of two mechanisms: first, emotions can be a motivational factor that causes rumour, that is, rumours are initially created because of certain emotions. Second, rumour and rumour communications can evoke certain emotions. Previously discussed rumour types are more about motivating-rumour-emotions (that is, situational-induced emotions) and (rumour-induced emotions)

Rumour theories have recognised the role of anxiety in rumouring behaviours. Anxiety refers to “an affective state- acute or chronic- that is produced by, or associated with, apprehension about and impending, potentially disappointing outcome” (Difonzo & Bordia 2002, p. 3). Anxiety has been widely recognised as closely related to rumouring behaviours (Oh et al. 2013; Rosnow 1991). First, rumour theories suggest anxiety tend to generate rumour activities

(Difonzo and Bordia 2002). For example, rumours that evoke the public's feeling of anxiety, such as rumours about campus murder or food poisoning, tend to have high level of transmission. Second, anxiety moderates the relationship between belief and transmission. Previous study shows that belief in rumour has the strongest influence on rumour transmission when the level of anxiety is moderate, while the relationship between belief and transmission is relatively weak when the level of anxiety is both low and high (Rosnow et al. 1988). People who do not feel anxious may not have enough motivation to evaluate the credibility of rumour; however, people who are overly anxious may employ whatever information available without the consideration of its credibility. Therefore, people may transmit implausible rumours under either low or high level of anxiety; moderate level of anxiety, however, provides enough motivation yet not too much to assess the rumour.

However, apart from anxiety, there are other emotional motivations behind rumours which tend to evoke more diverse emotions. Wish rumours or pipe-dream rumours raise hope and desire, and sometimes relief (Allport & Postman 1946), and other rumours contain and communicate fear, disgust, disappointment, anger, and hate (Rosnow and Foster 2005). Therefore, the consideration of emotional effects should be extended. For example, rumour-induced disgust was found to increase rumour transmission. Therefore, instead of focusing solely on anxiety, this thesis seeks to understand the impacts of general emotional appeals on rumour dissemination.

Emotional appeals of online rumour messages can have emotion-related effects such as increased cognitive involvement and physiological arousal,

which consequently drive sharing behaviours (Stieglitz and Dang-Xuan 2013). Emotional valence and level of arousal are the most commonly used dimensions of emotions and are used to measure the emotional appeal of online rumour messages. Emotional valence is an influential factor for message sharing. Some scholars argue that negative information is more likely to be spread out (Godes et al. 2005), others' advocate that positive emotion will facilitate information propagation more. Eckler and Bolls (2011) studied the role of different emotional tones (pleasant, unpleasant, coactive) of video on consumer's attitudes and forwarding intentions. They found that pleasant emotional tone has the strongest effect on forming attitude and stimulate forwarding, while unpleasant emotional tone has the weakest effect. Considering that online content sharing often involves self-identity representation, the sharing of positive content tends to create positive effect on sender's identity, and thus are more likely to occur (Berger & Milkman 2012). Berger and Milkman (2012) also found that positive content is more likely to be viral. In the context of online rumour messages where people tend to be uncertain about some issues or events, they may care about their identity representation and are more willing to pass on messages with positive emotional appeals to create positive perception on their identities. Therefore, it is possible that online rumour messages with pleasant emotions are likely to be disseminated more than online rumour messages with unpleasant emotions.

*H3a: Rumour message with high level of emotional pleasantness is disseminated more than rumour message with unpleasant emotions.*

Apart from valence, the activation or the level of arousal is also likely to

influence sharing of messages in social media. Emotions with high level of activation, such as anxiety and amusement, have stronger effect in driving sharing behaviour than low arousal emotions like sadness and contentment (Berger 2011). Berger (2011) tested this using experiment where the manipulation of high arousal was to let students watch emotional videos. Emotional contagion theory suggests that emotional state can be transferred; that is, exposure to person who expresses emotions can cause emotional change of the observer (Pugh 2001). Although this contagion is original thought to occur in face-to-face human interactions (Hatfield & Cacioppo 1994), more recent studies also observe emotional contagion in computer-mediated communication (e.g. Kramer et al. 2014; Hancock et al. 2008). The emotion embedded in computer-mediated content reflects the emotion of the sender, and the receiver might experience such emotional contagion. According to Berger and Milkman (2012), “sharing positive content may help boost others’ mood” (pp. 2), whereas sharing upset information tend to make other people sad. Moreover, people’s emotion tends to influence their behaviour. Prior research suggests that high arousal emotion tend to elicit active behaviour. If emotional contagion occurs, receivers who are exposed to high arousal emotion from sender-generated content are more likely to pass along the message. On the contrary, receivers who are infected by low arousal emotion would probably not share. In the context of online rumour messages, messages with high level of emotional activation are likely to elicit high level of arousal from receivers, which further makes the messages more likely to be spread out. Therefore, it is reasonable to think the emotional activation of online rumour messages can influence sharing behaviour. Therefore, we propose that emotion of advertisement content will

influence the sharing of the message. To be more specific, we hypothesise that:

*H3b: Rumour message with high level of emotional activation is disseminated more than rumour message with low level of sentiment arousal.*

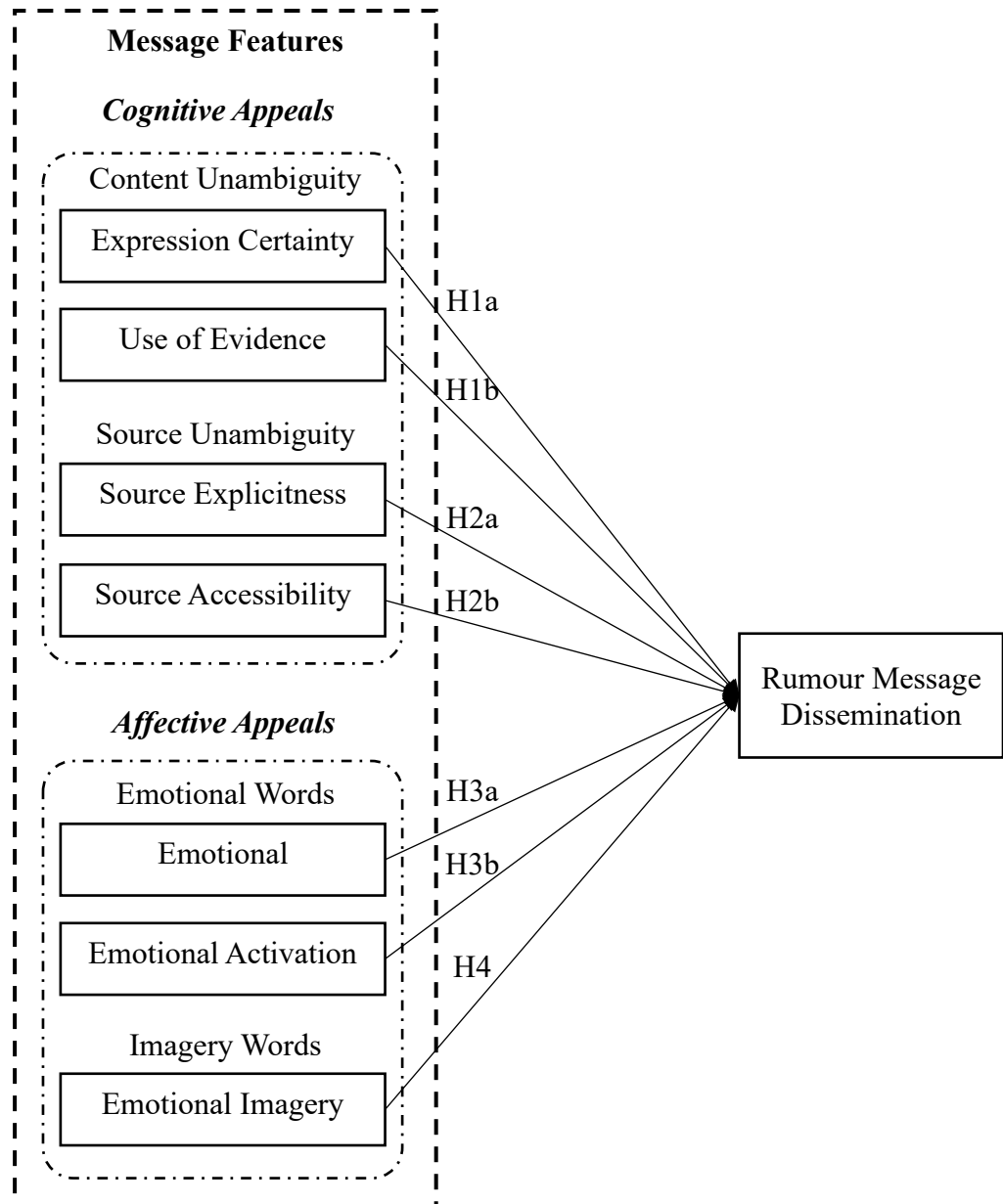
#### **3.1.4 Imagery Words**

Apart from emotional pleasantness and activation, in the literature of marketing and psychology, imagery word has been studied as a form of affective appeal which will increase the vividness of information. It is a common and yet intuitive belief that imagery words and vividness of message enhances the persuasiveness of communication (Kisielius and Sternthal 1984; Smith and Shaffer 2000). Persuasion literature theorises such belief. Persuasion theories expect high level of imagery words and vividness to increase persuasion effects from two approaches: first, imagery words can attract attention so that imagery and vivid message is more likely to be processed by the audience than pallid message (Smith and Shaffer 2000); second, increasing message vividness makes it easier to create and retrieve mental images, which facilitates message processing directly and through priming information in memory (Smith and Shaffer 2000). However, the findings from empirical research on the effect of message vividness are conflicting. While some research supported the positive influences of message vividness on persuasiveness, others found no enhancing or even inhibiting effects (Keller and Block 1997; Kisielius and Sternthal 1984). The plausible reasons given by literature are that imagery word and vivid presentation may evoke tangential imagery and disturb recipients' processing of

the message. However, the effects of imagery word have not been studied in the context of rumour. This study proposes messages with imagery presentation as an influential variable on rumour dissemination, such that:

*H4: Rumour message with high level of emotional imagery is disseminated more than rumour message presented pallidly.*

The hypotheses related to message features (i.e. expression certainty, use of evidence, source explicitness, source accessibility, emotional pleasantness, emotional activation and emotional imagery) are summarised in Figure 5.



**Figure 5 Effects of Message Features (Hypotheses 1 - 4)**

### 3.2 Communicator Features

Marketing and advertising practitioners share the belief that a communicator's characteristics have significant influences on the persuasiveness of the message. Studies have considered the effects of a communicator's credibility, trustworthiness, expertise, attractiveness, likeability, similarity with receiver and



sociability (Iyer and Debevec 1991). For example, communicator's physical attractiveness has been found to shape receiver's attitudes and level of compliance. Previous research also shows that the similarity between message communicator and receiver have positive influence on the message persuasiveness (Berscheid, 1996). Moreover, trustworthiness and credibility of communicator are also key factors that affect persuasiveness (Iyer and Debevec 1991). Communicators with high level of credibility have been shown to be more persuasive than communicators with low level of credibility. In some studies, communicator credibility is used as equivalent to trustworthiness, whereas other studies conceptualise credibility as consisting of two dimensions, namely trustworthiness and knowledge (or expertise). In computer-mediated communication, it can be difficult to demonstrate and prove communicator knowledge, expertise and attractiveness perceived by the receivers. However, communicator trustworthiness, social influence and likeability are made visible and can be important variables that influence dissemination of online rumour messages.

### **3.2.1 Communicator Trustworthiness**

Communicator trustworthiness refers to "the degree of confidence in the communicator's intention to communicate the assertions he considers most valid" (Ohanian 1990). In social media, communicator trustworthiness is often associated with the verification of the user account. A trustworthy communicator is considered as being motivated to provide accurate information (Tormala and Petty 2004). Sharing accurate information is an important social norm in

communications. According to social reputation theories, sharing rumours that are false is likely to bring untrue hopes and raise false alarms which will hurt the transmitter's social reputation, credibility and social standing (Difonzo and Bordia 2007). Previous work also supported the influence of rumour communicator's trustworthiness on rumour transmission (Brooks et al. 2013; Cheung and Thadani 2012; Ohanian 1990). Because rumour is unsubstantiated, the trustworthiness and expertise of the individual or organisation that generate or communicate the rumour are of great importance in shaping the believability of the rumour (Iyer and Debevec 1991). Therefore, in this thesis, communicator trustworthiness is considered to serve as a potential communicator feature that can influence dissemination of online rumour messages. Specifically, it is hypothesised that:

*H5: Rumour message communicated by user with high level of trustworthiness is disseminated more than rumour message communicated by user with low level of trustworthiness.*

### **3.2.2 Communicator Social Influence**

Although proactive searching is an important information retrieval mechanism during a public-concerned event, communicator's social influence can still determine the reach of the rumour message within one's own social network. Social influence can be measured by communicator's degree centrality (Li et al. 2015). High level of indegree (large number of followers the communicator has) indicates that the communicator has large number of audiences in his or her network that can receive the message. Moreover, the social influence of source

can also reflect the level of source credibility. Empirical research has supported the effects of social influence on message dissemination (Suh et al. 2010). Bampo et al. (2008) studied the process of viral marketing and found that ‘social structure of digital networks plays a critical role in the spread of a viral message’. Therefore, this study hypothesises that:

*H6: Rumour message communicated by user with high level of social influence is disseminated more than rumour message communicated by user with low level of social influence.*

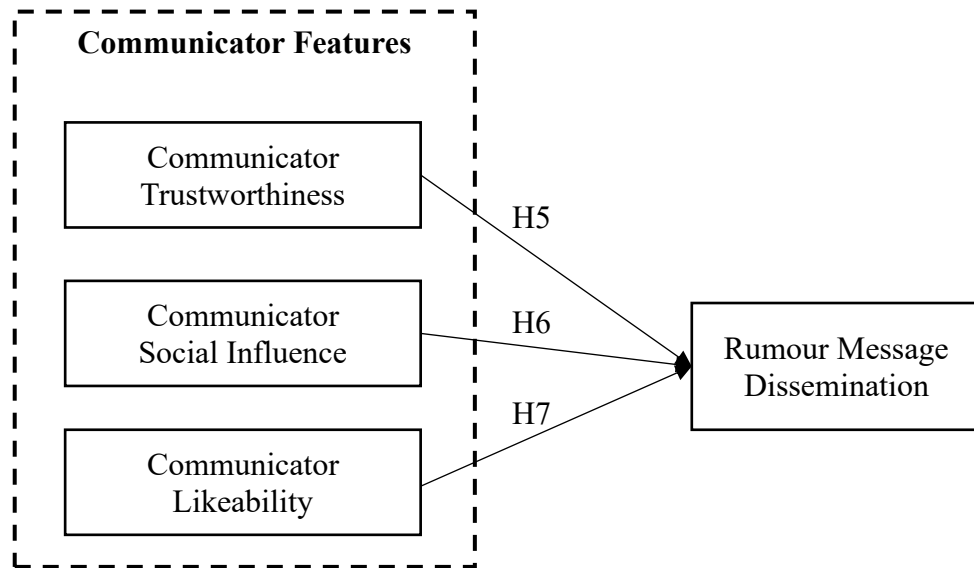
### **3.2.3 Communicator Likeability**

Likeability towards a message communicator can be defined as an affective evaluation related to the communicator. Roskos-Ewoldsen & Fazio (1992) studied recipient’s attitudes about the source and expected that level of source likeability in recipient’s memory affects attitude change. Moreover, the influence of attitude change on information processing is likely to be influenced by likeability. When the people perceive high likeability of a communicator, they tend to display highly biased processing of information (Roskos-Ewoldsen and Fazio 1992). Communicator likeability or attractiveness are commonly used peripheral cues in elaboration likelihood model, and has been proved to play significant role in persuasion (Karmarkar and Tormala 2010). Therefore, in this study, it is proposed that:

*H7: Rumour message communicated by user with high likeability is disseminated more than rumour message communicated by user with low*

*likeability.*

The research hypotheses related to communicator features (i.e. communicator trustworthiness, communicator social influence and communicator likeability) are summarised in Figure 6.



**Figure 6 Effects of Communicator Features (Hypothesis 5, 6, 7)**

### 3.3 Communication Channel Features

A communication channel refers to the means, method or medium by which messages travel from one individual (the sender or communicator) to another (the receiver) (Rogers 1995). Although in social media, there could be different platforms such as blogs, microblogging services, etc., this study focuses on the communication channels within one platform.

There are different approaches to get communication to others within one social media platform, and one of the approaches is direct communication. On Twitter, for example, mentions (@) can be used for selecting and notifying specific targeted individuals directly. Usually, individuals are mentioned because the communicator wants the message to be received directly by them which leads to direct communication. In other cases, a communicator may mention other users because the communicator retweeted their posts. This kind of mentions are referred as indirect mentions in this thesis, and will not be considered as direct communication. Apart from direct communication, another common approach by which communicators delivery their messages to receivers is community communication, for example, the use of hashtags (#) on Twitter. Hashtags can mark out and attract specific targeted communities that are interested in the tagged post.

Rumour activity “involves collective talking, interactive information sharing, and social support among like-minded groups of people” (Oh et al. 2013, p. 411). Individuals are more likely to share information with those they are familiar with, and are more likely to believe the information shared from people they know (Garrett 2011). Therefore, direct communication is more likely to occur between acquaintances and thus leads to higher likelihood of belief. People who believe a piece of rumour are considered more likely to transmit the rumour message (Pezzo and Beckstead 2006; Zhu and Smith 2016). Moreover, collective concerns tend to drive rumour mongering (Difonzo et al. 1994). Mediated communication and community communication both are likely to be relevant to the rumour context. These channels connect the communicator with

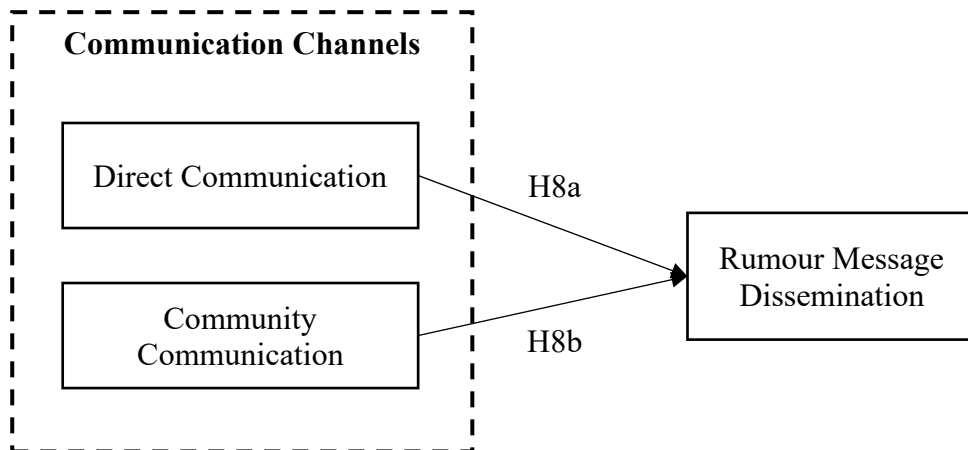
others who also perceive the rumour context important and uncertain, and importance and uncertainty are two important factors for rumour transmission.

Based on the above, this study hypothesises that:

*H8a: Rumour message communicated using direct communication channel is disseminated more than rumour message communicated without direct communication channel.*

*H8b: Rumour message communicated using community communication channel is disseminated more than rumour message communicated without community communication channel.*

The research hypotheses related to communication channel features (i.e. direct communication channel and community communication channel) are shown in Figure 7.



**Figure 7 Effects of Communication Channel Features (Hypothesis 8a, 8b)**

### **3.4 Rumour Features**

Rumour features can be considered as the contextual factor that shapes the communication. While message emotional appeals capture the emotions of individual rumour messages, rumour valence and target reputation capture the overall evaluation and emotion associated with the rumour situation. The situational anxiety is the most prevalently studied context features in previous rumour literature. Research on situational anxiety's influence on rumour transmission shows inconsistent findings. While some studies found that situational anxiety can facilitate rumour dissemination (Pezzo and Beckstead 2006; Walker and Beckerle 1987), others found the impact to be insignificant (Rosnow et al. 1988).

This thesis extends situational context features to go beyond the situational anxiety and explore the roles of the valence of rumour topic and the target reputation which represents the public emotions towards the rumour target are considered. Referent reputation is about the public's existing perceptions. It may influence the perception of the new information. In the case of online rumour messages, both the rumour topic and the rumour message can be considered as the new information. Specifically, the valence of rumour topic and the emotional appeal of the rumour message both can serve as the information newly received by receivers. To enhance the previous understanding on rumour context features, built on Schemas Theory, this thesis considers the congruence between rumour target reputation and rumour topic valence (i.e. the topic-congruence) and the congruence between rumour target reputation and rumour message emotional

appeals (i.e. the message-congruence). Moreover, previous research has pointed out the possibility of over-reporting the prevalence of negative information using self-reported approach. This is because respondents may find it easier to recall negative information than to recall positive information. This study using panel data avoids the over-reporting problem in surveys, and thus the finding will contribute to solving the controversy.

### **3.4.1 Rumour Topic-congruence**

Rumours help to handle the ambiguous and uncertain situation, while the resolution may involve positive or negative consequences (Kamins et al. 1997). Rumour valence refers to whether the consequences mentioned in the rumour is perceived as positive or negative; a rumour is positive if the consequences are positive, and a rumour is negative if the consequences are negative. Positive rumours are sometimes perceived as transmitted more. Positive rumours cause the feeling that uncertainty is resolved in a positive way. This offers an affective benefit for the recipient and reinforces individuals' preference for perceiving the positive instead of the negative in interpersonal communications (Kamins et al. 1997). In addition, research shows that individuals who convey favourable information and evaluations of people and products are liked more by the recipient of the information.

However, there are also evidences to support that negative rumours are more prevalent than positive rumours (Difonzo and Bordia 2007). For example, during the WWII, more than 90% of rumours reported are negative rumours with



negative consequences (Knapp 1944). Studies on word-of-mouth also support that negative information is spread more than positive information. Moreover, in online discussion, negative rumours have been found to be disseminated more than positive rumours (Bordia and Difonzo 2004). The biased impact on rumour transmission can be explained by Prospect Theory. According to prospect theory, “losses are felt more keenly than equivalent gains” and people are more alert to information on loss than on gain (Difonzo and Bordia 2007). Based on the theory, negative rumours that mentions the possible loss would attract more attention and stimulate recipients to transmit more than a positive rumour that mentions same level of possible benefit.

One possible reason for the inconsistent findings is that the valence of rumour sometimes is within the expectation of the public based on target reputation, whereas other cases the valence of rumour is against the existing expectation. Therefore, understanding the congruence between rumour topic valence and target reputation can help understand the factors that shape rumour message dissemination. Target reputation refers to the evaluation of the public towards the referent of rumour (i.e. the target of the rumour). Because rumour is unverified, when the new information incongruent with the existing knowledge, the attention of the receivers are likely to be caught. Moreover, when receiving incongruent information, receivers are likely to experience cognitive dissonance and may stimulate the dissemination of rumour messages to seek for further information and get confirmation. Therefore, in this thesis it is proposed that:

*H9a: Rumour message whose rumour target reputation is incongruent*

*with rumour topic valence is disseminated more than those whose rumour target reputation is congruent with rumour topic valence.*

According to the Schemas Theory, schema-congruent material can work in different ways. Some scholars argue that schema-congruent material can grab attention, increase cognitive elaboration, and enhance memory (Mor and Inbar 2009), whereas others suggest that schema-incongruent information is more likely to stimulate cognitive evaluation and increase the number of elaborations (Um and Kim 2016). When receiving rumour, the unverified information may lead incongruent information to trigger high level of cognitive elaborations. On the contrary, congruent information may lead receivers to have low level of elaborations of rumour messages and rely more on simple and obvious affective appeals. Therefore, when there is topic-congruence, the effects of cognitive appeals tend to be weakened whereas the effects of affective appeals tend to be strengthened. Based on the above, the following hypotheses are proposed:

The message-congruence can also influence the extent of cognitive processing on arguments. Ambiguous information has higher likelihood of being accepted when there is affective trust instead of cognitive distrust (Oh et al. 2013). When the emotion expressed in a rumour message is congruent with the target reputation, receivers of the rumour message is likely to trust the information. In this case, low level of cognitive elaboration is used to process the arguments. Therefore, the impacts of cognitive appeals on rumour dissemination tend to be weakened, whereas the effects of affective appeals on

rumour dissemination are likely to be strengthened. Based on the above, we hypothesize that:

*H9b: Rumour topic-congruence moderates the effect of expression certainty on rumour dissemination.*

*H9c: Rumour topic-congruence moderates the effect of use of evidence on rumour dissemination.*

*H9d: Rumour topic-congruence moderates the effects of source explicitness on rumour dissemination.*

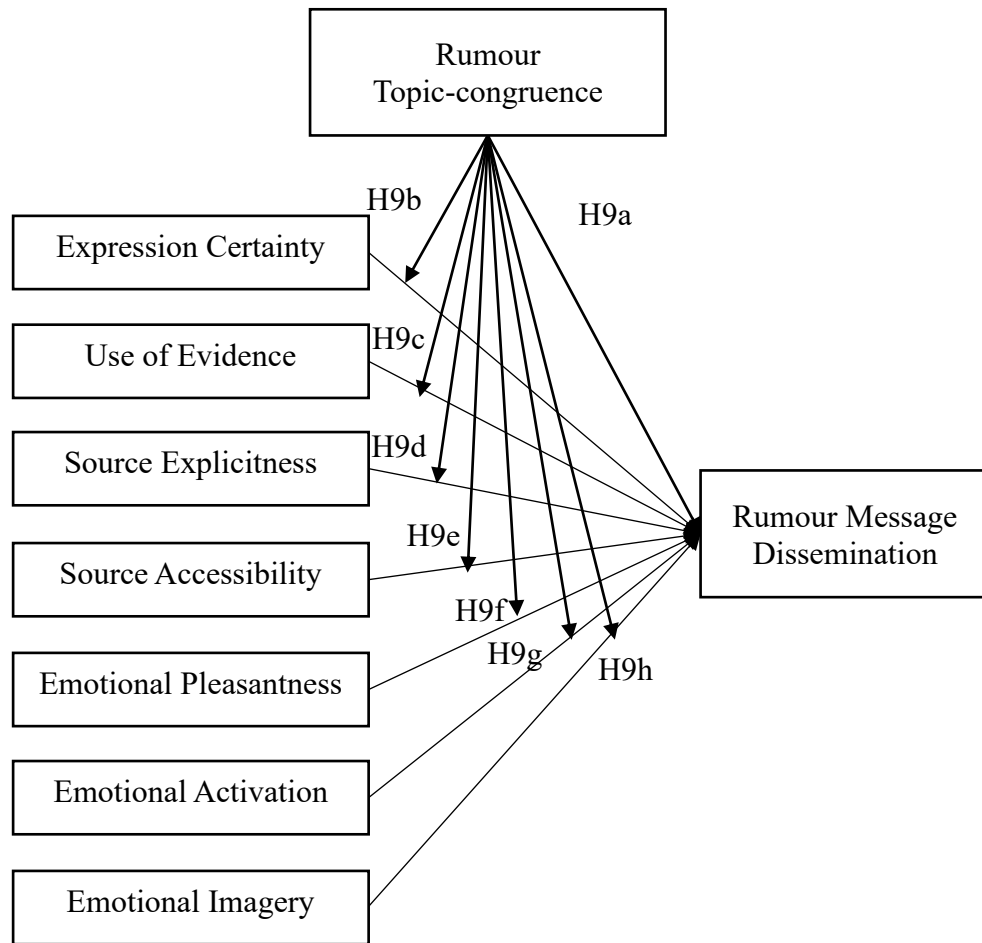
*H9e: Rumour topic-congruence moderates the effects of source accessibility on rumour dissemination.*

*H9f: Rumour topic-congruence moderates the effects of emotional pleasantness on rumour dissemination.*

*H9g: Rumour topic-congruence moderates the effects of emotional activation on rumour dissemination.*

*H9h: Rumour topic-congruence moderates the effects of emotional imagery on rumour dissemination.*

The research hypotheses related to rumour topic-congruence are specified in Figure 8.



**Figure 8 Effects of Rumour Topic-congruence (Hypothesis 9a - 9h)**

### 3.4.2 Rumour Message-congruence

In online rumour dissemination, the rumour message serves as the new information whereas the target reputation represents the existing perception related to the rumour context. When the emotion of the new information is incongruent with the existing perception and evaluation, the new information can attract high level of attention. Moreover, because there is a gap in the perception, people are more likely to pass along the information to seek confirmation and explanation. Therefore, when the rumour target reputation is

incongruent with the rumour message emotional appeal, the dissemination of rumour messages tends to boost. Therefore, it is proposed that:

*H10a: Rumour message whose rumour target reputation is incongruent with rumour message emotional appeal is disseminated more than those whose rumour target reputation is congruent with rumour message emotional appeal.*

Emotion-congruent information processing suggests that emotions influence perception such that emotion-congruent stimuli are perceived more efficiently than emotional incongruent stimuli (Innes-Ker and Niedenthal 2002; Niedenthal and Setterlund 1994). In terms of rumour communication, the target reputation is likely to shape people's emotional state towards the context of rumour in general, while emotional appeal in rumour message is the emotional stimuli to be processed. When the two emotions are congruent, the receivers tend to display more efficient information processing such that both cognitive appeals and affective appeals can be interpreted better which may further stimulate the dissemination of rumour messages. Based on the above, the following hypotheses were proposed:

*H10b: Rumour message-congruence moderates the effect of expression certainty on rumour dissemination.*

*H10c: Rumour message-congruence moderates the effect of use of evidence on rumour dissemination.*

*H10d: Rumour message-congruence moderates the effects of source explicitness on rumour dissemination.*

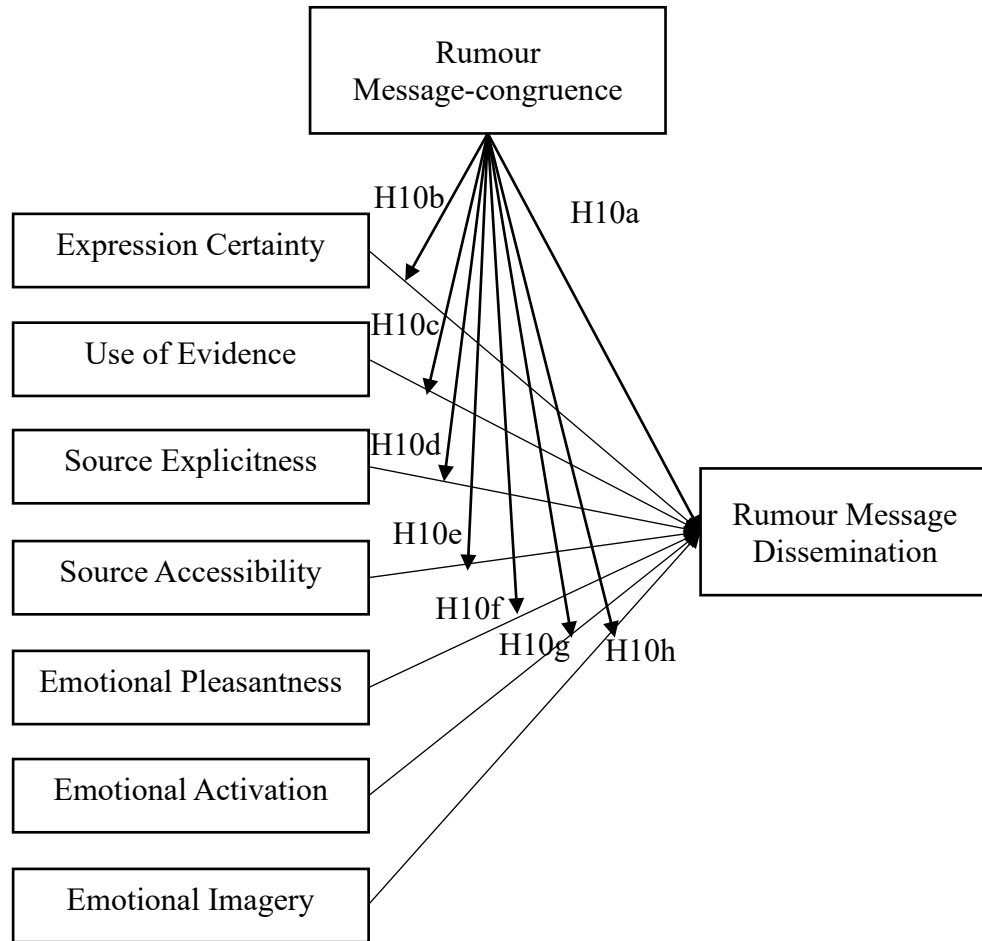
*H10e: Rumour message-congruence moderates the effects of source accessibility on rumour dissemination.*

*H10f: Rumour message-congruence moderates the effects of emotional pleasantness on rumour dissemination.*

*H10g: Rumour message-congruence moderates the effects of emotional activation on rumour dissemination.*

*H10h: Rumour message-congruence moderates the effects of emotional imagery on rumour dissemination.*

The research hypotheses related to rumour message-congruence are specified in Figure 9.



**Figure 9 Effects of Rumour Message-congruence (Hypotheses 10a - 10h)**

### 3.5 Control Variables

*Communicator's outdegree* is included as control variable. High level of outdegree (large number of followers the communicator has) indicates that the opportunity the communicator has to expose to information. Everything else being equal, more followings the communicator has, the more likely for the user to receive new and novel information (Li et al. 2015).

Moreover, previous studies also used the *number of status* was listed as social features to study retweeting behaviour. These features are also included as control variables. The overall conceptual framework is shown in Figure 10.



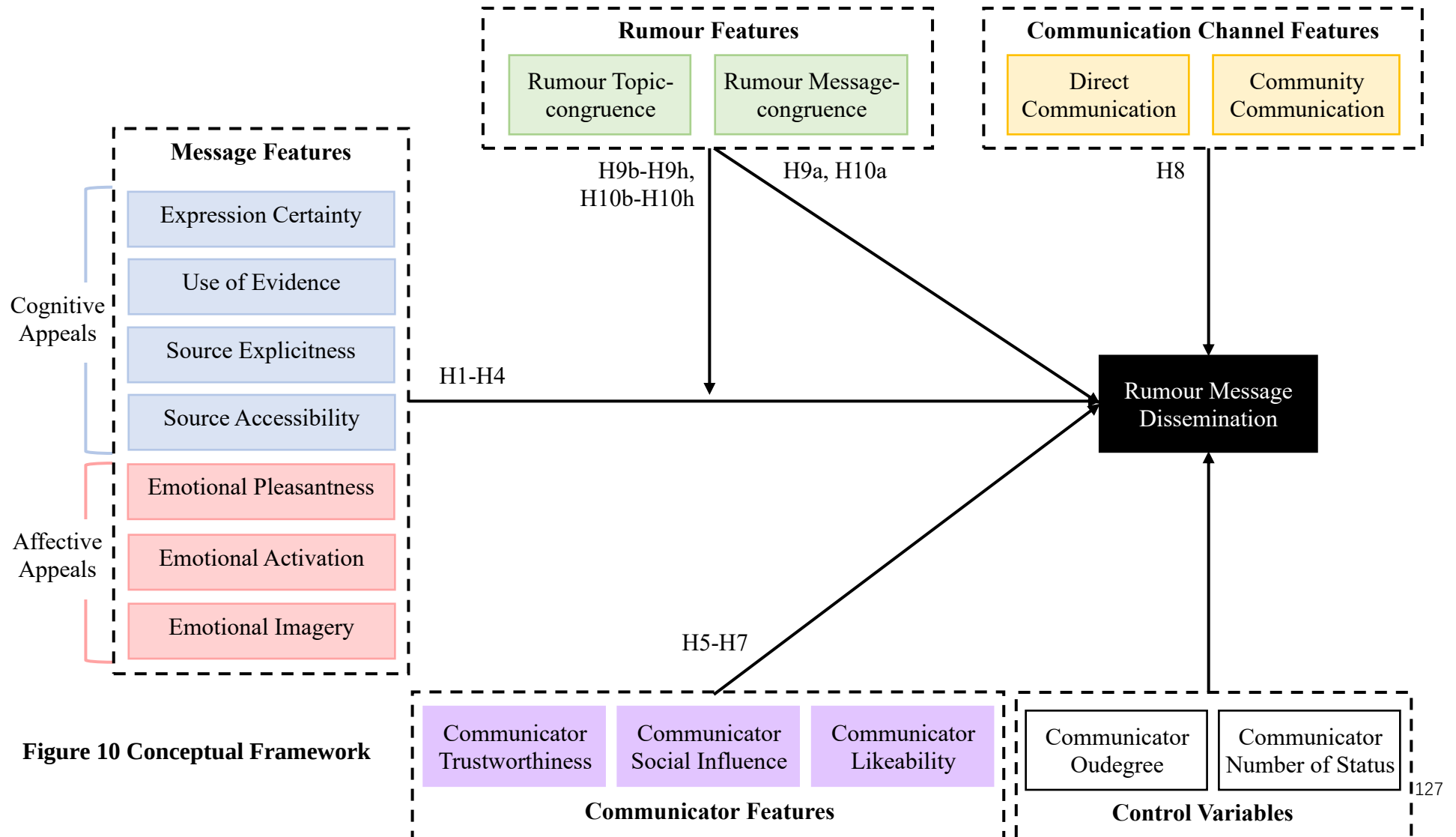


Figure 10 Conceptual Framework

## **CHAPTER 4 RESEARCH METHODOLOGY**

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Following the preliminary conceptual framework developed in the previous chapter, this chapter explains the research methodology of the thesis. This thesis incorporates qualitative and quantitative research and deductive and inductive approaches. Specifically, this thesis consists of three studies. First, guided by the preliminary conceptual framework, a qualitative and inductive content analysis was carried out to explore the thematic and communication features of online rumour topics and messages. The results of the first study were used to develop research model and hypotheses on the rumour message level. Second, a quantitative and deductive study based on social media data was carried out to test the hypotheses on the rumour message level. Then, a quantitative and inductive study based on social media data was carried out to investigate the patterns of rumour message dissemination on the rumour topic level.

Specifically, this chapter is organised as follows. The first section reviews the research philosophy and strategy. The second section provides an overview to the research design and presents the overall rationale behind such design. The research context and specifies the scope of research were specified in the third

section. The fourth section describes the data collection process. Following that, the fifth to seventh sections present the research design of three inter-related studies. The final section illustrates how ethical issue is addressed in this research.

## **4.1 Research Philosophy and Strategy**

Research is about addressing problems and searching for knowledge systematically (Kothari 2004). The philosophy of research refers to “the set of common beliefs and agreements shared between scientists” regarding the way in which problems are addressed and knowledge is sought (Kuhn 1962, p. 45). The philosophical choice determines the design and methodology of research.

Research philosophy is made up of ontology, epistemology, methodology and methods (Scotland, 2012). Ontology focuses on assumptions about what the reality is, that is, the nature of the investigated phenomenon. Common ontological assumptions include nominalist view which denies the existence of universals and abstract objects but affirms the existence of general or abstract terms and predicates, and realist view which believe that reality exists independently of observers. Ontological assumptions give rise to epistemological assumptions. Epistemology is about how reality is known, that is, the way knowledge is created, acquired and communicated. Common epistemological theories include rationalism and empiricism. Epistemological assumptions in turn determine the research methodologies and methods. Research approaches may vary according ontological and epistemological assumptions. The positivist approach considers that there exists a single reality that can be measured and that the researchers are independent of the observation. Different from positivist approach, the interpretivist approach considers that there is no single truth about how the reality needs to be interpreted and that the observer is included in the object of observation as well.

Research methodology refers to the approach to systematically solve the research problem and research questions. It is associated with how research is done scientifically. Research methodology covers various dimensions and research methods are important elements of the research methodology. Research methods refer to all the methods and techniques used by the researcher to conduct a research and to perform research operations to study the research question (Kothari 2004). Research methods include the methods used to collect data and the methods that are employed to analyse data. The latter data analytical methods further consist of the statistical techniques used to examine relationships and test hypotheses and the techniques used to evaluate the results (Kothari 2004). It is important to employ appropriate research methods and techniques, and it is also crucial to understand the logic behind the choice of those methods and techniques in regard to a particular research. Researchers should understand the assumptions underlying different research methods and techniques, which methods and techniques are relevant to her/his research problem, and why. These issues are all within the scope of research methodology.

Deductive reasoning moves from general principles and theories to specific hypotheses and conclusions (Rovai et al. 2014). Deductive reasoning usually begins with researchers having a theory in mind regarding the topic they are interested in. Researchers then narrow down the theory into specific hypotheses and further collect observations to test the hypotheses and conclude if the original theory is confirmed. Therefore, deductive reasoning is mainly concerned with testing hypotheses. Inductive reasoning, on the other hand, starts from specific observations to general theories. Inductive reasoning tends to be

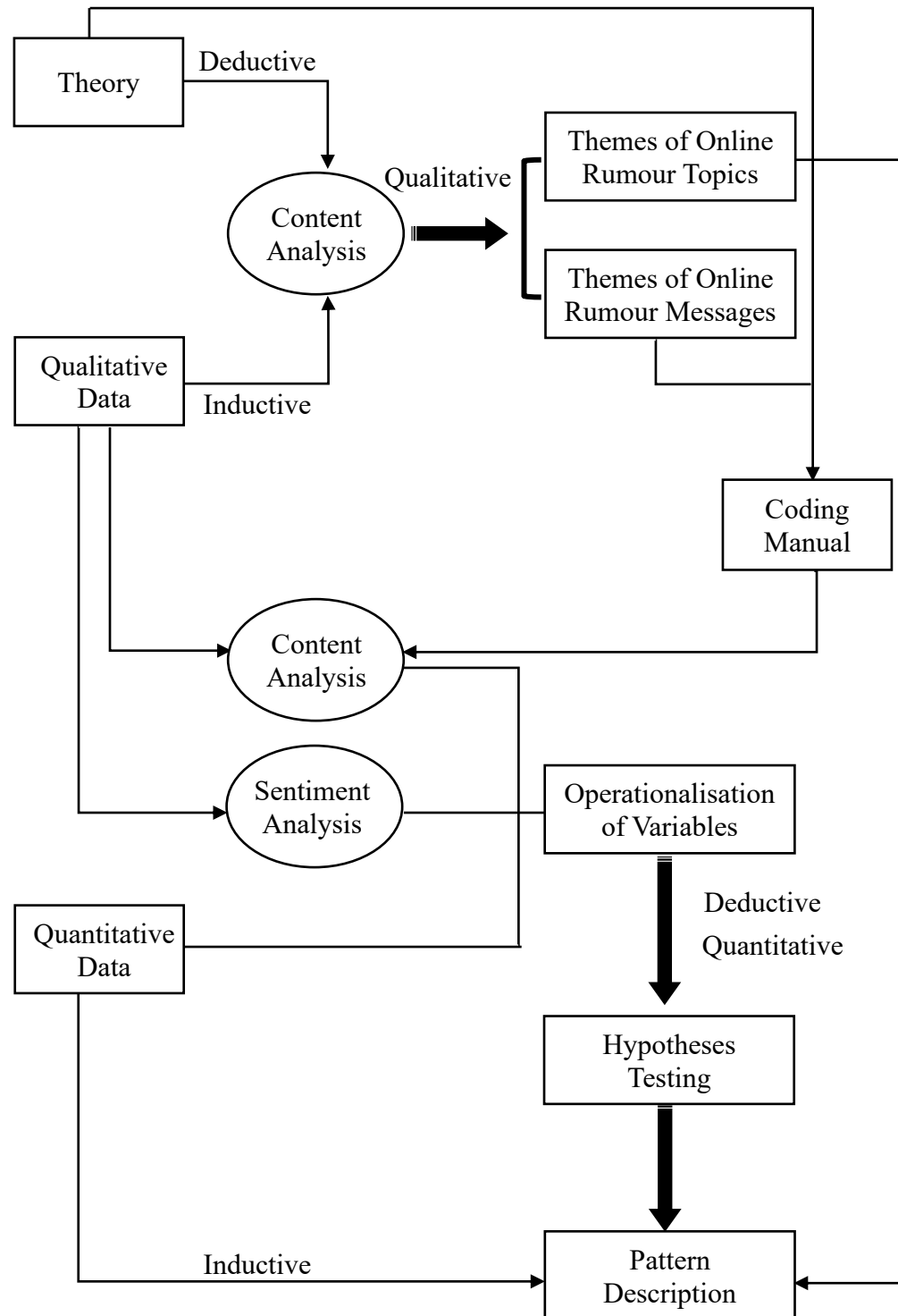
more exploratory than deductive reasoning. This thesis, like many research projects, involves both inductive and deductive reasoning processes.

Quantitative and qualitative research paradigms are built on different assumptions regarding the nature of the world and can be used to understand the world differently (Rovai et al. 2014). Researchers taking positivist position tend to use quantitative methods which involve objective assessment of the reality of the world. Researchers taking interpretivist approach are likely to employ qualitative methods which are concerned with subjective interpretation of attitudes, opinions and behaviour. Quantitative and qualitative paradigms use different types of data and entail different procedures, measurements and instruments to collect and analyse the data. The research design that combines both quantitative and qualitative paradigms- the mixed methods approach- is increasingly prevalent nowadays in information systems research. This thesis also adopts the mixed methods approach.

## **4.2 Research Design**

This thesis is designed to mainly adopt the positivist approach with deductive reasoning as its primary focus is to empirically test the hypotheses of the message-level rumour dissemination model. Such hypotheses testing can be objective with researcher separating from the observation. However, this thesis also incorporates subjective interpretation and theory development in the process, taking the interpretivist approach with an inductive reasoning. As illustrated in Figure 11, this thesis entails both inductive and deductive reasoning

and takes the mixed methods approach. The development of theory via the exploratory qualitative content analysis serves as an initial input to the generation of hypotheses. The exploratory quantitative analysis on topic-level rumour dissemination patterns also serves to seeks to measure the reality objectively yet there is no single truth to interpret the reality. As explained, in this thesis, a quantitative and deductive part following a qualitative and inductive part was incorporated with another quantitative yet inductive part. The qualitative research was used to facilitate and develop quantitative research. Such mixed-methods design can be referred to as the “dominant-less dominant” design. In the case of this thesis, the quantitative paradigm plays a more dominant role than the qualitative paradigm.



**Figure 11 Overview of Research Design**



To investigate the research questions and achieve the research objectives of this thesis, three studies were conducted. The overview of these studies is provided in Table 5. The details of these three studies will be explained in the subsequent sub-sections in this chapter.

**Table 5 Overview of Studies**

| <b>Study</b>   | <b>Main Research Purposes</b>                                                                                         | <b>Unit of Analysis</b> | <b>Method of Analysis</b>                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Study 1</b> | To identify key themes of online rumour topics                                                                        | Online rumour topic     | <ul style="list-style-type: none"> <li>• Inductive content analysis</li> </ul>                                 |
|                | To identify the features of online rumour messages                                                                    | Online rumour message   |                                                                                                                |
| <b>Study 2</b> | To test message-level dissemination model of online rumour messages on the message level                              | Online rumour message   | <ul style="list-style-type: none"> <li>• Sentiment analysis</li> <li>• Negative binomial regression</li> </ul> |
| <b>Study 3</b> | To compare message-level dissemination model of online rumour messages across different types of online rumour topics | Online rumour message   | <ul style="list-style-type: none"> <li>• Negative binomial regression</li> </ul>                               |

### 4.3 Research Context and Scope of Research

In addressing the research questions, this thesis focuses on business-related false rumours. Business-related false rumour in this thesis refers to the rumour that is related to one or several specific brands, companies, products, services and/or

business operations and that turns out to be false. Business-related false rumours are targeted because of their prevalence in social media and the potential harms they may cause. For example, such rumours may cause the public to misunderstand the situation and create unnecessary stress and tension. Business-related false rumours can also damage companies' reputations and cause decreases in sales and in stock market prices (Boland 2017; Bordia et al. 2005). Nevertheless, business related rumours are usually studied in offline context, focusing on the effects of workplace rumours on employee performance (Difonzo et al. 1994). Most previous studies on rumours in social media focuses on natural disasters such as the tornado struck, social crises such as the terrorist bombing attack, and social events such as the election (Liu et al. 2014; Mondal et al. 2018; Oh et al. 2013; Shin et al. 2018; L. Zhao et al. 2016), yet business related rumours are often overlooked. Moreover, false rumours are difficult to predict as they are not stimulated by certain facts. Because of this, understanding the dissemination is of great importance to control and combat false rumours. For true rumours, business crises or word-of-mouth complaints which are based on events or incidents that actually happened, companies may be able to prepare how to manage the spreading of such information beforehand. However, false rumours are usually beyond the rumour target's expectation. Because of this, it is almost impossible for companies to prepare an upfront handling mechanism prior to the spreading of false rumours. The control of such false rumours thus heavily relies on understanding the dissemination of rumour messages. With the knowledge regarding how business-related false rumour messages are disseminated, companies can develop their handling strategies quickly after the occurrence of rumours, and thus are likely to minimize the harms those rumours

may cause. Therefore, business-related false rumour serves as an important context to investigate the research questions.

This thesis studies the dissemination of false-related false rumour in social media. Social media have many forms. To clearly outline the research scope of this thesis, social media here are defined as the web or mobile based platforms on which users can establish a profile available to the public or within certain restricted network where they can connect to other users and create, co-create and exchange user-generated content highly interactively (Ellison 2008; Hollebeek et al. 2014; Kaplan and Haenlein 2010; Laroche et al. 2013; Majchrzak et al. 2013). Among various social media platforms, Twitter is chosen as the context of this thesis. Twitter is a type of social media service for short messages (Berthon et al. 2012). A tweet allows less than 140 characters with which users post their daily life and their opinions.

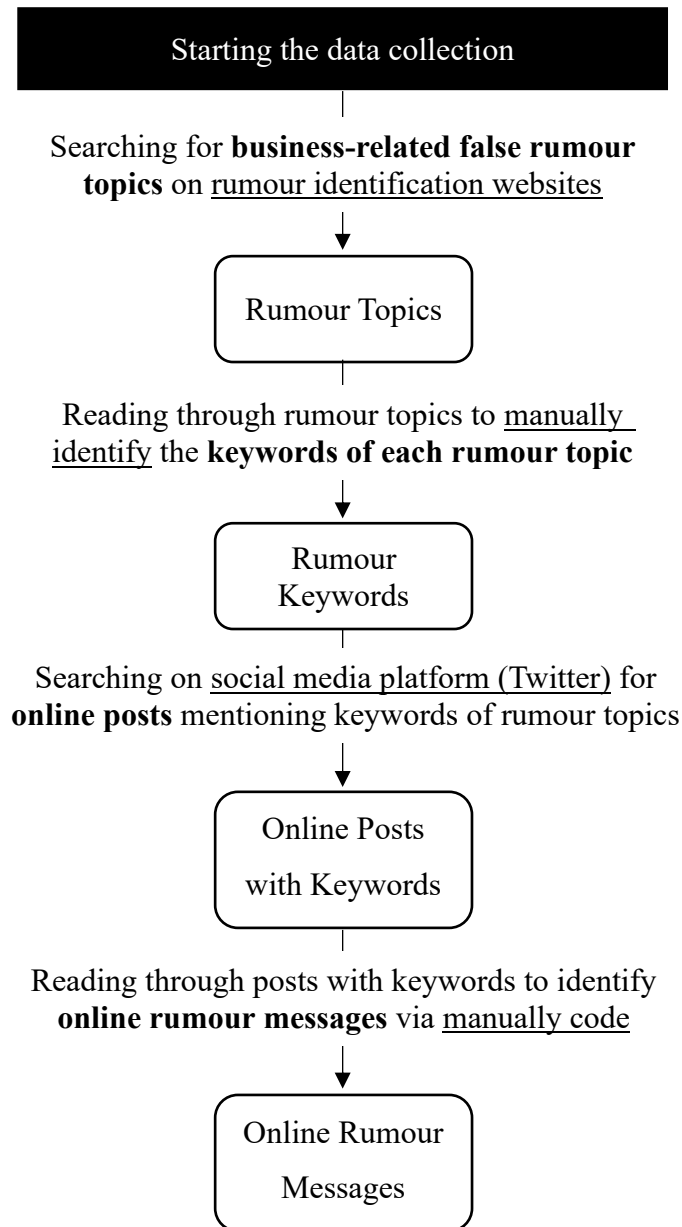
Twitter is chosen as the context for several reasons. First, it is one of the most commonly used social media (Stieglitz and Dang-Xuan 2013). Since its launch in 2006, to date Twitter holds approximately 330 million monthly active users (Aslam 2018). By 2010, Twitter has over 145 million users generating around 90 million tweets each day (Kietzmann et al. 2011). In December 2012, the number of Twitter's users reached 200 million (Stieglitz and Dang-Xuan 2013). By the end of 2017, Twitter's daily active users reached 100 million users, and every day there were around 500 million tweets posted (Aslam 2018). Second, previous reports and research studies have shown that false rumours are prevalent on Twitter and a large number of false rumours are being disseminated

(Oh et al. 2013; Stieglitz and Dang-Xuan 2013; Usui et al. 2015). Third, Twitter is a representative social media platform with a powerful information sharing function - “retweeting” (Stieglitz and Dang-Xuan 2013). Retweeting involves the resending someone else’s tweet. Because of this function, Twitter provides and is used as an ideal platform for users to disseminate information, and the information posted on Twitter mostly can be accessed by the public. Retweeting thus provides an explicit approach for researchers to track the diffusion of information, including the dissemination of online rumour messages (Stieglitz and Dang-Xuan 2013). Because of this, Twitter is widely accepted and commonly referred to by researchers who are interested in rumours and information dissemination in social media (Gunarathne et al. 2018; Liu et al. 2014; Oh et al. 2013). For example, Kumar and Shah (2018) reviewed previous research and found that Twitter is one of the most dominant social media platforms that study false information including false rumours. Twitter is also considered as being able to reflect collective emotive trends and public opinions (Stieglitz and Dang-Xuan 2013). Moreover, Twitter’s data is available to the public (Stieglitz and Dang-Xuan 2013). Therefore, in this thesis, Twitter is selected as an appropriate research context for studying online rumours and their disseminations. Carrying out research in the context of Twitter enables the addressing of listed research questions of this thesis. Specifically, in this thesis, the dissemination of online rumour messages is studied in terms of the scale of dissemination, that is, the quantity of triggered retweets.

#### **4.4 Method of Data Collection**

Generated by end users of social media who voluntarily share their views and information online, social media data usually is public and freely available (Chan et al. 2016). As digital technologies develop, researchers are enabled to collect valuable social media data online, generating new research opportunities. This thesis rests on social media data to solve the research questions.

This study first referred to rumour identification websites to identify rumour topics that are business related and turn out to be false. After this, the keywords of the rumour topics were listed, which were then used to search for social media posts that may carry rumours. The search results were then manually coded to identify online rumour messages which are the posts that actually contain rumours. The process of data collection is summarised in Figure 12.



**Figure 12 Processes of Data Collection**

#### **4.4.1 Identifying Rumour Topics**

This study mainly focuses on false rumours about companies and brands. This study refers to rumour-identification websites as the starting points to identify

rumour topics. *Snopes.com*, *TruthOrFiction.com*, *FactCheck.org* and *PolitiFact.com* are typical and most well-known rumour identification and verification websites. Such websites capture rumours and attempt to identify the truth of the rumours. However, *FactCheck.org* and *PolitiFact.com* intensively focus on political rumours which are beyond the scope of this study. *Snopes.com* and *TruthOrFiction.com* cover a wider range of topics including the business-related rumour topics that this study is interested in. Thus, this study chooses *Snopes.com* and *TruthOrFiction.com* as the sources of rumour topics.

*Snopes.com* was created by Daavid Mikkelsen in 1994, and it is one of the oldest and largest fact-checking websites online. Currently, the website is entirely owned and managed by the Snopes Media Group Inc. (SMG). The operation of *Snopes.com* is funded via the sales of digital advertising and contribution from readers and non-partisan organisations and advertisers and contributors have no influence on the content being published on the website. It has received two Webby awards and have been part of the “Best of the Web” list on a regular base. *Snopes.com* has been considered as an important reference for rumour-related research, and the website has been well recognised and referred to as sources of rumour (Degroot 2011; Swami et al. 2012). Regarding the information published, *Snopes.com* has several labels, including true, mostly true, mixture, mostly false, false, unproven, outdated, miscaptioned, correct attribution, misattributed, scam and legend (see Table 5). *Snopes.com* has a high variety of topics, such as business, crime, food, politics, science, college, entertainment, embarrassments, horror, humour.

*TruthOrFiction.com*, established in 1999, seeks to online hoaxes and scams and check the truth for rumours, interesting tales, virus requests for help and other unverified stories that people encounter when they surf on the Internet. The website has been referred to in many research papers for the verification of online information (Droge et al. 2010; Polonsky and Hyman 2007). It labels the information into several categories, including fiction, mostly fiction, truth, truth & fiction, truth & misleading, truth & outdated, disputed, commentary, unproven, and outdated (see Table 6). *TruthOrFiction.com* focuses on topics related to politics, religion, terrorism, nature, space/aviation, food/drink, household, crime/police, money/financial, health/medical.

**Table 6 Labels Used by Websites**

| <b>Snopes.com</b>              | <b>TruthOrFiction.com</b>                 |
|--------------------------------|-------------------------------------------|
| False                          | Fiction                                   |
| Mostly false                   | Mostly fiction; Reported as fiction       |
| Mixture                        | Truth & Fiction; Disputed                 |
| Mostly true                    | Truth & misleading; Truth & outdated      |
| True                           | Truth                                     |
| Correct attribution            | Confirmed authorship; Correct attribution |
| Unproven                       | Unproven                                  |
| Outdated                       | Outdated                                  |
| Misattributed; Miscalcaptioned | Inaccurate attribution                    |
| Scam; Legend                   | Scam; Commentary                          |



This study targets business-related rumours that turn out to be false. The two core components here are “business-related” and “false”. Business-related rumours in this study is defined as rumours are that are related to specific brands, companies, their products, services and/or business operations. If a rumour is related to the general business sector of the society or community, such as tax policy change, it will not be taken into account in this study. *Snopes.com* and *TruthOrFiction.com* both label the rumour based on their verification. Only the rumours that are labelled as “false” in *snopes.com* or “fiction” in *truthorfiction.com* are selected as rumour topics in this study. The label “false” indicates that the “primary elements of the claim are demonstrably false” (Snopes.com n.d.). Business-related rumours that are considered as mostly false, mixture, mostly true, disputed, misattributed or miscaptioned, although also carry certain level of misinformation, are not included in this study. This is because the complex combination of truth and fiction makes it difficult to accurately identify online social media messages that carry only the fiction parts of the rumour. If an online message focuses on the truth part of the rumour, then the message is not really related to the “false rumour” and thus would be outside the scope of this study.

In total 118 rumour topics that fit the above criteria were selected, covering 80 companies and brands. All the rumour topics were double-checked by two researchers to ensure they fit the definition of rumour in this study, that is, when being propagated, the rumour topics were unverified statements, propositions or beliefs that refer to certain circumstances, objects or people and were topically relevant to the social networks that they were passed along (Difonzo and Bordia

2007; Huang et al. 2015; Lee et al. 2015; Oh et al. 2013).

#### **4.4.2 Identifying Keywords of Rumour Topics**

Once the target rumour topics were selected, the rumours were manually coded to identify a list of keywords that most precisely describe the rumour topics. Two coders coded the keywords separately and then discussed together to reach an agreement.

#### **4.4.3 Collecting Online Posts**

Based on the identified keywords, the posts that carry the keywords were extracted using keyword search from one of the most popular social media platforms- Twitter. For each rumour topic, one or several keyword combinations were used in order to extract all possible tweet messages related to the rumour topic. To further improve the completeness of the dataset for each rumour topic, the user IDs and hashtags embedded in the tweet messages were also collected and used to further collect relevant tweets and retweets.

The tweet messages were crawled using Webometric Analyst, Easy Web Extract and Octoparse. The first one is a data collection tool developed by a group of academics (Thelwall et al. 2010), while the latter two are commercialised tool for crawling data. The data extraction relies on all three tools because Twitter API only allows the extraction of tweets from the most recent week, while the tweets that this study is interested in can be traced back up to several months or even years ago. Only English tweets were extracted

considering the capability to understand and analyse the tweets.

#### **4.4.4 Identifying Online Rumour Messages**

To identify rumour messages, this study adopted and adapted the coding scheme of Oh et al.'s (2013) to code the archived tweet messages into rumour messages, rumour combating messages and non-rumour related messages.

*Data preparation.* The 106,290 tweet messages were filtered and removed the repeated data. Weblinks, emoticons, pictures, drawings, and videos in those tweets were transformed into textual descriptions for coding.

*Unit of analysis.* According to Oh et al. (2013) and Bordia and Difonzo (1999), the unit of analysis needs to be specified before content coding. The unit of analysis can be “a word, phrase, sentence, paragraph, page, or entire narrative.” (Bordia et al. 1999, p. 23). Without a proper choice of the unit of analysis, there may be an overlap of multiple categories or lack of sufficient information for interpretation. Often, the unit of analysis is chosen to be the minimum unit that presents a complete thought. In this study, tweet messages often present the complete thoughts of twitter users. Moreover, the length of tweet messages has a limit of 140 characters, which is already a very limited number of words for sentences with complex meanings. In fact, according to O’Connor et al. (2010), on average, each tweet contains only 11 words. To ensure there is sufficient information for interpretation of one thought, each piece of tweet message was considered as the unit of analysis in this study.

*Coding scheme.* According to Oh et al.'s (2013) coding scheme, rumour is considered to have three dimensions, namely unverified proposition, referents of the statement and topical relevance, and is defined as:

*“A Twitter message which does NOT explicitly indicate a person (e.g., the prime minister of Indian government), source (e.g., BBC, NDTV, website, etc.), context or known data to serve as a proof or verification for the message. The message MUST be topically relevant to the incidents under this study (Mumbai terrorist attack, Toyota recalls, and Seattle shooting incident), and it MUST refer to an object, person, or situation rather than an idea or theory.”*

While Oh et al.'s (2013) coding scheme is to code rumour, this study pre-selected rumour topics and need to code the relevant rumour messages. Modified from Oh et al.'s (2013) coding scheme, four conditions were used in this study to code the tweet messages: (1) if the tweet message is relevant to the statement in the rumour topic; (2) if the tweet message refers to a referent related to the brand or company mentioned in the rumour topic; and (3) if the tweet message does not have an authoritative verification or clarification of the rumour topic. The coding scheme used in this study is detailed in Table 7. All tweet messages were coded into three categories, where 1 represents online rumour messages, 2 represents rumour combating messages, and 0 represents the tweet messages that are irrelevant to the rumour topic.

**Table 7 Coding Scheme of Rumour Messages**

| Coding Scheme in this Study |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                               |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Category                    | Definition                                                                                                                                                                                                                                                                                                                                                                                                   | Example                                                                       |
| Rumour Message              | A tweet message that is <b>RELEVANT</b> to the statement, hypothesis or belief of the rumour topic. The message <b>MUST</b> refer to a referent related to the brand or company mentioned in the rumour topic and does <b>NOT</b> explicitly indicate an authoritative person (e.g., the spokesperson of a company), source (e.g., BBC), or data to provide a valid proof or verification for the statement. | <i>Who knew abercrombie&amp;fitch had a pair of 'n-word brown' pants? 0_o</i> |
| Rumour Combating Message    | A tweet message that is <b>RELEVANT</b> to the statement, hypothesis or belief of the rumour topic. The message <b>MUST</b> refer to a referent related to the brand or company mentioned in the rumour topic and explicitly <b>HAVE</b> an authoritative person (e.g., the spokesperson of a company), source (e.g., BBC), or data to provide a valid proof or verification for the statement.              | <i>That is not real. Amazon did not cause Postal Service bankrupt!!</i>       |
| Non-Rumour Related Message  | A tweet message that is <b>IRRELEVANT</b> to the statement, hypothesis or belief of the rumour topic and/or have <b>NO</b> referent related to the brand or company mentioned in the rumour topic.                                                                                                                                                                                                           | <i>Still waiting for my delivery, bought it last Weds on Amazon...</i>        |

*Inter-Rater Reliability.* To ensure the reliability of coding, two coders independently labelled a small sample of the tweet messages – 150 tweets- as rumour messages and non-rumour messages following the coding scheme. After the sample was coded, Cohen's Kappa analysis was conducted to test the agreement between two coders on classification, via which to check the coding consistency. The resulted kappa ( $k$ ) was 0.788, while an agreement is significant when  $k$  is above 0.600. Therefore, there was sufficient level of agreement between coders and thus the coding based on the current coding scheme was consistent. Then, the two coders coded all the tweet messages individually and independently. The results of two coders were compared. The inter-coder reliability  $k$  was 0.862, which suggest that the coding was consistent. For the tweet messages with inconsistent labels, meetings between coders were organised to discuss. In total seven 1-hour meetings were organised to get both coders agree with all the labels. Based on the coding, 10,308 of the tweet messages were non-rumour related messages and were removed from the dataset, leaving 95982 rumour messages for further analysis.

#### **4.4.5 Collecting Supplementary Data**

Based on the data collected, Easy Web Extract was used to further obtain the sender's digital profile information, to extract tweets that mention the rumour-targeted company and to collect tweets that are relevant to the key issue mentioned in the rumour topic. With the username of senders of online rumour messages, their user profile pages were accessed and extracted. Those senders who restrict the public to access their profile pages were removed from the dataset. The sender's profile description, whether it is a verified account,

external links, geo-locations, date of join were collected, together with the sender's number of posts, number of followings, number of followers, number of favourites received. Referring to the rumour topics and the time stamp of online rumour messages, tweets that contained the company name and were posted three months prior to the emergence of rumour were collected, tweets that mentioned the key issue of interest and were posted three months prior to the emergence of rumour were also collected. Three months are appropriate for the data collection because it is long enough for sufficient amount of data and it is also short enough in the sense that the sentiments can reflect public's prior attitudes and may influence the evaluation of online rumour messages.

#### **4.5 Study One: Exploratory Study on Online Rumour Topics and Online Rumour Messages**

While there are a few studies that predict misinformation in social media, the understanding of online rumour topics and messages in social media is still limited and there lacks a strong theory to fully guide this thesis. Given the complication of online rumour messages in social media, it is necessary to interpret this phenomenon in-depth and systematically and portray the information carried in online rumour messages, in the context of social media. Qualitative paradigm allows the understanding of the complex online rumouring phenomenon and enables the seeking of deep and detailed insights. In order to address the first research question of the thesis, that is, to understand the themes of online rumours and online rumour messages, a qualitative and exploratory study was undertaken. Specifically, this qualitative study has the following

purposes. First, the rumour topics spread on social media may be different from the ones shared in face-to-face manner. To clarify the rumour topics, this study aims to identify key themes of online rumour topics and develops a typology accordingly. Second, online rumour messages and their dissemination in social media can be of great difference from those in offline context, yet there lacks insight on the information carried in online rumour messages. This study aims to portray the features of online rumour messages and compare the features of rumour messages across different types of rumour topics.

#### **4.5.1 Research Data**

In this study, rumour topics collected from rumour identification websites and online rumour messages extracted from Twitter are analysed to understand their themes and types.

#### **4.5.2 Qualitative Data Analysis**

Inductive content analysis (also known as thematic analysis) was used to analyse the data. By employing inductive qualitative content analysis, this study generated grounded codes, summarised the themes of online rumour topics and online rumour messages, categorised the rumour topics and messages, and compared the features across categories.

Content analysis is commonly used in studying information content and features. It is the research technique that analyses the content of spoken or written verbal materials or other meaningful matter (e.g., image, sound) to make valid and replicable inferences to their contexts of usage (Kothari 2004; Skalski



et al. 2017). Content analysis analyses the content of communicated materials using “classification, tabulation, and evaluation of its key symbols and themes” (Krippendorff 2004, p. xvii). Before the 1940s, content analysis was mainly about analysing the counted patterns of materials quantitatively. However, after the 1950s, content analysis has been dominated by qualitative analysis. Regardless of using quantitative or qualitative methods, content analysis can be used as the key technique whenever the focus is to understand the nature of verbal materials (Kothari 2004). Moreover, content analysis is appropriate when researchers would like to focus more on describing the phenomenon instead of interpreting the phenomenon like the grounded theory or hermeneutic phenomenology does. Content analysis is widely used to explore and subjectively interpret qualitative data sets. Compared to self-reported methods like questionnaires, content analysis based on information data itself is more direct and can reflect the complexity and heuristics that human find difficult to report (Brutus et al. 2013). Content analysis can serve as a key technique whenever a study focuses on understanding the nature of verbal materials such as texts. Content analysis enables researchers to enhance their understandings and to explore new insights regarding particular phenomena or contexts (Skalski et al. 2017).

Inductive content analysis can subjectively interpret the communication content, where the manual work reduces the mistaken interpretation based on face value by software and computers (Jaminson-Powell et al., 2012) This approach suits this study for several reasons. Understanding the online rumour topics disseminated on social media is difficult considering the topics are about

complicated issues and interpreting images and screenshots may be necessary to understand the situation. Understanding the online rumour messages posted on Twitter is difficult because the expression on Twitter tends to be informal and perhaps constrained by word limits. Moreover, the exact meaning of words or phrases in Tweets may be beyond their face value, being used as sarcastic. Furthermore, there are other information other than textual, such as emoticons, pictures, external sources, etc. that require subjective interpretation. Compared to self-reported methods, content analysis based on information itself is more direct and can reflect the complexity and heuristics that human find difficult to report (Brutus et al. 2013).

Combining the qualitative content analysis processes suggested by Elo and Kyngäs (2008) and by Zhang and Wildemuth (2009), the process of content analysis in this study is outline in Figure 13. This process was applied to analyse online rumour topics and online rumour messages separately. The detailed process is explained below:

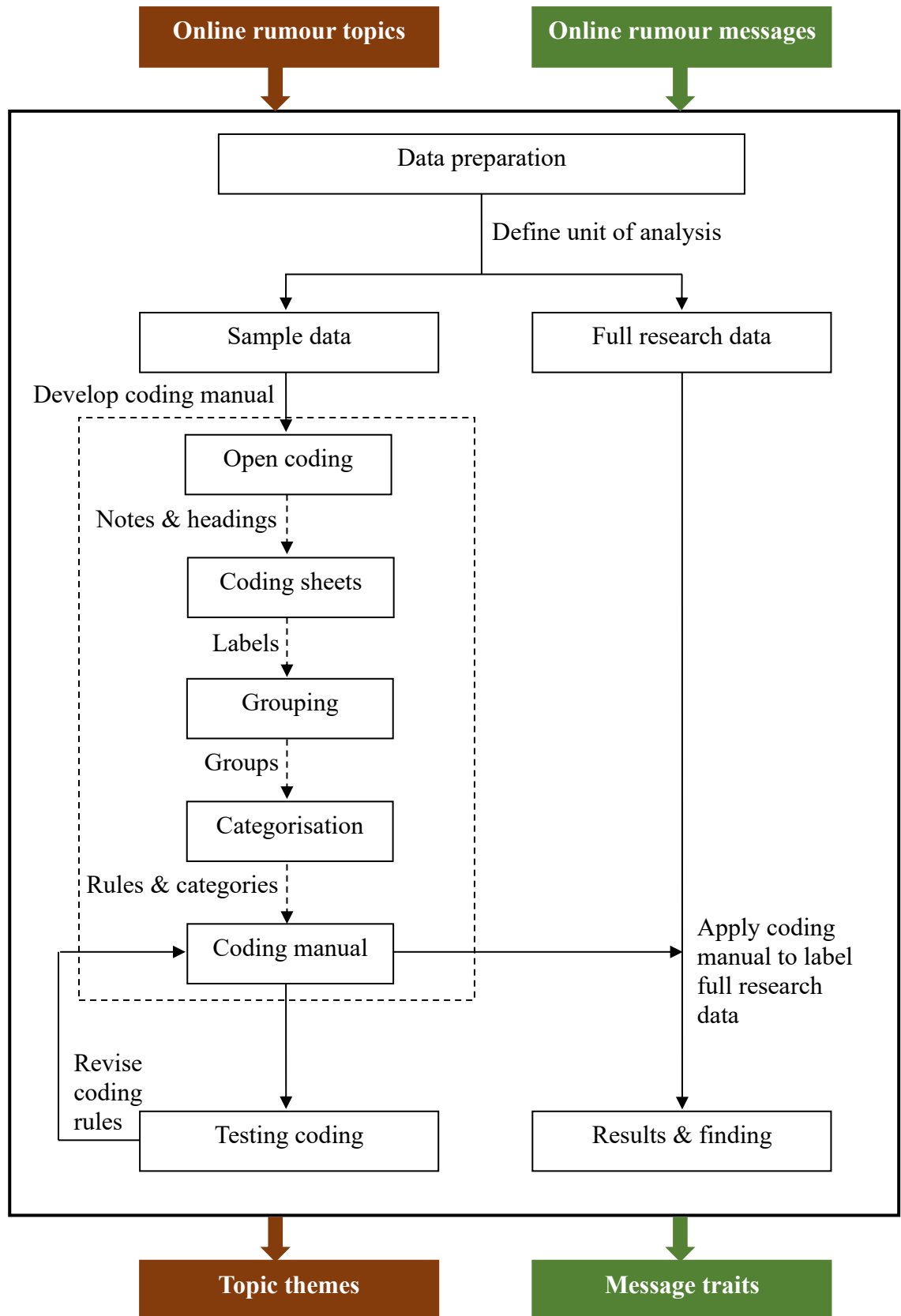


Figure 13 Process of Qualitative Content Analysis

- *Data preparation and unit of analysis:* All the non-textual information contained in sampled online rumour topics and online rumour messages was converted into textual descriptions. The unit of analysis was different for the analyses of rumour topics and rumour messages. When analysing rumour topics, the unit of analysis was the description of each rumour topic. For the analysis of rumour messages, the unit of analysis was defined as each individual rumour-related communication message, including the content of tweet and all associated information, such as the information of the sender and the timing of posting.
- *Open coding:* A sample of full research data was used to develop the coding manuals for rumour topics and rumour messages. For rumour topics, the sample incorporates 50 rumour topic descriptions, whereas for rumour messages, the sample consists of 500 rumour-related communication messages. Two coders (me and another PhD student) independently read through the sample, wrote down headings and notes that describe the information in text margin, re-read the sample to note down more notes to cover all aspects of the information.
- *Coding sheets:* The headings and notes generated by both coders were transferred into coding sheets, where only one item was kept if two different headings sharing the same meaning had been written by different coders. Initial categories were developed.
- *Grouping:* Items and initial categories in coding sheets were transferred onto post-it notes, one item or category per note. All notes were assembled and dissembled into groups using an open space and an

affinity diagramming technique.

- *Categorisation:* Groups were formulated into hierarchical categories; the upper hierarchy categories were more general themes while the lower level categories were more specific. Recapitulative phrases were used to name categories, and each category was defined in detail. The hierarchy of categories, their names, coding rules including definition and distinction, and examples together constructs the coding manual.
- *Testing coding manual:* The coding manual was tested for validity and consistency. The rest of rumour topics descriptions and a new set of 500 prepared rumour-related communication messages were coded following the coding manual by 2 coders, and the level of agreement was tested. The level of agreement was low ( $\kappa$  in average=0.504, lower than significant agreement level 0.600). After the discussion between coders, the coding manual was revised regarding categorisation and coding rules. Such testing and revising process repeated 3 rounds before finalizing the coding manual; axial and selective coding was used to validate or challenge categories at each iteration. The last round of testing produced the average  $\kappa$  of 0.887. This indicated a high level of coding consistency.
- *Applying the coding manual:* to understand the structure of themes and to estimate the proportion of content relating to each theme, the full research data were coded following the coding manual. Two full research

datasets were coded, one with all the rumour topic descriptions and the other with all the online rumour messages extracted from Twitter.

As the results of content analyses, themes, typologies and concepts of online rumour topics and messages were identified. The results of information feature of online rumour messages confirmed the message features proposed in the research framework. The detailed results and findings of study one is presented in Chapter 5.

#### **4.6 Study Two: Hypotheses Testing of Message-Level Dissemination Model of Online Rumour Messages**

As digital technologies develop, researchers are enabled to collect valuable data online, generating new research opportunities (Chan et al. 2016). However, such data are often qualitative and highly unstructured (Chan et al. 2016). More recently, content analysis is also increasingly used in various social media studies to convert codified information into a more usable format. In this thesis, content analysis was used to serve both purposes. To understand the qualitative and unstructured social media data and to quantify the data to test the hypotheses on rumour message dissemination, content analysis and sentiment analysis are used to operationalise the data, and regression analysis is used to examine the hypotheses.

Understanding the factors that influence dissemination of online rumour messages is of great importance to both academics and practitioners. The previous study (study one) lays the foundation to understand the nature and

informative insights of online rumour messages; however, little about dissemination can actually be known from the qualitative content analysis. This study (study two) thus aims to empirically investigate the rumour dissemination in social media on the message level. Specifically, with each piece of online rumour message as a unit of analysis, this study concerns what factors drive some online rumour messages to be shared more than others in social media. It uses social media data to examine the research model of rumour message dissemination on the message-level and to test the relevant hypotheses. The research model and hypotheses origins from a preliminary conceptual framework and is further developed with reference to the findings in study one.

#### **4.6.1 Research Data**

In this study, the online rumour message dataset was used as the key research data. Specifically, the dataset includes: the content of each tweet, the screen name and username of the sender, the number of retweets, replies and likes of the tweet. Such online rumour message data will be used to measure the dissemination of online rumour messages and to identify the message and channel variables. In addition, based on the username of sender from the online rumour message dataset, further data collection was conducted to get the sender's digital profile information (e.g. number of followers, number of followings, aggregate number of tweets the sender has posted). The sender's digital profile information is used to identify communicator variables. Moreover, online rumour topic dataset was also used to identify rumour valence. Referring to the rumour target in the dataset, tweets related to rumour target that were posted within 3 months prior to the initial sharing of the rumour were further

extracted using Easy Web Extract. Sentiment analysis was conducted on those tweets to identify target reputation. In the exploratory study, rumour topics were classified into four main types, including brand-oriented rumours, product/service-oriented rumours, people-oriented rumours and operation-oriented rumours. The themes were included as a control variable. Manual coding and sentiment analysis were employed to measure other variables. This will be explained in detail in the following section.

The sender's digital profile information is used to identify communicator variables. Moreover, online rumour topic dataset was also used to identify rumour valence. Referring to the rumour target in the dataset, tweets related to rumour target that were posted within 3 months prior to the initial sharing of the rumour were further extracted using Easy Web Extract.

#### **4.6.2 Variable Operationalisation**

The dependent variable in this study- the dissemination of online rumour messages- is measured using number of retweets. The independent variables examined can be divided into four dimensions, namely the message features, the communicator features, the channel features and the rumour features. Some variables were identified from the data collected whereas others were measured using sentiment analysis and manual coding. To operationalise the variables, data collected was pre-processed first. Raw data was cleaned, reformat and integrated into usable format. In addition, to anonymize the data, usernames in the tweets were replaced with pseudonyms 'uN', where N is the unique number assigned to a username; this number was reused to see find tweets from the same



author, but whilst keeping the author anonymous. After that, actions were taken to identify or measure the variables. Table 8 summarises the operationalisation of the variables.

**Table 8 Operationalisation of Variables**

| Variable                         | Operationalisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Dependent Variable</i></b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Rumour message dissemination     | Measured by the number of retweets each message has.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b><i>Message Features</i></b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Expression certainty             | <p>Binary coded: 1 represents high certainty expressed, and 0 represents low certainty.</p> <ul style="list-style-type: none"> <li>High expression certainty: assertive or affirmative message</li> </ul> <p>Example: <i>Taco Bell Announcing Bankruptcy, All Stores Closing By 2018</i></p> <ul style="list-style-type: none"> <li>Low expression certainty: unassertive or interrogative message</li> </ul> <p>Example: <i>@tacobell are y'all really going bankrupt? If so, I'll save you!!</i></p> |
| Use of evidence                  | <p>Binary coded: 1 represents having used evidence, and 0 represents using no evidence.</p> <ul style="list-style-type: none"> <li>Use of evidence</li> </ul> <p>Example: <i>Ford Foundation Gave Black Lives</i></p>                                                                                                                                                                                                                                                                                  |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p><i>Matter \$100,000,000</i></p> <ul style="list-style-type: none"> <li>• Use no evidence</li> </ul> <p>Example: <i>Let us not forget @Ford who donated a large sum to Black Lives Matter! They need to feel the pinch also! No more Fords for me! All I've owned</i></p>                                                                                                                                                                                                                                                                                                                                                                                       |
| Source explicitness  | <p>Binary coded: 1 represents a clear rumour source is mentioned explicitly, and 0 represents no clear rumour source mentioned.</p> <ul style="list-style-type: none"> <li>• High source explicitness: the name or website of the source is specified</li> </ul> <p>Example: <i>Ford Foundation sponsored Black Lives Matter</i> <a href="https://theroot.com/black-lives-matter">theroot.com/black-lives-ma...</a> via <a href="#">@TheRoot</a></p> <ul style="list-style-type: none"> <li>• Low source explicitness: neither the name nor the website of the source is specified</li> </ul> <p>Example: <i>Ford Foundation sponsored Black Lives Matter</i></p> |
| Source accessibility | <p>Binary coded: 1 represents the message's recipient can have direct access to the rumour source, for example, there is a direct hyperlink or mentioning on the rumour source. 0 represents that there is no such direct link to the source.</p> <ul style="list-style-type: none"> <li>• High source accessibility: there is a direct link to the source</li> </ul> <p>Example: <i>Ford Foundation sponsored Black Lives Matter</i> <a href="https://theroot.com/black-lives-matter">theroot.com/black-lives-ma...</a> via</p>                                                                                                                                  |

|                                     |                                                                                                                                                                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p><i>@TheRoot</i></p> <ul style="list-style-type: none"> <li>• Low source accessibility: there is no direct link to the source</li> </ul> <p>Example: <i>The Root: Ford Foundation sponsored Black Lives Matter</i></p>                                                          |
| Emotional pleasantness              | Measured using sentiment analysis with Whissell Dictionary of Affect in Language. The result is a decimal.                                                                                                                                                                        |
| Emotional activation                | Measured using sentiment analysis with whissell Dictionary of Affect. The result is a decimal.                                                                                                                                                                                    |
| Emotional imagery                   | Measured using sentiment analysis with Whissell Dictionary of Affect in Language. The result is a decimal.                                                                                                                                                                        |
| <b><i>Communicator Features</i></b> |                                                                                                                                                                                                                                                                                   |
| Communicator trustworthiness        | Binary coded: 1 represents a high level of trustworthiness and is coded if the communicator's Twitter account is a verified account. 0 represents a low level of trustworthiness and is coded if the communicator's account is not a verified account.                            |
| Communicator social influence       | Measured by the number of followers that the communicator has at the time of data extraction. Although this does not accurately capture the social influence of the communicator at the time of rumour dissemination, it can serve as a potential good approximant.               |
| Communicator likeability            | Measured by the number of likes that the communicator receives on his or her profile page at the time of the data extraction. Although this does not accurately capture the likeability of the communicator at the time of rumour dissemination, it can serve as a potential good |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | approximant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b><i>Communication Channel Features</i></b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Direct communication channel                 | Binary coding: 1 represents having direct communication and is coded if the message contains direct communications, i.e. @ the message receiver. 0 represents no direct communication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Community communication channel              | Binary coding: 1 represents having community communication and is coded if the message contains hashtag(s). 0 represents no community communication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b><i>Rumour Features</i></b>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Topic congruence                             | <p>Binary coded: 1 represents there is congruence between rumour target reputation and the rumour topic valence (i.e. same polarity of emotion), whereas 0 represents there is no such congruence.</p> <p>Target reputation is the sentiment polarity (i.e. positive, negative and neutral) identified using SentiStrength and NLTK Toolkit on all messages related to the rumour target within a certain period of time (i.e., the most recent three-months). 0 represents neutral, 1 represents positive and 2 represents negative.</p> <p>Rumour topic valence is the consequence of a rumour topic. It is manually labelled, with 0 represents neutral, 1 represents positive and 2 represents negative.</p> |
| Message congruence                           | <p>Binary coded: 1 represents there is congruence between rumour target reputation and the rumour message sentiment (i.e. same polarity of emotion), whereas 0 represents there is no such congruence.</p> <p>Target reputation is the sentiment polarity (i.e. positive,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <p>negative and neutral) identified using SentiStrength and NLTK Toolkit on all messages related to the rumour target within a certain period of time (i.e., the most recent three-months). 0 represents neutral, 1 represents positive and 2 represents negative.</p> <p>Rumour message sentiment is the sentiment polarity of the individual online rumour message (i.e. positive, negative and neutral). It is measured using SentiStrength and NLTK Toolkit, with 0 represents neutral, 1 represents positive and 2 represents negative.</p> |
| <b><i>Control Variables</i></b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Communicator<br>oudegree                   | Measured by the number of followings a communicator has at the time of data extraction. Although this does not accurately capture the social influence of the communicator at the time of rumour dissemination, it can serve as a potential good approximant.                                                                                                                                                                                                                                                                                    |
| Communicator<br>number of social<br>status | Measured by the total number of posts a communicator has at the time of data extraction. Although this does not accurately capture the social influence of the communicator at the time of rumour dissemination, it can serve as a potential good approximant.                                                                                                                                                                                                                                                                                   |

Measuring the variables based on archival Twitter data can be difficult because the textual data is highly unstructured. Moreover, the expression on Twitter tends to be informal and the limited space means that abbreviations are

common. In addition, the exact meaning of words or phrases in tweets may be beyond their face value. For example, with the phrase ‘Right... like I was right at the scene’, the written text may be interpreted as that the sender was indeed at the scene; however, given the contexts and the manner of language organisation, it is likely to mean that the sender was actually not at the scene. Although computers and software can be sophisticated in understanding the face value of texts, they may not be able to interpret the contexts as good as human beings. Therefore, qualitative content analysis is necessary for interpreting complex information to reduce the potential mistakes (Jaminson-Powell et al., 2012).

In terms of the manual coding, two PhD students, including me, were involved in the data coding process. The two coders were first trained to have general background knowledge regarding rumour and the rumour topics under investigation. Tweets were coded in terms of several variables, namely expression certainty, use of evidence, source explicitness, source accessibility and communicator proximity.

Although interpreting textual data is in general difficult, sentiment analysis is relatively well established and there are many tools and algorithms that focus on identifying sentiments of short communication messages such as tweets. Sentiment analysis is the research technique that systematically analyses written and spoken content through computers or qualitative methods to extract the attitudes or emotions expressed in the content (Kothari 2004; Stieglitz and Dang-Xuan 2013). It is the computational treatment to classify reviews into positive

and negative polarity or other emotional dimensions (Pang and Lee, 2008). It is increasingly popular to understand user generated content on the Internet (Archak et al., 2011; Pang and Lee, 2008).

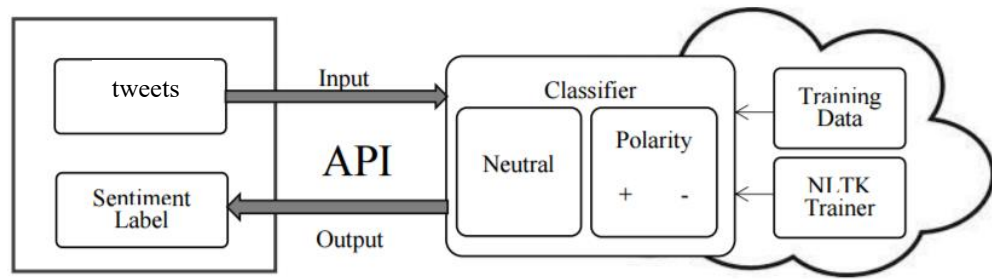
In this thesis, several sentiment analysis tools were used to identify the emotions. The Whissell Dictionary of Affect in Language was used to measure the affective appeals of online rumour messages, including the level of pleasantness, the level of activation and the level of imagery.

SentiStrength was used to analyse the polarity of the target reputation and the rumour message emotional appeal when measuring topic congruence and message-congruence. The polarity is used (instead of level of pleasantness or strength of sentiment) is because it is the simplest approach of doing sentiment analysis and thus is most likely to be accurate; also it matches the valence of online rumour topic which concerns the consequence related to a rumour topic. SentiStrength is a tool that has been proved useful for analysing short and informal messages such as tweets (Stieglitz and Dang-Xuan 2013). Comparing to standard machine learning approach of analysing sentiment strength, algorithms used by SentiStrength shows to be more accurate. SentiStrength refers to a lexicon of emotional words and terms and has certain linguistic rules for issues including spelling corrections, negations (e.g., “not good”), booster words (e.g., “very sad”), amplifications (e.g., “haaaaaaaappy”), word weighting and emoticons (Stieglitz and Dang-Xuan 2013). SentiStrength not only evaluates sentiment polarity- emotional valence (i.e., positive, neutral, negative), but also analyses texts for different sentiment strength- valence strengths with 5

represents strongly positive, -5 represents strongly negative, and 1 and -1 represent neutral. The sentiment polarity is calculated as *positive sentiment score* + *negative sentiment score*, and the sentiment strength is calculated as  $(\text{positive sentiment score} - \text{negative sentiment score}) - 2$ . Note that the *negative sentiment score* is a negative number, and the -2 is used to convert the starting value of the sentiment strength as 0 to avoid confusion.

NLTK API was used to verify the accuracy of SentiStrength's results. As shown in Figure 14, for each object, we wrote a Python script to do HTTP POST to Natural Language Processing Application Programming Interface (API) and the returned response contained the probabilities of different sentiment classes and final labelling. Based on the probabilities, sentiment was labelled positive, negative or neutral using hierarchical classification, and was coded as 1, -1 and 0 correspondingly. In the classification process, the neutrality is identified first. The probability of neutrality is ranged from 0 to 1. If the probability of labelling as neutral is greater than 0.5, the object (i.e. text of reviews) will be labelled as 'neutral'. However, if the probability of neutrality is smaller than 0.5, the texts are classified as not neutral, and then we will determine which polarity the sentiment is. The sum of probabilities for positive label and negative is 1, and the one with higher probability will be used to label the review sentiment. NLTK trainer was used to train the classifiers, and the training data are from several data sets by Pang and Lee (2008). The returned results were cross-checked with the returned results of SentiStrength, and these results were over 91% consistent. For those who are not consistent, manual checking was conducted to ensure the polarity value and the subsequent congruence were labelled accurately.





**Figure 14 Sentiment Analysis Using NLTK API**

## 4.7 Study Three: Descriptive Analysis on Topic-Level Dissemination Patterns

### 4.8.1 Aim and Objectives of Study Three

While it is important to understand the rumour dissemination on the message level, apart from the variance in number of retweets among individual online rumour messages, the dissemination pattern of rumour messages may vary across rumour topics. Therefore, exploring the descriptive patterns of rumour dissemination on the topic level can provide additional interesting results and supplement the message-level research model. This study (study three) thus aims to describe the topic-level dissemination patterns of online rumour messages.

### 4.8.2 Research Data and Research Methods

In this study, several analyses were conducted to describe the online rumour message dissemination patterns on a topic level using previous online rumour

message dataset. First, all the tweets were aggregately analysed to seek general understandings of online rumour messages without distinguishing rumour topics. Subsequently, all the tweets related to one single topic were collectively grouped as a unit of analysis and were analysed correspondingly across rumour topics. Text mining and linguistic analytics provide possible solutions for getting meaningful information out of unstructured textual data (He et al. 2013). Two leading tools in textual data analysis and mining were used in this study, namely WordStat and Linguistic Inquiry and Word Count (LIWC).

#### **4.8 Research Ethics**

With ethical approval obtained prior to the conduction of research, this thesis follows the ethical principles that govern the research. The data extracted, stored and analysed in this thesis are public data that are available on the Internet with no access restriction.

This thesis does not involve people to actually participate in some activities or surveys; instead, it uses the content generated by people as key research data. If considering the online social media data as secondary data, there will be low risk of disclosing individual identities and the project will follow the requirements and policies of those social media, i.e. the data suppliers. This is because: 1) the data captured will be those that are already available to the public, and the authors of those content are aware of such openness; 2) the identities or other personal sensitive information will be anonymised in this project, and the information published in this thesis cannot be used to trace back to personal

identity or other sensitive information. Although no participant was actually approached and contacted in this thesis, data of this thesis rests on the Internet where people might be identified. Procedures were carefully designed to anonymise personal identity and protect personal information. Specifically, no account name of user, their e-mail addresses, or any other personal information will be revealed in this research. IDs of posts from search API will be stored separately with other information of the posts. Images and names contained in the posts will be checked for user's personal information, and all @name will be replaced with new unique code @1-@n, while the personal images will not be revealed in the thesis. All the data will be kept confidentially and anonymously and only the researchers of this research will have the access to the data. If any post is found deleted by the user who sent it, it will also be deleted from the research data and will not be used. The dataset collected will never be shared following those social media's terms and conditions and the Data Protection Act 1998.

Although the online data extraction is not intelligent enough to rule out people that are under 16 years old or that have other vulnerability, this thesis will simply use the data as an external 'observer' and will not have influence on anyone's well-beings.

## **CHAPTER 5 DATA ANALYSIS AND FINDINGS**

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This chapter presents the data analysis and the research findings of this thesis. The first section of this chapter presents the results of the exploratory qualitative study. It first identifies the themes emerged from online rumour topics and concludes a typology of rumour topics. Then, the themes and features of online rumour messages are reported. Following that, the themes and features of online rumour messages are compared across different types of rumour topics. The second section of this chapter reports the quantitative data analysis on the research model and the results on the rumour-message level. The third section of this chapter shows the analysis and the results on the rumour-topic level.

## **5.1 Findings of the Exploratory Study**

### **5.1.1 Thematic Results of Online Rumour Topics**

The results of the exploratory study show that, online rumour topics can be classified based on its referent and its focal point. Table 9 summarises the key themes of online rumour topics. Two major themes emerge from the content analysis, namely the object of rumour and the focal point of rumour. The results show that, all business-related online rumour topics have a key object, being it a brand, a service or product offering, a human being related to the business, or a business operation. The object of rumour describes which aspect of the business that the rumour focuses on. Regarding the object, rumour topics may concern different focal points. The focal point can be about describing the issues related to the object (i.e. what happened), explaining the reasons behind what happened (i.e. why) or discussing the impacts of what happened (i.e. with what impact). The key object and the focal point together form rumour topics.

Referring to the two major themes of online rumour topics, namely the object and the focal point of the rumour, online rumour topics were categorised into 12 types (as shown in Table 9 and Table 10). The details and examples of these rumour categories are presented below:

- 1) Brand-related description-oriented rumour. This type of rumour describes the issue or event related to the brand of the focal business, focusing on what happened. An example is:

*“Chef Boyardee was a fictional creation whose name was taken from the given names of the company’s three founders: Boyd, Art, and Dennis.”*

2) Brand-related cause-oriented rumour. This kind of rumour focuses on showing or explaining the reasons behind what happened with the brand of the focal business. An example is:

*“The Canada Dry beverage company got its name because it produced alcohol-free beverages during Prohibition.”*

3) Brand-related impact-oriented rumour. This kind of rumour focuses on presenting or arguing the impacts of what happened with the brand of the focal business. An example is:

*“The ‘Come Alive With Pepsi!’ slogan was interpreted in China as ‘Pepsi Brings Your Ancestors Back From the Dead.’”*

4) Offerings-related description-oriented rumour. This type of rumour describes the nature of the product offerings or service offerings of the focal business or describes certain issue or event related to the offerings, focusing on what are the offerings and what happened with those offerings. An example is:

*“McDonald’s McRib sandwiches made from inverted pork rectums.”*

5) Offerings-related cause-oriented rumour. This kind of rumour focuses on showing or explaining the reasons behind the issue or event with the product offerings or service offerings of the focal business. An example is:

*“People get “hooked” on Carmex Lip Balm because it contains ground glass (or other irritants), necessitating its continued use.”*

6) Offerings-related impact-oriented rumour. This kind of rumour focuses on presenting or arguing the impacts of the issue or event related to the offerings of the focal business. An example is:

*“A university student who drank too many Cokes died from carbon dioxide poisoning.”*

7) People-related description-oriented rumour. This type of rumour describes the issue or event related to any person that is connected to the focal business, focusing on what happened with those people and/or what they did. An example is:

*“Workers at the Corona brewery urinate into the beer.”*

8) People-related cause-oriented rumour. This kind of rumour focuses on showing or explaining the reasons behind the issue or event related to the people of the focal business and/or what the people did. An example is:

*“Continental Airlines CEO Gordon Bethune once personally threw a passenger off an airplane for being rude to a flight attendant.”*

9) People-related impact-oriented rumour. This kind of rumour focuses on presenting or arguing the impacts of what is done by the people connected to the focal business. An example is:

*“Richard Blum, the husband of Senator Dianne Feinstein, obtained a contract to broker sales of USPS facilities due to his wife's influence.”*

10) Operation-related description-oriented rumour. This type of rumour describes the issue or event related to the operation of the focal business, focusing on what happened. An example is:

*“Chick-fil-A restaurants have announced they will be reversing their longstanding policy and start opening on Sundays.”*

11) Operation-related cause-oriented rumour. This kind of rumour focuses on showing or explaining the reasons behind the operation or the operation decision of the focal business. An example is:

*“Chipotle has stopped offering pork at its restaurants in order to appease Muslim customers.”*

12) Operation-related impact-oriented rumour. This kind of rumour focuses on presenting or arguing the impacts of the operation of the focal business. An example is:

*“Amazon's use of USPS for package delivery services has resulted in severe financial loss to the Postal Service.”*



Table 9 Themes of Online Rumour Topics

| High level themes       | Primary themes                       | Definition                                                                                                                  | Typical Categories         | Examples                                                                                                      |
|-------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Object of rumour</b> | Brand-related issues<br>(27.12%)     | Various aspects of the focal business's brand, including but are not limited to issues related to brand name and ownership. | Brand name                 | "The Canada Dry beverage company got its name because it produced alcohol-free beverages during Prohibition." |
|                         |                                      |                                                                                                                             | Brand ownership            | "Costco warehouse club stores are owned by China."                                                            |
|                         |                                      |                                                                                                                             | Other brand-related issues | "The Red Lobster chain is about to close its doors and go out of business for good. "                         |
|                         | Offerings-related issues<br>(26.27%) | Various aspects of the focal business's product and service offerings and the delivery of the offerings.                    | Offerings                  | "A university student who drank too many Cokes died from carbon dioxide poisoning."                           |
|                         |                                      |                                                                                                                             | Delivery of offerings      | "Starbucks adds semen to its drinks."                                                                         |
|                         | People-related issues                | Issues related to the people who are somewhat relevant to the focal                                                         | Managers                   | "Continental Airlines CEO Gordon Bethune once personally threw a                                              |

|                                      |                                                                                                                                                                                                                                       |                                                  |                                                                                                                            |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| (12.71%)                             | business. People can be recognisable individuals or unspecified individuals, including but are not limited to the executive managers of the company, the employees of the company and the family or friend of managers and employees. |                                                  | passenger off an airplane for being rude to a flight attendant.”                                                           |
|                                      |                                                                                                                                                                                                                                       | Employees                                        | “Workers at the Corona brewery urinate into the beer.”                                                                     |
|                                      |                                                                                                                                                                                                                                       | Other people (e.g. family of managers/employees) | “The wife of the owner of a Thailand Starbucks told customers that the particularly trendy coffee spot is not for Asians.” |
| Operation-related issues<br>(33.90%) | The key object concerned by the rumour is about the internal operation of the focal business, including but are not limited to issues related to marketing, endorsement, and other management issues such as strategic decision.      | Marketing                                        | “Budweiser is ditching their famous Clydesdale ads in an effort to appeal to millennials.”                                 |
|                                      |                                                                                                                                                                                                                                       | Sales                                            | “In June 2012, 79% of General Motors’ sales were made to the U.S. federal government.”                                     |
|                                      |                                                                                                                                                                                                                                       | Human resource                                   | “Macy’s refused to hire an applicant because she was a veteran who had served in Afghanistan.”                             |
|                                      |                                                                                                                                                                                                                                       | Endorsement                                      | “Coors financially supports Nazi groups.”                                                                                  |

|                              |                            |                                                                                        |                          |                                                                                                                         |
|------------------------------|----------------------------|----------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
|                              |                            | Other management issues                                                                |                          | “Nestle is negotiating with the Brazilian government to privatize the Guarani Aquifer in South America.”                |
| <b>Focal point of rumour</b> | What happens (71.19%)      | Describing what is the issue, what happened, what is happening and/or what will happen | What is the issue        | “The U.S. government prohibits high mileage Volkswagen Passat automobiles from being purchased by U.S. customers.”      |
|                              |                            |                                                                                        | What is/has/will be done | “Chick-fil-A restaurants have announced they will be reversing their longstanding policy and start opening on Sundays.” |
|                              | Why (18.64%)               | Interpreting the reasons behind the issue and/or the purposes of what happens          | Reasons                  | “The Canada Dry beverage company got its name because it produced alcohol-free beverages during Prohibition.”           |
|                              |                            |                                                                                        | Purpose                  | “Chipotle has stopped offering pork at its restaurants in order to appease Muslim customers.”                           |
|                              | With what impacts (10.17%) | Explaining the impacts of the issue and/or of what happens                             | Impacts on the business  | “Richard Blum, the husband of Senator Dianne Feinstein, obtained a contract to                                          |

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broker sales of USPS facilities due to his wife's influence.”

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Impacts on other parties

“A university student who drank too many Cokes died from carbon dioxide poisoning.”

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**Table 10 Categorisation of Online Rumour Topics**

|                              |                          | Object of Rumour                                |                                                     |                                                  |                                                     |
|------------------------------|--------------------------|-------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
|                              |                          | Brand                                           | Offerings                                           | People                                           | Operation                                           |
| <b>Focal Point of Rumour</b> | <b>What happens</b>      | Brand-related<br>description-oriented<br>rumour | Offerings-related<br>description-oriented<br>rumour | People-related<br>description-oriented<br>rumour | Operation-related<br>description-oriented<br>rumour |
|                              | <b>Why</b>               | Brand-related<br>cause-oriented<br>rumour       | Offerings-related<br>cause-oriented<br>rumour       | People-related<br>cause-oriented<br>rumour       | Operation-related<br>cause-oriented<br>rumour       |
|                              | <b>With what impacts</b> | Brand-related<br>impact-oriented<br>rumour      | Offerings-related<br>impact-oriented<br>rumour      | People-related<br>impact-oriented<br>rumour      | Operation-related<br>impact-oriented<br>rumour      |

### **5.1.2 Thematic Results of Online Rumour Messages**

Performing qualitative content analysis on online rumour messages reveals two top-level themes: i) the features of online rumour messages, and ii) the motivations of posting online rumour messages. Online rumour messages are found to consist of three general features, namely content unambiguity, source unambiguity and affective appeal. Specifically, content unambiguity includes expression uncertainty and use of evidence, source unambiguity consists of source explicitness and source accessibility, and affective appeal includes the emotional words and the imagery words. These themes of online rumour message features were used to support the development of the conceptual framework of this thesis. Apart from the features of online rumour messages, the results of qualitative content analysis suggest that the online rumour messages are communicated for different purposes, including to seek the fact, to share information, to combat rumour and to express oneself. The detailed summaries of themes are shown in Table 11.

**Table 11 Themes of Rumour Communication Message on Twitter***Message Features*

| High level themes   | Primary themes         | Definition & categories                                                             | Example                                                                                                                                                |
|---------------------|------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Content unambiguity | Expression uncertainty | confidence or conviction expressed in a message (high uncertainty; low uncertainty) | “I hear Ford has announced that they are going to fund BLM.. <b>So they support hate groups now?</b> Never buy a Ford again!”                          |
|                     | Use of evidence        | inclusion of evidence (statistical evidence; testimonial evidence; no evidence)     | “George Soros and Ford Foundation have put <b>\$500M</b> into BLM. (Ford as in Ford cars/trucks:they supplied Nazi vehicles during WW2)Rothchilds-Gas” |

|                   |                 |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source            | Explicitness    | clear and explicit rumour source is mentioned in the rumour message (explicit; non-explicit)                                |  <p><b>Rumor:</b> LeBron told Tristan Thompson that Khloe Kardashian's not allowed to come around the team:</p> <p><b>LeBron tells Tristan Thompson no Kardashians h...</b><br/>Whether it's true or not.... Khloe Kardashian's relationship with Tristan Thompson has triggered quite the feud with his Cavaliers teammate LeBron James...<br/>legionhoops.com</p> |
|                   | Accessibility   | URL or other sorts of link that can directly lead to the rumour source (direct path; no direct path)                        | “What Ford Is Doing For BLM Will Make You Never Want To Buy Their Cars Again: <a href="http://...">http://...</a> ”                                                                                                                                                                                                                                                                                                                                    |
| Affective appeals | Emotional words | Words that communicate emotions such as anger, fear, disgust, hate, sadness, disappointment, wish, happiness, relief        | “@Ford Will never buy another FORD again IF you endorse BLM .. <b>Disgusting.</b> ”                                                                                                                                                                                                                                                                                                                                                                    |
|                   | Imagery words   | High imagery words such as house, train, hat elicit vivid images, whereas low imagery words include idea, skill, wish, etc. | “Those green lines on <b>monster drink</b> says 666 in Hebrew also you have the cross on the letter o next to the letter m .”                                                                                                                                                                                                                                                                                                                          |



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***Communication Motivation***

| Themes              | Definition & categories                                                      | Example                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fact seeking        | Messages are posted mainly to seek the fact and truth                        | “Is it true that Adidas stands for all day I dream about sex ?”                                                                                        |
| Information sharing | Messages are posted mainly to spread and share the rumour information        | “Check this out: Coca-Cola Recalls Dasani Water! Clear Parasite Worm Was Found In Bottles... <a href="http://goo.gl/89FOtF">http://goo.gl/89FOtF</a> ” |
| Rumour combating    | Messages are posted mainly to stop or combat the rumour information          | “it's not real. RT @Ti*y: O.o RT @Sa*h And in other news, Abercrombie & Fitch now sell n-word pants. -_-...”                                           |
| Self-expression     | Messages are posted mainly to express opinions or share personal experiences | “#Trump is an idiot. If the #USPS loses the #Amazon account they will be like his companies. #Bankrupt ...”                                            |

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## **5.2 Quantitative Analysis of Research Model**

Since the number of retransmissions (i.e. the number of retweets) is discrete count and most of the counts are 0 or 1, there is an overdispersion of the dependent variable. Overdispersion in the data means the variance of dependent variable exceeds the mean. One possible solution to the overdispersion issue is to refer to a distribution that permits less restricted variance function. Therefore, negative binomial regression was applied to analyse the data using Stata (Chua et al. 2017; Lee et al. 2015; Sutton et al. 2015). The communicator's social influence (i.e. number of followers of the communicator) and communicator's likeability (i.e. number of likes received by the communicators) were log-transformed in order to reduce the violations of the assumptions of normality. The results of negative binomial regression are shown in Table 12.

Results show that expression certainty and emotional activation are positively associated with the number of retweets (coef.= 0.330,  $p<0.001$ ; coef.= 1.263,  $p<0.001$ ), supporting H1a and H3b. The source explicitness, source accessibility, emotional pleasantness and emotional imagery are also found to have significant effects on the number of retweets (coef.= -0.216,  $p<0.001$ ; coef.= -0.292,  $p<0.001$ ; coef.= -0.465,  $p<0.01$ ; coef.= -0.694,  $p<0.001$ ). However, different from the hypothesised positive effects, these variables are found to have negative relationships with the number of retweets. Although the effects are different from the hypotheses, the results are not beyond expectation.

**Table 12 Results of Hypotheses Testing**

|                               | Coef.  | Std.Err | P>z   | Sig. |
|-------------------------------|--------|---------|-------|------|
| Expression Certainty          | 0.330  | 0.042   | 0.000 | ***  |
| Use of Evidence               | 0.112  | 0.089   | 0.204 | n.s. |
| Source Explicitness           | -0.216 | 0.066   | 0.001 | ***  |
| Source Accessibility          | -0.292 | 0.036   | 0.000 | ***  |
| Emotional Pleasantness        | -0.465 | 0.154   | 0.003 | **   |
| Emotional Activation          | 1.263  | 0.176   | 0.000 | ***  |
| Emotional Imagery             | -0.694 | 0.131   | 0.000 | ***  |
| Direct Communication          | -0.171 | 0.031   | 0.000 | ***  |
| Community Communication       | 0.464  | 0.031   | 0.000 | ***  |
| Communicator Trustworthiness  | 0.600  | 0.076   | 0.000 | ***  |
| Communicator Social Influence | 0.684  | 0.007   | 0.000 | ***  |
| Communicator Likeability      | 0.041  | 0.005   | 0.000 | ***  |
| Topic-congruence (TC)         | -1.585 | 0.275   | 0.000 | ***  |
| Message-congruence (MC)       | -0.424 | 0.155   | 0.006 | **   |
| TC × Expression Certainty     | 0.323  | 0.177   | 0.069 | n.s. |
| TC × Use of Evidence          | -0.068 | 0.250   | 0.784 | n.s. |
| TC × Source Explicitness      | 0.019  | 0.228   | 0.932 | n.s. |
| TC × Source Accessibility     | 0.271  | 0.117   | 0.021 | *    |
| TC × Pleasantness             | 2.221  | 0.714   | 0.002 | **   |
| TC × Activation               | 2.427  | 0.907   | 0.007 | **   |
| TC × Imagery                  | -3.865 | 0.441   | 0.000 | ***  |
| MC × Expression Certainty     | -0.050 | 0.087   | 0.568 | n.s. |
| MC × Use of Evidence          | 0.427  | 0.199   | 0.032 | *    |
| MC × Source Explicitness      | -0.039 | 0.137   | 0.777 | n.s. |
| MC × Source Accessibility     | 0.266  | 0.078   | 0.001 | ***  |
| MC × Emotional Pleasantness   | -1.051 | 0.401   | 0.009 | **   |
| MC × Emotional Activation     | 0.413  | 0.482   | 0.391 | n.s. |
| MC × Emotional Imagery        | 0.961  | 0.289   | 0.001 | ***  |
| R <sup>2</sup> =15.23%        |        |         |       |      |

Note: \*\*\* p<0.001, \*\* p<0.01, \* p<0.05

Given the nature of rumour, the ambiguous source may drive people to share online rumour messages for fact checking or to combat the rumour. The online rumour messages with less pleasant emotions may be shared more because of effects of negative bias. When emotions communicated in a message is negative, the receivers may perceive the effects to be more serious and become more nervous and more motivated to share the message. The actual surprising result is, the use of evidence is found to have no significant influence on the number of retweets. This may be because that the effectiveness of use of evidence differs across different types of rumour, or that the effects of use of evidence is moderated by congruence. Both possibilities are supported in the follow-up analyses.

In general, the results indicate that most cognitive appeals and affective appeals are significantly associated with the dissemination of online rumour messages. Expression certainty is positively associated with the dissemination of online rumour messages, supporting the persuasion literature which asserts that in general a communication is more likely to be accepted if the argument is clear and strong (Corner and Hahn 2009). Although use of evidence does not directly relate to online rumour dissemination, its interaction effects with message-congruence is significant. Source unambiguity (i.e. source explicitness and source accessibility) is negatively associated with the dissemination of online rumour messages and this may be because that rumour in nature are unverified information and the uncertainty plays an important role in stimulating the rumouring behaviour (Rosnow et al. 1988). In the context of online rumour

dissemination, the importance of uncertainty is found to rest more on the source of message instead of the communication content itself.

The results also reveal that, the communicator features (i.e., communicator trustworthiness, communicator social influence and communicator likeability) are all positively associated with the number of retweets of online rumour messages (coef.= 0.600,  $p<0.001$ ; coef.= 0.684,  $p<0.001$ ; coef.= 0.041,  $p<0.001$ ), supporting H5, H6 and H7. Regarding the effects of communication channel, community communication is positively associated with the number of retweets, supporting H8b; however, the direct communication is found to be negatively associated with the number of retweets. One possible reason for the negative relationship between direct communication and number of retweets is that, unlike in the offline context where direct communication generates strong impact, in social media direct communication may limit the scope of receivers and lose attention from those who are not directly involved in the communication. As a result, comparing to the online rumour messages that are not communicated directly to someone, there may be fewer retransmissions for an online rumour message directly communicated between the communicator and receiver.

In terms of the congruence effects, both topic-congruence (i.e. the congruence between target reputation and rumour topic valence) and message-congruence (i.e. the congruence between target reputation and rumour message valence) are negatively associated with the number of retweets (coef.= -1.585,

$p < 0.001$ ;  $\text{coef.} = -0.424$ ,  $p < 0.01$ ), supporting H9a and H10a. Apart from the direct effects, both types of congruence moderate the effects of message features on the number of retweets. Specifically, topic-congruence significantly moderates the effects of affective appeals on number of retweets. It also significantly moderates the effect of source accessibility on number of retweets. However, it is not a significant moderator for the effects of other cognitive appeals on number of retweets. In terms of message-congruence, it significantly moderates the effects of use of evidence, source accessibility, emotional pleasantness and emotional imagery on number of retweets. However, message-congruence does not moderate the effects of expression certainty, source explicitness and emotional activation on number of retweets. The results of hypotheses testing are summarised in Table 13.

**Table 13 Summary of Hypotheses Testing**

| <b>Hypothesis</b> |                                                                                                                            | <b>Remark</b>                                     |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>H1a</b>        | Rumour message with high expression certainty is disseminated more than rumour message with low expression certainty       | <b>Supported</b>                                  |
| <b>H1b</b>        | Rumour message with evidence is disseminated more than rumour message without evidence                                     | <b>Not Supported</b><br>(Not significant)         |
| <b>H2a</b>        | Rumour message with explicit rumour source is disseminated more than rumour message with no explicit source                | <b>Not Supported</b><br>(Significantly less than) |
| <b>H2b</b>        | Rumour message with direct link to rumour source is disseminated more than rumour message with no direct link              | <b>Not Supported</b><br>(Significantly less than) |
| <b>H3a</b>        | Rumour message with high level of emotional pleasantness is disseminated more than rumour message with unpleasant emotions | <b>Not Supported</b><br>(Significantly less than) |
| <b>H3b</b>        | Rumour message with high level of emotional activation is disseminated more than rumour message with low level of          | <b>Supported</b>                                  |

|            |                                                                                                                                                                          |                                                   |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|            | sentiment arousal                                                                                                                                                        |                                                   |
| <b>H4</b>  | Rumour message with high level of emotional imagery is disseminated more than rumour message presented pallidly                                                          | <b>Not Supported</b><br>(Significantly less than) |
| <b>H5</b>  | Rumour message communicated by user with high level of trustworthiness is disseminated more than rumour message communicated by user with low level of trustworthiness   | <b>Supported</b>                                  |
| <b>H6</b>  | Rumour message communicated by user with high level of social influence is disseminated more than rumour message communicated by user with low level of social influence | <b>Supported</b>                                  |
| <b>H7</b>  | Rumour message communicated by user with high likeability is disseminated more than rumour message communicated by user with low likeability                             | <b>Supported</b>                                  |
| <b>H8a</b> | Rumour message communicated using direct communication channel is disseminated more than rumour message                                                                  | <b>Not Supported</b><br>(Significantly less than) |



|            |                                                                                                                                                                                              |                                           |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
|            | communicated without direct communication channel                                                                                                                                            |                                           |
| <b>H8b</b> | Rumour message communicated using community communication channel is disseminated more than rumour message communicated without community communication channel                              | <b>Supported</b>                          |
| <b>H9a</b> | Rumour message whose rumour target reputation is incongruent with rumour topic valence is disseminated more than those whose rumour target reputation is congruent with rumour topic valence | <b>Supported</b>                          |
| <b>H9b</b> | Rumour topic-congruence moderates the effect of expression certainty on rumour dissemination                                                                                                 | <b>Not Supported</b><br>(Not significant) |
| <b>H9c</b> | Rumour topic-congruence moderates the effect of use of evidence on rumour dissemination                                                                                                      | <b>Not Supported</b><br>(Not significant) |
| <b>H9d</b> | Rumour topic-congruence moderates the effects of source explicitness on rumour dissemination                                                                                                 | <b>Not Supported</b><br>(Not significant) |

|             |                                                                                                                                                                                                                    |                                           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>H9e</b>  | Rumour topic-congruence moderates the effects of source accessibility on rumour dissemination                                                                                                                      | <b>Supported</b>                          |
| <b>H9f</b>  | Rumour topic-congruence moderates the effects of emotional pleasantness on rumour dissemination                                                                                                                    | <b>Supported</b>                          |
| <b>H9g</b>  | Rumour topic-congruence moderates the effects of emotional activation on rumour dissemination                                                                                                                      | <b>Supported</b>                          |
| <b>H9h</b>  | Rumour topic-congruence moderates the effects of emotional imagery on rumour dissemination                                                                                                                         | <b>Supported</b>                          |
| <b>H10a</b> | Rumour message whose rumour target reputation is incongruent with rumour message emotional appeal is disseminated more than those whose rumour target reputation is congruent with rumour message emotional appeal | <b>Supported</b>                          |
| <b>H10b</b> | Rumour message-congruence moderates the effect of expression certainty on rumour dissemination                                                                                                                     | <b>Not Supported</b><br>(Not significant) |

|             |                                                                                                   |                                           |
|-------------|---------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>H10c</b> | Rumour message-congruence moderates the effect of use of evidence on rumour dissemination         | <b>Supported</b>                          |
| <b>H10d</b> | Rumour message-congruence moderates the effects of source explicitness on rumour dissemination    | <b>Not Supported</b><br>(Not significant) |
| <b>H19e</b> | Rumour message-congruence moderates the effects of source accessibility on rumour dissemination   | <b>Supported</b>                          |
| <b>H10f</b> | Rumour message-congruence moderates the effects of emotional pleasantness on rumour dissemination | <b>Supported</b>                          |
| <b>H10g</b> | Rumour message-congruence moderates the effects of emotional activation on rumour dissemination   | <b>Not Supported</b><br>(Not significant) |
| <b>H10h</b> | Rumour message-congruence moderates the effects of emotional imagery on rumour dissemination      | <b>Supported</b>                          |

Apart from testing online rumour dissemination using number of retweets, to provide a comprehensive understanding on the online rumour dissemination, this thesis also measured online rumour dissemination as a binary outcome: either has been retweeted or has not been retweeted. The dependent variable in this sense is a dichotomous variable with two mutually exclusive and also exhaustive categories. The independent variables are still measured in the same way as before. Logistic regression is used because it is relatively flexible with no assumption about the normal distribution of variables. Independent variables in logistic regression do not have to be discrete variables. Logistic regression also does not assume for a linear relationship. It can work well for non-normal and non-linear data, and it also allows independent variables to be continuous variables, discrete variables or dichotomous variables or a combination of them. The results are shown in Table 14.

**Table 14 Results of Testing with Binary Measurement of Dissemination**

|                               | Coef.  | Std.Err | P>z   | Sig. |
|-------------------------------|--------|---------|-------|------|
| Expression Certainty          | 0.212  | 0.036   | 0.000 | ***  |
| Use of Evidence               | -0.001 | 0.072   | 0.993 | n.s. |
| Source Explicitness           | -0.096 | 0.057   | 0.090 | n.s. |
| Source Accessibility          | -0.093 | 0.032   | 0.003 | **   |
| Emotional Pleasantness        | -0.455 | 0.127   | 0.000 | ***  |
| Emotional Activation          | 0.703  | 0.156   | 0.000 | ***  |
| Emotional Imagery             | -0.281 | 0.103   | 0.007 | **   |
| Direct Communication          | -0.231 | 0.026   | 0.000 | ***  |
| Community Communication       | 0.454  | 0.024   | 0.000 | ***  |
| Communicator Trustworthiness  | 0.177  | 0.060   | 0.003 | **   |
| Communicator Social Influence | 0.416  | 0.006   | 0.000 | ***  |
| Communicator Likeability      | 0.047  | 0.004   | 0.000 | ***  |
| Topic-congruence (TC)         | -0.742 | 0.234   | 0.002 | **   |
| Message-congruence (MC)       | -0.064 | 0.130   | 0.624 | n.s. |
| TC × Expression Certainty     | -0.078 | 0.135   | 0.566 | n.s. |
| TC × Use of Evidence          | 0.302  | 0.201   | 0.133 | n.s. |
| TC × Source Explicitness      | 0.443  | 0.185   | 0.016 | *    |
| TC × Source Accessibility     | -0.559 | 0.113   | 0.000 | ***  |
| TC × Pleasantness             | 2.877  | 0.582   | 0.000 | ***  |
| TC × Activation               | -2.537 | 0.766   | 0.001 | ***  |
| TC × Imagery                  | 0.223  | 0.382   | 0.561 | n.s. |
| MC × Expression Certainty     | -0.087 | 0.072   | 0.230 | n.s. |
| MC × Use of Evidence          | 0.515  | 0.147   | 0.000 | ***  |
| MC × Source Explicitness      | 0.184  | 0.111   | 0.099 | n.s. |
| MC × Source Accessibility     | -0.088 | 0.068   | 0.195 | n.s. |
| MC × Emotional Pleasantness   | -1.051 | 0.311   | 0.105 | n.s. |
| MC × Emotional Activation     | 0.413  | 0.381   | 0.866 | n.s. |
| MC × Emotional Imagery        | 0.961  | 0.226   | 0.007 | **   |
| R <sup>2</sup> =15.23%        |        |         |       |      |

Note: \*\*\* p<0.001, \*\* p<0.01, \* p<0.05

The results are mostly in line with the results when testing dissemination using number of retweets. Expression certainty and emotional activation also have positive effects on the dissemination of online rumour messages. Source accessibility, emotional pleasantness and emotional imager also have negative effects on the dissemination. Directed communication also is found to have negative influence on dissemination whereas community communication is found to have positive influence. All three communicator features are again found to have positive effects on dissemination, and the topic-congruence is found to have negative effect. However, source explicitness, use of statistical evidence and message-congruence are not significantly associated with the dissemination. In terms of the interaction effects, the interactions between topic-congruence and both types of source unambiguities (i.e. source explicitness and source accessibility) are significant. Topic-congruence also significantly interacts with emotional pleasantness and emotional activation. For message-congruence, its interaction effects with use of evidence and with emotional imagery are significant.

### **5.3 Patterns Across Different Types of Rumour Topics**

When the focal point of communication is different, the motivations of communication and the elaborations are likely to vary. To understand the online rumour message dissemination on rumour topic level, the conceptual framework is compared across different types of rumour topics. The rumour

topics are first classified based on the focal point of the rumour topic, that is, whether the rumour topic focuses on describing what happens, explaining the reasons or describing the impacts. Table 15 shows the results of the comparisons across rumour topics classified based on focal point. Then the topics are also classified based on the object, that is, whether the rumour topic focuses on the rumour target's brand, offerings, people or operations. However, when classifying rumour topics based on the object, in the category of "*people-focused rumour topics*" too few brands are involved. This potentially affects the reliability, validity and accuracy of topic-congruence. Therefore, topic-congruence and related interaction effects are removed from the framework in this comparison. The results of comparisons across rumour topics classified based on rumour object are shown in Table 15.

**Table 15 Comparing Research Framework across Different Types of Rumour Topics (Based on Focal Point of Rumour)**

|                        | Description-oriented |      | Reason-oriented |      | Impact-oriented |      |
|------------------------|----------------------|------|-----------------|------|-----------------|------|
|                        | Coef.                | Sig. | Coef.           | Sig. | Coef.           | Sig. |
| Expression Certainty   | 0.170                | ***  | 0.709           | ***  | -0.008          | n.s. |
| Use of Evidence        | 0.203                | **   | -1.852          | *    | 0.020           | n.s. |
| Source Explicitness    | 0.017                | n.s. | -0.050          | n.s. | -0.442          | *    |
| Source Accessibility   | -0.474               | ***  | -0.140          | n.s. | -0.606          | ***  |
| Emotional Pleasantness | 0.650                | ***  | -5.641          | ***  | 0.873           | n.s. |
| Emotional Activation   | -0.612               | **   | 7.402           | ***  | 0.251           | n.s. |
| Emotional Imagery      | -0.480               | ***  | -0.753          | n.s. | -0.770          | *    |
| Direct Communication   | -0.403               | ***  | -0.197          | *    | 0.024           | n.s. |

|                               |                             |      |                        |      |                        |      |
|-------------------------------|-----------------------------|------|------------------------|------|------------------------|------|
| Community Communication       | 0.410                       | ***  | 0.810                  | ***  | 0.568                  | ***  |
| Communicator Trustworthiness  | 0.160                       | n.s. | 2.129                  | ***  | 0.519                  | **   |
|                               | <b>Description-oriented</b> |      | <b>Reason-oriented</b> |      | <b>Impact-oriented</b> |      |
|                               | Coef.                       | Sig. | Coef.                  | Sig. | Coef.                  | Sig. |
| Communicator Social Influence | 0.695                       | ***  | 0.710                  | ***  | 0.554                  | ***  |
| Communicator Likeability      | 0.011                       | *    | 0.103                  | ***  | 0.019                  | n.s. |
| Topic-congruence (TC)         | -1.445                      | ***  | -2.239                 | n.s. | -1.691                 | ***  |
| Message-congruence (MC)       | -0.845                      | ***  | -0.561                 | n.s. | -0.687                 | *    |
| TC × Expression Certainty     | 0.104                       | n.s. | -0.100                 | n.s. | 0.560                  | n.s. |
| TC × Use of Evidence          | -0.298                      | n.s. | -13.956                | n.s. | 0.181                  | n.s. |
| TC × Source Explicitness      | 0.501                       | n.s. | -2.095                 | n.s. | -0.180                 | n.s. |
| TC × Source Accessibility     | -0.340                      | n.s. | -0.139                 | n.s. | 1.807                  | ***  |
| TC × Pleasantness             | 3.140                       | **   | 1.885                  | n.s. | -4.538                 | **   |
| TC × Activation               | -1.094                      | n.s. | -1.839                 | n.s. | 8.804                  | ***  |
| TC × Imagery                  | -0.849                      | n.s. | -0.118                 | n.s. | -3.635                 | **   |
| MC × Expression Certainty     | 0.043                       | n.s. | 0.047                  | n.s. | 0.562                  | **   |
| MC × Use of Evidence          | 0.620                       | **   | 1.289                  | n.s. | -0.291                 | n.s. |
| MC × Source Explicitness      | -0.143                      | n.s. | 0.108                  | n.s. | 0.143                  | n.s. |
| MC × Source Accessibility     | 0.363                       | ***  | 1.023                  | **   | 0.626                  | ***  |
| MC × Emotional Pleasantness   | -1.466                      | ***  | 3.396                  | n.s. | -1.531                 | n.s. |
| MC × Emotional Activation     | 1.475                       | **   | -6.327                 | **   | 1.566                  | n.s. |
| MC × Emotional Imagery        | 0.682                       | *    | 3.369                  | **   | -0.082                 | n.s. |

Note: \*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , \*  $p < 0.05$



The results suggest that, for description-oriented rumour topics, apart from use of evidence and communicator trustworthiness, all features have significant influences on number of retweets. For most of the cognitive and affective appeals, topic-congruence does not moderate their effects on number of retweets, whereas the message-congruence tends to moderate many of their effects. For reason-oriented rumour topics, content unambiguity is positively and significantly associated with number of retweets, whereas source unambiguity shows no significant influence. Both emotional pleasantness and activation have significant influences on number of retweets, though emotional imagery does not. Communication channels and communicator attributes are all found to have significant effects on number of retweets. Topic-congruence and message-congruence are not significantly associated with number of retweets. Topic-congruence does not moderate any effects of cognitive and affective appeals. Message-congruence significantly moderates the effects of source accessibility, emotional activation and emotional imagery on number of retweets. Rumour message dissemination in the context of impact-oriented rumour topics is very different from the dissemination in the context of description-oriented rumour topics. For impact-oriented rumour topics, the number of sharing is significantly associated with source unambiguity, emotional imagery, communication channels and the trustworthiness and social influence of the communicator, and the congruences. Also, topic-congruence shows significant moderating effect on the relationship between all the affective appeals and the number of retweets. Message-congruence significantly moderates the effect of expression certainty

and source accessibility on number of retweets. Considering the effects of variables across different types of rumour topics, community communication and communicator social influence are the two factors that works consistently.

As can be seen from Table 16, for both brand-focused rumour topics and operations-focused rumour topics, all cognitive appeals and most affective appeals have significant influences on number of retweets. Communication channels and most communicator attributes are also influential. However, the moderating effect of message-congruence is very different between the two types of rumour topics. In the context of brand-focused rumour topics, message-congruence significantly moderates the effects of content unambiguity and emotional words on number of retweets; however, it does not moderate the effects of source unambiguity or imagery words. In the context of operations-focused rumour topics, message congruence significantly source unambiguity's effects on number of retweets. It also significantly moderates the effects of use of evidence, emotional pleasantness and emotional imagery, yet it does not moderate expression certainty and emotional activation. For offering-focused rumour topics, source unambiguity is found to be important, but content unambiguity is not. Although emotional pleasantness does not significantly influence number of retweets, emotional activation and emotional imagery both are influential. Community communication again is associated with the number of sharing, but direct communication is not. Message-congruence play significant moderating roles in the effects of expression certainty, source accessibility, emotional activation and emotional imagery on number of sharing.

In terms of people-focused rumour topics, only source accessibility and communicators show significant influences on number of retweets. It is found that source accessibility and communicator social influence are the two factors that works consistently across different types of rumour topics.

**Table 16 Comparing Research Framework across Different Types of Rumour Topics (Based on Object of Rumour)**

|                               | Brand-focused |      | Offerings-focused |      | People-focused |      | Operations-focused |      |
|-------------------------------|---------------|------|-------------------|------|----------------|------|--------------------|------|
|                               | Coef.         | Sig. | Coef.             | Sig. | Coef.          | Sig. | Coef.              | Sig. |
| Expression Certainty          | 0.397         | ***  | 0.052             | n.s. | 0.197          | n.s. | 0.570              | ***  |
| Use of Evidence               | -0.656        | ***  | -0.238            | n.s. | -0.284         | n.s. | 0.544              | ***  |
| Source Explicitness           | -0.314        | ***  | -0.301            | *    | -1.030         | n.s. | 0.412              | **   |
| Source Accessibility          | -0.186        | ***  | -0.216            | **   | -0.540         | *    | -0.672             | ***  |
| Emotional Pleasantness        | -1.011        | ***  | -0.151            | n.s. | -0.071         | n.s. | 3.308              | ***  |
| Emotional Activation          | 1.288         | ***  | 1.997             | ***  | -1.122         | n.s. | -1.659             | **   |
| Emotional Imagery             | -0.061        | n.s. | -1.771            | ***  | 1.278          | n.s. | -1.631             | ***  |
| Direct Communication          | -0.223        | ***  | -0.054            | n.s. | 0.066          | n.s. | -0.260             | **   |
| Community Communication       | 0.345         | ***  | 0.537             | ***  | -0.080         | n.s. | 0.988              | ***  |
| Communicator Trustworthiness  | 0.035         | n.s. | 1.667             | ***  | 0.888          | **   | 0.072              | n.s. |
| Communicator Social Influence | 0.665         | ***  | 0.638             | ***  | 0.401          | ***  | 0.843              | ***  |
| Communicator Likeability      | 0.062         | ***  | 0.010             | n.s. | 0.181          | ***  | -0.053             | ***  |
| MC × Expression Certainty     | -0.259        | *    | 0.539             | ***  | 0.152          | n.s. | -0.346             | n.s. |
| MC × Use of Evidence          | 0.817         | *    | 0.142             | n.s. | -1.244         | n.s. | 0.683              | *    |
| MC × Source Explicitness      | 0.237         | n.s. | -0.063            | n.s. | -0.859         | n.s. | -0.745             | *    |
| MC × Source Accessibility     | 0.084         | n.s. | 0.364             | **   | 1.154          | n.s. | 0.461              | *    |
| MC × Emotional Pleasantness   | -1.573        | **   | 0.175             | n.s. | -4.417         | n.s. | -2.467             | *    |
| MC × Emotional Activation     | 2.049         | ***  | -2.232            | *    | 6.292          | n.s. | -0.396             | n.s. |
| MC × Emotional Imagery        | 0.267         | n.s. | 1.921             | ***  | -4.028         | n.s. | 3.491              | ***  |

Note: \*\*\* p&lt;0.001, \*\* p&lt;0.01, \* p&lt;0.05

In addition to the comparison of research models between different types of rumour topics, the linguistic characteristics of different types of rumour topics are also summarised. Linguistic Inquiry and Word Count (LIWC) was used to analyse the text for linguistic differences between types of rumour topics. All the online rumour messages that belong to the same type of rumour topic are grouped together for analysis. The results of the LIWC analyses are shown in Table 17.

**Table 17 Linguistic Analysis Results of Rumour Topics**

|                          | <b>i</b> | <b>we</b> | <b>you</b> | <b>She/he</b> | <b>they</b> | <b>anx</b> | <b>anger</b> | <b>sad</b> | <b>social</b> | <b>family</b> | <b>friend</b> |
|--------------------------|----------|-----------|------------|---------------|-------------|------------|--------------|------------|---------------|---------------|---------------|
| Brand-focused rumour     | 1.87     | 0.22      | 0.92       | 0.26          | 0.45        | 0.08       | 0.41         | 0.13       | 4.5           | 0.14          | 0.13          |
| Offerings-focused rumour | 3.26     | 0.25      | 1.24       | 0.32          | 0.39        | 0.1        | 0.56         | 0.15       | 5.86          | 0.18          | 0.18          |
| People-focused rumour    | 0.76     | 0.43      | 0.5        | 0.66          | 0.27        | 0.12       | 0.44         | 0.22       | 6.59          | 0.12          | 0.29          |
| Operation-focused rumour | 2.99     | 0.17      | 2.52       | 0.23          | 2.16        | 0.05       | 0.42         | 1.37       | 12.33         | 0.57          | 0.07          |

Brand-focused rumour messages are relatively less personal, less social and communicates less anxiety, anger and sadnesses comparing to rumour messages of other types of topics. The offerings-focused rumour messages tend to communicate with high level of anxiety and anger. They often refer to the communicator himself or herself and also the other party of the conversation. In

general, people-focused rumour messages tend to be less personal but more concerned about the community. Those rumour messages tend to be highly social and friend-related. Rumour messages about operations-focused topics tend to express much higher level of sadness and have uses words that are more social-and family-oriented than rumour messages about other types of topics do.

Apart from the linguistic features of online rumour messages, the perceptions of the rumour targeted may also be captured using the brand personality (see Table 18). Those personalities' influences on the brand reputation and the dissemination of online rumour messages were analysed. Interestingly, the results do not show significant relationships between rumour target's brand personality and the dissemination of online rumour messages, in general and for each type of rumour topics. One possible reason is that the brand personality does not have direct influence on dissemination but may serve certain moderating roles. The dynamics between rumour target's brand personality and dissemination of online rumour messages needs to be further investigated in the future.

**Table 18 Measurement of Brand Personality**

| <b>Brand reputation</b> |       | <b>Measurements</b>                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target's Sincerity      | Brand | <p>Measured with linguistic analysis by counting the aggregate number of keywords mentioned in <i>all messages related to the referent (i.e., the rumour target)</i> within a certain period of time (i.e., the most recent three-months).</p> <p>Typical keywords used include: down-to-earth, family oriented, small-town, honest, sincere, wholesome, cheerful, sentimental, friendly.</p>     |
| Target's Excitement     | Brand | <p>Measured with linguistic analysis by counting the aggregate number of keywords mentioned in <i>all messages related to the referent (i.e., the rumour target)</i> within a certain period of time (i.e., the most recent three-months).</p> <p>Typical keywords used include: daring, trendy, exciting, spirited, cool, young, imaginative, unique, up-to-date, independent, contemporary.</p> |
| Target's Competence     | Brand | <p>Measured with linguistic analysis by counting the aggregate number of keywords mentioned in <i>all messages related to the referent (i.e., the rumour target)</i> within a certain period of time (i.e., the most recent three-months).</p>                                                                                                                                                    |

|                               |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Typical keywords used include: reliable, hard-working, secure, intelligent, technical, successful, leader, confident.                                                                                                                                                                                                                                               |
| Target's Brand Sophistication | <p>Measured with linguistic analysis by counting the aggregate number of keywords mentioned in <i>all messages related to the referent (i.e., the rumour target)</i> within a certain period of time (i.e., the most recent three-months).</p> <p>Typical keywords used include: upper class, glamorous or glamorous, good-looking, charming, feminine, smooth.</p> |
| Target's Brand Ruggedness     | <p>Measured with linguistic analysis by counting the aggregate number of keywords mentioned in <i>all messages related to the referent (i.e., the rumour target)</i> within a certain period of time (i.e., the most recent three-months).</p> <p>Typical keywords used include: outdoorsy, masculine, tough, rugged.</p>                                           |

In addition, the number of online rumour messages of each rumour topic was investigated over time. It is found that the number of rumour messages and duration of survival vary across rumour topics. Furthermore, the overall



dissemination pattern of rumour messages show that the messages accelerated dramatically and then faded quickly. Most rumour topics have only one such tipping point, though a few rumour topics have several rounds of bursts. Further investigation was conducted to understand the information contained in online rumour messages of each rumour topic over time. The results suggest that, the content of online rumour message rarely distorts or exaggerated. For most rumour topics, the majority of the online rumour messages contain only a small amount of unique information and the content of information rarely changes along the dissemination. This finding is contradicted with traditional rumour theory which suggest that rumour content changes and exaggerated as it flows through the community (Bordia et al. 1999). This is probably related to the features of the social media and computer-mediated communication. The commonly used functions such as copying and pasting, sharing from website directly to social media and forwarding buttons within the social media tend to enable messages to be disseminated as it originally was (De Fleub 1962).

## **CHAPTER 6    CONCLUSION**

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This chapter summarises the findings of this thesis and shows how the findings realise the research objectives. The theoretical contribution as well as practical implications are also discussed. Finally, some potential limitations of this thesis are presented, followed with suggestions for future research.

## **6.1 Research Objective Realisation and Research Findings**

Social media stimulate the dissemination of online rumour messages and may have adverse impacts on individuals, companies and the society. Therefore, this thesis seeks to what drives the dissemination of online rumour messages. Taking the perspectives of rumour theories, communication theories, emotion theories and congruence theories, this thesis develops and tested a message-level conceptual framework of online rumour message dissemination. Specifically, message features, communicator features, communication channel features and rumour features are integrated to provide a comprehensive understanding on the phenomenon of rumour dissemination in social media.

To address research question 1, content analysis and linguistic analysis have been done to explore the typology of online rumour topics and the communication features of online rumour messages. This thesis finds that online rumour topics can be mainly categorised into brand-related rumour, offerings-related rumour, people-related rumour and operations-related rumour based on the object of rumour, and into description-oriented rumour, reason-oriented rumour and impact-oriented rumour based on the focal point of rumour. Online rumour messages are found to consist of three general themes, namely content unambiguity, source unambiguity and affective appeal. Specifically, content unambiguity includes expression uncertainty and use of evidence, source unambiguity consists of source explicitness and source accessibility, and affective appeal comprises emotional words and imagery words. Based on

persuasion theory, content unambiguity and source unambiguity can be considered as the cognitive appeals of online rumour messages. Both cognitive appeals and affective appeals can influence message persuasiveness and message receivers' behaviours. The findings of the themes of online rumour messages are used to support the development of a message-level research framework of online rumour message dissemination.

To address research question 2, social media data was analysed and manual coded to examine the research framework. Previous studies have examined rumour transmission between two persons, often on individual level and sometimes on group level (Brooks et al. 2013). With the Internet and social media, people are involved in open and broadcasting communications which are traceable. In this context, rumour messages tend to be broadcasted to communicator's social network and are available to any proactive information seeker. Because the potential social and financial damages online rumours can cause, there is a major significance to conduct message-level rumour study under this new context and update rumour theory. Based on such motivation, this study helps researchers and practitioners to understand how rumour disseminate in social media.

The results suggested that communicator features are all positively associated with the dissemination of online rumour messages. Community communication channel is positively associated with the dissemination of online rumour messages, whereas the direct communication channel is negatively

associated with the dissemination. For the cognitive and affective appeals of message features, the findings indicate that expression certainty and emotional activation are positively associated with the dissemination of online rumour messages, whereas source ambiguity (including source explicitness and source accessibility), emotional pleasantness and emotional imagery are negatively associated with the dissemination of online rumour messages. The use of evidence is found relevant to the dissemination of online rumour messages. Both topic-congruence and message-congruence show negative association with the dissemination of online rumour messages. They all moderate the effects of message features on the dissemination of online rumour messages in some way.

In addition, to solve research question 3, the data was analysed on the rumour topic level to understand the patterns of disseminations across rumour topics. The comparisons of the research framework across different types of rumour topics suggest that, the variables that drive the dissemination of online rumour messages can be very different across types of rumour topics. Through counting the number of online rumour messages and looking into their content over time, this thesis finds that the number of rumour messages and duration of survival vary dramatically rumour topics. However, there are some patterns that the dissemination of rumour messages tends to have a tipping point where the number of rumour messages accelerated dramatically and then drops quickly. Moreover, there is limited information distortion and exaggeration as online rumour messages survive over time. The findings address the third research question and provides extra topic-level insights to supplement the message-level

dissemination framework. Together they offer a comprehensive understanding on what drives the dissemination of online rumour messages in social media.

## **6.2 Theoretical Contribution**

This thesis contributes to previous literature in the following ways. First, this thesis contributes to rumour theory by extending the understanding of rumour dissemination to the online social media context. Instead of looking at what makes individuals transmit rumours in a face-to-face manner, this thesis takes the social media context into consideration and extends traditional rumour theory by introducing new dimensions to its key concepts- content unambiguity and source unambiguity. Specifically, expression certainty and use of evidence are considered as the new dimensions of content unambiguity, and source explicitness and source accessibility are considered as the new dimensions of source unambiguity. The effects of expression certainty, use of evidence, source explicitness and source accessibility on online rumour message dissemination are investigated in this thesis. By doing so, this thesis enhances the understanding of how content unambiguity and source unambiguity influence rumour message dissemination in the context of social media.

Second, this thesis provides new insights to rumour theory by investigating how emotional content plays a role in shaping the dissemination of online rumour messages. While the studies of rumour in both offline and online contexts have mainly focused on understanding the role of anxiety in rumouring,

this thesis goes beyond anxiety and considers emotions from more general aspects. Combining rumour theory with emotion theory, this thesis considers emotional valence, level of arousal and message vividness as three general aspects of emotional content. Specifically, through analysing how pleasantness, activation and imagery influence online rumour message dissemination, this thesis enhances the understanding of emotional content's role in shaping rumour dissemination.

Third, this thesis provides new insights to rumour theory by considering the congruences of rumour contexts and between rumour context and rumour message. Previous rumour research has paid attention to rumour target reputation and rumour valence, yet they are rarely investigated together. This study proposes topic-congruence to capture target reputation and rumour valence simultaneously. Moreover, although the effects of emotional appeal in information dissemination have been examined in other contexts (e.g. electronic word-of-mouth), the effects of situational emotions are often overlooked. This thesis uses message-congruence to consider both message-level emotional appeals and situation-level target reputation simultaneously. Building on congruence theory and persuasion theory, this thesis takes topic-congruence and message-congruence as moderators and clarifies how these two moderators influence the effects of cognitive appeals and affective appeals on rumour dissemination. This highlights the importance of congruence effects and enhances the current understanding of rumour dissemination.

Fourth, this thesis enriches existing rumour research through comparing rumour dissemination across rumour topic types. Although some studies about rumour combating strategies have taken the rumour type into consideration, the type of rumour topics is rarely considered when understanding the dissemination of rumours, and there lacks a detailed classification of online rumour topics. In this thesis, a typology of online rumour topics is proposed, classifying online rumour topics based on their objects (i.e., brand-related, offerings-related, people-related and operations-related) and focal points (i.e. description-oriented, reason-oriented and impact-oriented). The effects of message features, communicator features, channel features and rumour context features on online rumour message dissemination are compared for different types of online rumour topics.

Moreover, this thesis uses social media data to track rumours and their disseminations. Identifying rumours and tracing rumour disseminations have long been difficult for rumour researchers, and most rumour studies use self-reported experiments and questionnaire surveys to understand rumour dissemination. This thesis extends rumour research by investigating the actual social media data to trace and study rumour dissemination that was traditionally untraceable. Compared with self-reported research methods, this thesis measures the actual communication message, communicator features and dissemination behaviours reflected from the social media data, so that it avoids the potential problems with experimental settings and limitations of self-reporting methodology.



### **6.3 Practical Implication**

Practically, the findings of this thesis can be used to identify rumour messages that are likely to become viral so that practitioners can monitor and control rumours efficiently and effectively. First, since unambiguity and vividness can boost the dissemination of online rumour messages, practitioners should pay attention to messages with evidence, source link and pictures. Second, if an online rumour message communicates the same sentiment as the company's current reputation, practitioners should monitor and control such online rumour message as they are most likely to be processed carefully by the receivers. Third, practitioners should quickly develop combating strategies for online rumour messages that communicate the same sentiment as the company's current reputation, as they are likely to be disseminated more in social media.

Moreover, rumour combating strategies should be tailored according to the types of the rumour topic. Instead of preparing a universal rumour combating strategy, companies should at least develop a bank of strategies to fit different types of rumour topics in different situations. However, no matter what types of rumour topics the practitioners are dealing with, they should always train users with huge social influence on social media to be cautious of false rumours. Such users with huge social influence can also be good places to start launching rumour-combating campaigns, and they can serve as a good choice for seeding strategy. Moreover, it is important for practitioners to keep an eye on their reputation and to consider the congruence between the valence of rumour topics

and the reputation of their companies. Additionally, the research findings can be used to guide the design of emerging technologies and information systems to blunt enablers of rumour message dissemination. Artificial Intelligence (AI) technologies such as text mining and deep learning can be deployed to automatically identify the rumour messages that are likely to be viral in real time so that strategies can be deployed timely.

## **6.4 Limitation and Future Research**

Despite the potential contributions of this research, the research has some limitations and can be improved in future studies. Apart from the two-dimensional emotion model, the appraisal of certainty (or uncertainty) is also an important aspect to consider because uncertainty is closely related to rumouring phenomenon. The appraisal of certainty evaluates “the degree to which events are predictable and comprehensible” (Yin et al. 2014). People with uncertain feelings about the subsequent situations tend to use high level of cognitive effort to cope with the situation. Therefore, uncertain emotions stimulate people to intentionally evaluate and assess the rumour message, and the content and source credibility will attract more attention and be valued more. The effects of discrete dimensions should be explored and examined in the future.

Moreover, the current research did not consider the cultural differences in online rumour disseminations. However, social media allow people all over the world to share information and communicate, the same rumour may be received

and distributed by people with different cultural backgrounds. Future research can compare the research model in different cultural backgrounds or include cultural dimension into the research framework to examine the potential influences of culture on forwarding behaviour of rumour in social media.

In addition, future research can further build on this study to test the effectiveness of different rumour combating strategies. Previous research has examined the effects of crisis communication strategies or service recovery strategies. However, relative limited attention has been paid to identify and evaluate the effects of different rumour combating strategies. In fact, inappropriate rumour combating strategy may even enhance the public's belief in the rumour and stimulate the transmission of rumour. Future research can explore this issue and use social media data and questionnaire survey data to empirically test the effectiveness of different rumour combating strategies.

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## Appendix

### List of Rumour Topics That are Labelled as False or Fiction

| No. | Rumour Claim                                                                                                              | Source                                                                                                                                                                              | Key Company Involved | Keywords                                               |
|-----|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------|
| 1   | Abercrombie & Fitch is offering “n****r brown” pants for sale.                                                            | <a href="https://www.snopes.com/fact-check/brown-pants/">https://www.snopes.com/fact-check/brown-pants/</a>                                                                         | Abercrombie & Fitch  | Abercrombie & Fitch; A&F; N****r Brown; N-word         |
| 2   | The athletic shoe Adidas gained its name from an acronym for ‘All Day I Dream About Sex.’                                 | <a href="https://www.snopes.com/fact-check/adidas/">https://www.snopes.com/fact-check/adidas/</a>                                                                                   | Adidas               | Adidas; All Day I Dream About Sex                      |
| 3   | Coors financially supports Nazi groups.                                                                                   | <a href="https://www.snopes.com/fact-check/coors-and-nazis/">https://www.snopes.com/fact-check/coors-and-nazis/</a>                                                                 | Adolph Coors Company | Adolph Coors; Coors; Nazi                              |
| 4   | Alpha ZXT is endorsed by Denzel Washington, Jamie Foxx, Tiger Woods                                                       | <a href="https://www.snopes.com/fact-check/alpha-zxt-not-endorsed/">https://www.snopes.com/fact-check/alpha-zxt-not-endorsed/</a>                                                   | Alpha ZXT            | Alpha ZXT; Denzel Washington; Jamie Foxx; Tiger Woods  |
| 5   | Amazon's use of the USPS for package delivery services has resulted in a severe financial loss to the Postal Service.     | <a href="https://www.snopes.com/fact-check/amazon-bankrupting-united-states-postal-service/">https://www.snopes.com/fact-check/amazon-bankrupting-united-states-postal-service/</a> | Amazon               | Amazon; USPS; Financial loss; Bankrupt; Postal service |
| 6   | The UK hardware store B&Q warned its staff about a surge in rope, duct tape, and tie down sales before the release of the | <a href="https://www.snopes.com/fact-check/fifty-shades-of-hardware/">https://www.snopes.com/fact-check/fifty-shades-of-hardware/</a>                                               | B&Q                  | B&Q; Rope; Duct Tape; Fifty Shades of Grey             |

|    |                                                                                                                                |                                                                                                                                 |               |                                                  |
|----|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------|
|    | Fifty Shades of Grey movie.                                                                                                    |                                                                                                                                 |               |                                                  |
| 7  | Budweiser is ditching their famous Clydesdale ads in an effort to appeal to millennials.                                       | <a href="https://www.snopes.com/fact-check/dont-spare-the-horses/">https://www.snopes.com/fact-check/dont-spare-the-horses/</a> | Budweiser     | Budweiser; Clydesdale; Millennials               |
| 8  | Burger King created a fake ‘discontinued’ sandwich from the 1970s for nostalgia purposes                                       | <a href="https://www.snopes.com/fact-check/yumbo-drop/">https://www.snopes.com/fact-check/yumbo-drop/</a>                       | Burger King   | Burger King; Yumbo; False; Fake; Exist; Remember |
| 9  | Men’s underwear is referred to as ‘BVDs’ because the term is an abbreviation of the phrase ‘Boy’s Ventilated Drawers.’         | <a href="https://www.snopes.com/fact-check/top-drawers/">https://www.snopes.com/fact-check/top-drawers/</a>                     | BVD           | BVD; Boy’s Ventilated Drawers                    |
| 10 | The Canada Dry beverage company got its name because it produced alcohol-free beverages during Prohibition.                    | <a href="https://www.snopes.com/fact-check/canada-dry/">https://www.snopes.com/fact-check/canada-dry/</a>                       | Canada Dry    | Canada Dry; Name; Prohibition; Alcohol Free      |
| 11 | People get “hooked” on Carmex Lip Balm because it contains ground glass (or other irritants), necessitating its continued use. | <a href="https://www.snopes.com/fact-check/carmex-addiction/">https://www.snopes.com/fact-check/carmex-addiction/</a>           | Carmex        | Carmex; Addicted                                 |
| 12 | Chef Boyardee was a fictional creation whose name was taken from the given names of the company’s three                        | <a href="https://www.snopes.com/fact-check/chef-boyardee/">https://www.snopes.com/fact-check/chef-boyardee/</a>                 | Chef Boyardee | Chef Boyardee; Fictional; Not Real               |

|    |                                                                                                                       |                                                                                                                                             |                        |                                     |
|----|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------|
|    | founders: Boyd, Art, and Dennis.                                                                                      |                                                                                                                                             |                        |                                     |
| 13 | A leper was found working in a Chesterfield cigarette factory.                                                        | <a href="https://www.snopes.com/fact-check/the-leper-who-changes-spots/">https://www.snopes.com/fact-check/the-leper-who-changes-spots/</a> | Chesterfield Cigarette | Chesterfield Cigarette; Leper       |
| 14 | The Chevrolet Nova sold poorly in Spanish-speaking countries because its name translates as "doesn't go" in Spanish.  | <a href="https://www.snopes.com/fact-check/dont-go-here/">https://www.snopes.com/fact-check/dont-go-here/</a>                               | Chevrolet Nova         | Chevrolet Nova; Spanish; Doesn't go |
| 15 | Chipotle has stopped offering pork at its restaurants in order to appease Muslim customers                            | <a href="https://www.snopes.com/fact-check/pulled-pork/">https://www.snopes.com/fact-check/pulled-pork/</a>                                 | Chipotle               | Chipotle; Pork; Muslim              |
| 16 | Chick-fil-A restaurants have announced they will be reversing their longstanding policy and start opening on Sundays. | <a href="https://www.snopes.com/fact-check/chick-fil-open-sundays/">https://www.snopes.com/fact-check/chick-fil-open-sundays/</a>           | Click-fil-A            | Click-fil-A; Sunday                 |
| 17 | A university student who drank too many Cokes died from carbon dioxide poisoning.                                     | <a href="https://www.snopes.com/fact-check/cosub2sub-fast-2-furious/">https://www.snopes.com/fact-check/cosub2sub-fast-2-furious/</a>       | Coca-Cola              | Coca-Cola; Die                      |
| 18 | An early experiment in subliminal advertising at a movie theater substantially increased sales of popcorn and Coke.   | <a href="https://www.snopes.com/fact-check/subliminal-advertising/">https://www.snopes.com/fact-check/subliminal-advertising/</a>           | Coca-Cola              | Coca-Cola; subliminal               |

|    |                                                                                                                       |                                                                                                                                                                                                                                 |           |                                                              |
|----|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------|
| 19 | Coca-Cola and Nestle are negotiating with the Brazilian government to privatize the Guarani Aquifer in South America. | <a href="https://www.snopes.com/fact-check/nestle-coca-cola-trying-to-privatize-guarani-aquifer-in-south-america/">https://www.snopes.com/fact-check/nestle-coca-cola-trying-to-privatize-guarani-aquifer-in-south-america/</a> | Coca-Cola | Coca-Cola; Privatize; Guarani Aquifer; Brazil; South America |
| 20 | Coca-Cola Has Recalled Dasani Water Due to Parasite Worms.                                                            | <a href="https://www.truthorfiction.com/coca-cola-recalled-dasani-water-due-parasite-worms/">https://www.truthorfiction.com/coca-cola-recalled-dasani-water-due-parasite-worms/</a>                                             | Coca-Cola | Coca-Cola; Dasani; Worms                                     |
| 21 | Coca-Cola is an effective contraceptive.                                                                              | <a href="https://www.snopes.com/fact-check/killer-sperm/">https://www.snopes.com/fact-check/killer-sperm/</a>                                                                                                                   | Coca-Cola | Coca-Cola; Contraceptive; Kill Sperm                         |
| 22 | Coca-Cola spiked with monosodium glutamate (MSG) is an aphrodisiac or soporific                                       | <a href="https://www.snopes.com/fact-check/msg-mickey/">https://www.snopes.com/fact-check/msg-mickey/</a>                                                                                                                       | Coca-Cola | Coca-Cola; Monosodium Glutamate; MSG                         |
| 23 | Coca-Cola was originally green.                                                                                       | <a href="https://www.snopes.com/fact-check/how-green-was-my-ballyhoo/">https://www.snopes.com/fact-check/how-green-was-my-ballyhoo/</a>                                                                                         | Coca-Cola | Green                                                        |
| 24 | Coca-Cola's original diet cola drink, TaB, took its name from an acronym for "totally artificial beverage."           | <a href="https://www.snopes.com/fact-check/keeping-tab/">https://www.snopes.com/fact-check/keeping-tab/</a>                                                                                                                     | Coca-Cola | Coca-Cola; TaB; Origins of Name; Totally Artificial Beverage |
| 25 | Combining Coca-Cola and aspirin will get you high (or kill you)                                                       | <a href="https://www.snopes.com/fact-check/painkiller-combination/">https://www.snopes.com/fact-check/painkiller-combination/</a>                                                                                               | Coca-Cola | Coca-Cola; Aspirin                                           |
| 26 | Diet Coke contains more calories than claimed, but the company gets away with the deception by paying a yearly fine.  | <a href="https://www.snopes.com/fact-check/diet-riot/">https://www.snopes.com/fact-check/diet-riot/</a>                                                                                                                         | Coca-Cola | Diet Coke; Calories; Deception; Fine                         |

|    |                                                                                                                                                                                                                            |                                                                                                                               |                      |                                                 |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------|
| 27 | Pouring Coca-Cola onto a piece of raw pork will cause worms to come crawling out of the meat.                                                                                                                              | <a href="https://www.snopes.com/fact-check/coke-adds-life/">https://www.snopes.com/fact-check/coke-adds-life/</a>             | Coca-Cola            | Coca-Cola; Pork; Worm                           |
| 28 | Substituting Diet Coke for regular Coca-Cola in a popular ham recipe can cause an explosion.                                                                                                                               | <a href="https://www.snopes.com/fact-check/spiral-sliced-blam/">https://www.snopes.com/fact-check/spiral-sliced-blam/</a>     | Coca-Cola            | Diet Coke; Ham; Explode                         |
| 29 | The Mormons own the Coca-Cola Company.                                                                                                                                                                                     | <a href="https://www.snopes.com/fact-check/lds-sense/">https://www.snopes.com/fact-check/lds-sense/</a>                       | Coca-Cola            | Mormons; Own                                    |
| 30 | When tipped on its side, Coca-Cola's distinctive script logo shows a figure snorting a line of cocaine, and this image was deliberately placed there by the artist as a sly reference to the product's containing cocaine. | <a href="https://www.snopes.com/fact-check/derisive-snort/">https://www.snopes.com/fact-check/derisive-snort/</a>             | Coca-Cola            | Logo; Cocaine                                   |
| 31 | Continental Airlines CEO Gordon Bethune once personally threw a passenger off an airplane for being rude to a flight attendant.                                                                                            | <a href="https://www.snopes.com/fact-check/continental-divide/">https://www.snopes.com/fact-check/continental-divide/</a>     | Continental Airlines | Continental Airlines; Gordon Bethune; Passenger |
| 32 | Workers at the Corona brewery urinate into the beer.                                                                                                                                                                       | <a href="https://www.snopes.com/fact-check/urine-in-corona-beer/">https://www.snopes.com/fact-check/urine-in-corona-beer/</a> | Corona               | Corona; Urinate                                 |
| 33 | Costco warehouse club stores are owned by China.                                                                                                                                                                           | <a href="https://www.snopes.com/fact-check/costco/">https://www.snopes.com/fact-check/costco/</a>                             | Costco               | Costco; Owned by China                          |
| 34 | Crocs is closing down and will no                                                                                                                                                                                          | <a href="https://www.msn.com/en-">https://www.msn.com/en-</a>                                                                 | Crocs                | Crocs; Close Down; Shut                         |

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|    | longer be making and selling shoes.                                                                                                                                    | us/news/other/crocs-is-closing-all-of-its-manufacturing-facilities/ar-BBLGrPR                                                                                                       |                | Down; Out of Business; No Longer                         |
| 35 | DEW brand bottled water sold in Nigeria and other parts of Africa has caused at least 180 deaths and has been known to infect those who drink it with the Ebola virus. | <a href="https://www.truthorfiction.com/dew-bottled-water-has-caused-180-deaths-in-nigeria/">https://www.truthorfiction.com/dew-bottled-water-has-caused-180-deaths-in-nigeria/</a> | DEW            | DEW; Ebola; Virus                                        |
| 36 | Domino's Pizza financially supports Operation Rescue, an anti-abortion group.                                                                                          | <a href="https://www.snopes.com/fact-check/domino-theory/">https://www.snopes.com/fact-check/domino-theory/</a>                                                                     | Domino's Pizza | Domino's Pizza; Support; Operation Rescue; Anti-abortion |
| 37 | Domino's Pizza ended its "30 minutes or it's free" guarantee because a speeding delivery driver hit and killed a child.                                                | <a href="https://www.snopes.com/fact-check/mushroom-pizza/">https://www.snopes.com/fact-check/mushroom-pizza/</a>                                                                   | Domino's Pizza | 30 Minutes; Killed a Child                               |
| 38 | The popular Dr Pepper soft drink is made from prune juice.                                                                                                             | <a href="https://www.snopes.com/fact-check/belle-pepper/">https://www.snopes.com/fact-check/belle-pepper/</a>                                                                       | Dr Pepper      | Dr Pepper; Prune Juice                                   |
| 39 | Dunkin' Donuts refused to contribute to a reward offered for finding the killer of a police officer who was shot in one of its stores.                                 | <a href="https://www.snopes.com/fact-check/dunkin-donuts-shooting/">https://www.snopes.com/fact-check/dunkin-donuts-shooting/</a>                                                   | Dunkin' Donuts | Dunkin; Dunkin' Donuts; Find Killer                      |
| 40 | The discount chain E.J. Korvette took its name from a                                                                                                                  | <a href="https://www.snopes.com/fact-check/ej-korvette/">https://www.snopes.com/fact-check/ej-korvette/</a>                                                                         | E.J. Korvette  | E.J. Korvette; Eight Jewish Korean War Veterans          |

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|    | shortening of 'eight Jewish Korean War veterans' in honor of its founding partners.       |                                                                                                                                                               |                                              |                                         |
| 41 | Bottled water giant Evian named its product as a sly dig at consumers.                    | <a href="https://www.snopes.com/fact-check/evian-and-jaron/">https://www.snopes.com/fact-check/evian-and-jaron/</a>                                           | Evian                                        | Evian; Name; Naive                      |
| 42 | Fanta was invented by the Nazis.                                                          | <a href="https://www.snopes.com/fact-check/the-reich-stuff/">https://www.snopes.com/fact-check/the-reich-stuff/</a>                                           | Fanta                                        | Fanta; Nazis                            |
| 43 | After a bank failure, the FDIC has 99 years to pay back insured deposits.                 | <a href="https://www.snopes.com/fact-check/fdic-you/">https://www.snopes.com/fact-check/fdic-you/</a>                                                         | Federal Deposit Insurance Corporation (FDIC) | FDIC; 99 Years                          |
| 44 | Presenting a rare 1943 copper cent to Ford entitled the bearer to a brand-new automobile. | <a href="https://www.snopes.com/fact-check/the-cented-garden/">https://www.snopes.com/fact-check/the-cented-garden/</a>                                       | Ford Motor                                   | Ford; 1943 Copper; Brand-New Automobile |
| 45 | The Ford Motor Company donated \$100 million to the Black Lives Matter movement.          | <a href="https://www.snopes.com/fact-check/ford-motors-funds-black-lives-matter/">https://www.snopes.com/fact-check/ford-motors-funds-black-lives-matter/</a> | Ford Motor                                   | Ford; Black Lives Matter                |
| 46 | The Fox News Channel has been banned in Canada because they report false information.     | <a href="https://www.snopes.com/fact-check/canadian-fox/">https://www.snopes.com/fact-check/canadian-fox/</a>                                                 | Fox News                                     | Fox News; Banned; Rejected; Canada      |
| 47 | The Fox News Channel won a 2004 court case allowing the cable channel to lie to viewers.  | <a href="https://www.snopes.com/fact-check/fox-skews/">https://www.snopes.com/fact-check/fox-skews/</a>                                                       | Fox News                                     | Fox News; Court; Right to Lie           |
| 48 | General Motors is resurrecting the                                                        | <a href="https://www.snopes.com/fact-check/">https://www.snopes.com/fact-check/</a>                                                                           | General Motors                               | General Motors;                         |

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|    | Pontiac Firebird Trans Am for 2015                                                                                                    | check/2015-pontiac-firebird-trans-am/                                                                                         |                    | Pontiac Firebird Trans Am                      |
| 49 | In June 2012, 79% of General Motors' sales were made to the U.S. federal government.                                                  | <a href="https://www.snopes.com/fact-check/purchasing-agreement/">https://www.snopes.com/fact-check/purchasing-agreement/</a> | General Motors     | General Motors; 79%; Sales; Government         |
| 50 | The Chevy Volt "costs more than 7 times as much to run and takes 3 times as long to drive across country" than a standard automobile. | <a href="https://www.snopes.com/fact-check/chevy-revolt/">https://www.snopes.com/fact-check/chevy-revolt/</a>                 | General Motors     | General Motors; Chevy Volt; Cost More          |
| 51 | Girl Scout cookies fund Planned Parenthood.                                                                                           | <a href="https://www.truthorfiction.com/afag-susa-cookies/">https://www.truthorfiction.com/afag-susa-cookies/</a>             | Girl Scout Cookies | Girl Scout Cookies; Planned Parenthood         |
| 52 | Baby products company Graco took its name from an acronym for "God Rewards All Christian Organizations."                              | <a href="https://www.snopes.com/fact-check/graco/">https://www.snopes.com/fact-check/graco/</a>                               | Graco              | Graco; God Rewards All Christian Organizations |
| 53 | The H.J. Heinz company adopted the slogan '57 Varieties' because its product line included exactly 57 different items.                | <a href="https://www.snopes.com/fact-check/57-varieties/">https://www.snopes.com/fact-check/57-varieties/</a>                 | H.J. Heinz Company | H.J. Heinz Company; 57 Varieties; Product Line |
| 54 | Harley-Davidson's 'Fat Boy' motorcycle was designed to represent the dropping of atomic bombs on Japan.                               | <a href="https://www.snopes.com/fact-check/fat-boy/">https://www.snopes.com/fact-check/fat-boy/</a>                           | Harley-Davidson    | Harley-Davidson; Fat Boy; Atomic Bombs         |



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| 55 | Heinz marketed swastika-shaped pasta in Germany before and during World War II.                                                                                             | <a href="https://www.snopes.com/fact-check/uh-oh-swastik-os/">https://www.snopes.com/fact-check/uh-oh-swastik-os/</a>                   | Heinz              | Heinz; Marketed Swastika-Shaped Pasta;      |
| 56 | A pact with the Devil to save a cancer-stricken child resulted in the creation of the Hello Kitty brand.                                                                    | <a href="https://www.snopes.com/fact-check/hello-kitty-good-bye-soul/">https://www.snopes.com/fact-check/hello-kitty-good-bye-soul/</a> | Hello Kitty        | Hello Kitty; Devil; Evil                    |
| 57 | The jilted wife of the owner of Home Depot created the Lowe's chain of home improvement stores.                                                                             | <a href="https://www.snopes.com/fact-check/the-lowes-down/">https://www.snopes.com/fact-check/the-lowes-down/</a>                       | Home Depot; Lowe's | Home Depot; Lowe's; Jilted wife             |
| 58 | In accordance with a requirement of their original Royal Charter, the Hudson's Bay Company of Canada makes annual payments of elk and beaver pelts to the Queen of England. | <a href="https://www.snopes.com/fact-check/fur-the-queen/">https://www.snopes.com/fact-check/fur-the-queen/</a>                         | Hudson's Bay Co.   | Hudson's Bay; Pelts; Fur; Queen of England  |
| 59 | "No tears" baby shampoos work because they include anesthetics such as novocaine                                                                                            | <a href="https://www.snopes.com/fact-check/the-caine-mutiny/">https://www.snopes.com/fact-check/the-caine-mutiny/</a>                   | Johnson            | Johnson; No Tears Baby Shampoo; Anesthetics |
| 60 | A 3-year-old girl was ordered to leave a KFC outlet because her facial injuries were disturbing to customers.                                                               | <a href="https://www.snopes.com/fact-check/chicken-little/">https://www.snopes.com/fact-check/chicken-little/</a>                       | KFC                | KFC; Girl; Leave; Disturb                   |
| 61 | Colonel Sanders left instructions requiring KFC to                                                                                                                          | <a href="https://www.snopes.com/fact-check/">https://www.snopes.com/fact-check/</a>                                                     | KFC                | KFC; Colonel Sanders; Ku Klux Klan          |

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|    | donate money to the Ku Klux Klan or feed the homeless for free.                                                                                                   | check/no-colonel-of-truth/                                                                                                                  |                         |                                       |
| 62 | Kool menthol cigarettes are owned by the KKK (Ku Klux Klan).                                                                                                      | <a href="https://www.snopes.com/fact-check/menthol-things-considered/">https://www.snopes.com/fact-check/menthol-things-considered/</a>     | Kool Menthol Cigarettes | Kool menthol cigarettes; Ku Klux Klan |
| 63 | Kool menthol cigarettes contain fiberglass.                                                                                                                       | <a href="https://www.snopes.com/fact-check/menthol-things-considered/">https://www.snopes.com/fact-check/menthol-things-considered/</a>     | Kool Menthol Cigarettes | Kool Menthol Cigarettes; Fiberglass   |
| 64 | Symbols embossed on the exterior surfaces of Oreo cookies link the product to the alleged conspiratorial activities of the Knights Templar and Freemasons.        | <a href="https://www.snopes.com/fact-check/secret-symbols-oreo-cookies/">https://www.snopes.com/fact-check/secret-symbols-oreo-cookies/</a> | Kraft Foods             | Oreo; Knights Templar; Freemasons     |
| 65 | Life Savers candy was so named because its inventor's daughter died from choking on a non-holed mint.                                                             | <a href="https://www.snopes.com/fact-check/life-savers/">https://www.snopes.com/fact-check/life-savers/</a>                                 | Life Savers             | Life Savers; Inventor's Daughter; Die |
| 66 | Designer Liz Claiborne appeared on a popular television talk show and announced that she doesn't design clothes for black women because "their hips are too big." | <a href="https://www.snopes.com/fact-check/liz-biz/">https://www.snopes.com/fact-check/liz-biz/</a>                                         | Liz Claiborne           | Liz Claiborne; For Black Women        |
| 67 | Richard Blum, the husband of Senator Dianne Feinstein, obtained a contract to broker                                                                              | <a href="https://www.snopes.com/fact-check/liz-biz/">https://www.snopes.com/fact-check/liz-biz/</a>                                         | USPS                    | Richard Blum; USPS                    |

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|    | sales of USPS facilities due to his wife's influence.                                                                  |                                                                                                                                                             |                 |                                           |
| 68 | Lucky Strike brand cigarettes are so named because some packs contained marijuana cigarettes.                          | <a href="https://www.snopes.com/fact-check/lucky-strikes/">https://www.snopes.com/fact-check/lucky-strikes/</a>                                             | Lucky Strike    | Lucky Strike; Cigarettes; Name; Marijuana |
| 69 | Macy's refused to hire an applicant because she was a veteran who had served in Afghanistan.                           | <a href="https://www.snopes.com/fact-check/macys-wont-hire-veteran/">https://www.snopes.com/fact-check/macys-wont-hire-veteran/</a>                         | Macy's          | Macy's; Hire; Job; Afghanistan            |
| 70 | Harley-Davidson's chief executive officer Matthew Levatich called President Trump "a moron."                           | <a href="https://www.snopes.com/fact-check/did-ceo-harley-davidson-trump-moron/">https://www.snopes.com/fact-check/did-ceo-harley-davidson-trump-moron/</a> | Marley-Davidson | Davidson; Matthew Levatich; Trump; Moron  |
| 71 | The Milky Way and 3 Musketeers candy bars were given the wrong names when their wrappers were inadvertently switched.  | <a href="https://www.snopes.com/fact-check/3-musketeers/">https://www.snopes.com/fact-check/3-musketeers/</a>                                               | Mars            | Milky Way; 3 Musketeers; Name             |
| 72 | Cabbage Patch dolls were designed to get people accustomed to the appearance of mutants following a thermonuclear war. | <a href="https://www.snopes.com/fact-check/nuclear-family/">https://www.snopes.com/fact-check/nuclear-family/</a>                                           | Mattel          | Cabbage Patch Dolls; Mutants              |
| 73 | Cabbage Patch dolls were modeled upon mentally defective children.                                                     | <a href="https://www.snopes.com/fact-check/nuclear-family/">https://www.snopes.com/fact-check/nuclear-family/</a>                                           | Mattel          | Cabbage Patch Dolls; Defective Children   |

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| 74 | McDonald's uses worm meat fillers.                                                                                                                      | <a href="https://www.truthorfiction.com/mcdonalds-worm-burgers/">https://www.truthorfiction.com/mcdonalds-worm-burgers/</a>             | McDonald's | McDonald's; Worm Meat Filler                      |
| 75 | McDonald's buys their meat from a company called '100% Beef,' which allows them to legally but deceptively claim they use 100% beef in their hamburgers | <a href="https://www.snopes.com/fact-check/100-beef/">https://www.snopes.com/fact-check/100-beef/</a>                                   | McDonald's | McDonald's; 100% Beef; Company                    |
| 76 | Persistent claims that McDonald's McRib sandwiches made from inverted pork rectums surfaced in December 2016 and again in August 2017.                  | <a href="https://www.truthorfiction.com/mcdonalds-mcrib-pork-rectum/">https://www.truthorfiction.com/mcdonalds-mcrib-pork-rectum/</a>   | McDonald's | McDonald's; McRib; Pork Rectums                   |
| 77 | Sarajevo and Pyongyang are the only two world capitals without McDonald's restaurants.                                                                  | <a href="https://www.snopes.com/fact-check/your-kind-of-place/">https://www.snopes.com/fact-check/your-kind-of-place/</a>               | McDonald's | Two World Capitals Without McDonald's Restaurants |
| 78 | The McDonald's fast food chain has implemented a policy barring their customers from buying food for homeless people.                                   | <a href="https://www.snopes.com/fact-check/mcdonalds-homeless-policy/">https://www.snopes.com/fact-check/mcdonalds-homeless-policy/</a> | McDonald's | McDonald's; Bar/Ban; Food; Homeless               |
| 79 | List details food companies owned by Monsanto.                                                                                                          | <a href="https://www.snopes.com/fact-check/monsanto-owned-companies/">https://www.snopes.com/fact-check/monsanto-owned-companies/</a>   | Monsanto   | Monsanto Own                                      |
| 80 | Monsanto has purchased or merged with the                                                                                                               | <a href="https://www.snopes.com/fact-check/monsanto-">https://www.snopes.com/fact-check/monsanto-</a>                                   | Monsanto   | Monsanto; Purchase; Merge                         |

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|    | Whole Foods Market grocery chain.                                                                                                                                        | buys-whole-foods/                                                                                                                                                                                                               |                                 |                                                           |
| 81 | Monster brand energy drink uses a Hebrew version of 666 in their logo                                                                                                    | <a href="https://www.snopes.com/fact-check/monster-energy-drink-666/">https://www.snopes.com/fact-check/monster-energy-drink-666/</a>                                                                                           | Monster                         | Monster; Hebrew; 666                                      |
| 82 | Munilla Construction Management, the construction firm that built the bridge at Florida International University that collapsed in March 2018, is an all-female company. | <a href="https://www.snopes.com/fact-check/was-all-female-responsible-bridge-collapse/">https://www.snopes.com/fact-check/was-all-female-responsible-bridge-collapse/</a>                                                       | Munilla Construction Management | Munilla Construction; Collapsed                           |
| 83 | Woman gains revenge after being charged \$250 by Neiman Marcus for a cookie recipe.                                                                                      | <a href="https://www.snopes.com/fact-check/costs-a-fortune-cookie/">https://www.snopes.com/fact-check/costs-a-fortune-cookie/</a>                                                                                               | Neiman Marcus                   | Neiman Marcus; \$250; Cookie Recipe                       |
| 84 | Coca-Cola and Nestle are negotiating with the Brazilian government to privatize the Guarani Aquifer in South America.                                                    | <a href="https://www.snopes.com/fact-check/nestle-coca-cola-trying-to-privatize-guarani-aquifer-in-south-america/">https://www.snopes.com/fact-check/nestle-coca-cola-trying-to-privatize-guarani-aquifer-in-south-america/</a> | Nestle                          | Nestle; Privatize; Guarani Aquifer; Brazil; South America |
| 85 | A Nike shoe advertisement uses imagery from a terrorist bombing.                                                                                                         | <a href="https://www.snopes.com/fact-check/shoe-bomb/">https://www.snopes.com/fact-check/shoe-bomb/</a>                                                                                                                         | Nike                            | Nike; Shoe; Terrorist                                     |
| 86 | Nike makes their sneakers with elephant skins.                                                                                                                           | <a href="https://www.snopes.com/fact-check/nike-elephant-shoes-rumor/">https://www.snopes.com/fact-check/nike-elephant-shoes-rumor/</a>                                                                                         | Nike                            | Nike; Elephant; Skin; Sneaker; Shoes                      |

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| 87 | Tiger Woods named his restaurant “The Woods Jupiter” because Nike legally owns the rights to his name. e.g. Tiger Woods does not own the rights to the name Tiger Woods — Nike does. (2015/03) | <a href="https://www.snopes.com/fact-check/the-woods-jupiter/">https://www.snopes.com/fact-check/the-woods-jupiter/</a>                           | Nike                  | Nike; Own; Tiger Woods; Name                              |
| 88 | You can get a new pair of shoes from Nike by just mailing them any old, worn-out pair of sneakers.                                                                                             | <a href="https://www.snopes.com/fact-check/free-shoes-from-nike/">https://www.snopes.com/fact-check/free-shoes-from-nike/</a>                     | Nike                  | Nike; New; Worn-out                                       |
| 89 | Marlboro is owned by the KKK (Ku Klux Klan).                                                                                                                                                   | <a href="https://www.snopes.com/fact-check/welcome-to-marlboro-kkkcountry/">https://www.snopes.com/fact-check/welcome-to-marlboro-kkkcountry/</a> | Philip Morris Company | Marlboro; Owned; Own; KKK                                 |
| 90 | The president of Procter & Gamble announced on a popular talk show that he donates a portion of the company’s profits to the Church of Satan.                                                  | <a href="https://www.snopes.com/fact-check/trademark-of-the-devil/">https://www.snopes.com/fact-check/trademark-of-the-devil/</a>                 | Procter & Gamble      | Procter & Gamble; P&G; President; Donate; Church of Satan |
| 91 | Ralph Lauren clothiers is offering “nigger brown” shirts for sale.                                                                                                                             | <a href="https://www.snopes.com/fact-check/brown-shirts/">https://www.snopes.com/fact-check/brown-shirts/</a>                                     | Ralph Lauren          | Ralph Lauren; N-word; N****r Brown; Shirts                |
| 92 | Red Bull energy drinks contain bull sperm.                                                                                                                                                     | <a href="https://www.truthorfiction.com/red-bull-sperm/">https://www.truthorfiction.com/red-bull-sperm/</a>                                       | Red Bull              | Red Bull; Sperm                                           |
| 93 | The Red Lobster chain is about to close its doors and go out of business for good.                                                                                                             | <a href="https://www.snopes.com/fact-check/red-lobster-closing/">https://www.snopes.com/fact-check/red-lobster-closing/</a>                       | Red Lobster           | Red Lobster; Close                                        |

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| 94  | Sam's Club and Walmart donated \$10,000 to the Darren Wilson defense fund.                                                        | <a href="https://www.snopes.com/fact-check/sams-flub/">https://www.snopes.com/fact-check/sams-flub/</a>                                                                                                   | Sam's Club | Sam's Club; Darren Wilson       |
| 95  | Snapple donates millions of dollars to anti-marijuana efforts.                                                                    | <a href="https://www.snopes.com/fact-check/drug-bust/">https://www.snopes.com/fact-check/drug-bust/</a>                                                                                                   | Snapple    | Snapple; Donate; Anti-Marijuana |
| 96  | Snapple is affiliated with the Ku Klux Klan (or donates money to the anti-abortion group Operation Rescue).                       | <a href="https://www.snopes.com/fact-check/snapple-dragon/">https://www.snopes.com/fact-check/snapple-dragon/</a>                                                                                         | Snapple    | Snapple; Ku Klux Klan; KKK      |
| 97  | The Japanese corporation known as Sony based its name on an acronym formed from 'Standard Oil of New York.'                       | <a href="https://www.snopes.com/fact-check/sony/">https://www.snopes.com/fact-check/sony/</a>                                                                                                             | Sony       | Sony; Standard Oil of New York  |
| 98  | A woman named Shanell Rivers put blood and dog feces into various products at a Starbucks in Atlanta                              | <a href="https://www.snopes.com/fact-check/shanell-rivers-put-blood-dog-feces-into-starbucks-products/">https://www.snopes.com/fact-check/shanell-rivers-put-blood-dog-feces-into-starbucks-products/</a> | Starbucks  | Starbucks; Shanell Rivers       |
| 99  | Starbucks adds semen to its drinks.                                                                                               | <a href="https://www.truthorfiction.com/starbucks-semen-for-flavor/">https://www.truthorfiction.com/starbucks-semen-for-flavor/</a>                                                                       | Starbucks  | Starbucks; Semen                |
| 100 | Starbucks refused free product to Marines serving in Iraq, saying the company didn't support the war or anyone taking part in it. | <a href="https://www.snopes.com/fact-check/gi-joe/">https://www.snopes.com/fact-check/gi-joe/</a>                                                                                                         | Starbucks  | Starbucks; Refuse; Marines      |
| 101 | The wife of the owner of a Thailand                                                                                               | <a href="https://www.snopes.com/fact-check/">https://www.snopes.com/fact-check/</a>                                                                                                                       | Starbucks  | Starbucks; Not for Asians       |

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|     | Starbucks told customers that the particularly trendy coffee spot is not for Asians.                                                                                                                                    | check/ill-starred-bucks/                                                                                                                                                      |             |                                                         |
| 102 | Taco Bell has filed for bankruptcy and will close all Taco Bell locations in 2017 and 2018.                                                                                                                             | <a href="https://www.truthorfiction.com/taco-bell-files-bankruptcy/">https://www.truthorfiction.com/taco-bell-files-bankruptcy/</a>                                           | Taco Bell   | Taco Bell; Bankrupt; Bankruptcy                         |
| 103 | A photograph depicts a sign at Target informing customers purchasing pork or alcohol to choose another lane to accommodate Muslim employees.                                                                            | <a href="https://www.snopes.com/fact-check/targets-sharia-compliant-lanes/">https://www.snopes.com/fact-check/targets-sharia-compliant-lanes/</a>                             | Target      | Target; Offend; Muslim; Lane                            |
| 104 | An image shows a curt response from a Target customer service representative. Note: Customer Service (@Customer_Assist) is not the official customer service account of Target, the true one is AskTarget (@AskTarget). | <a href="https://www.snopes.com/fact-check/target-customer-service-transgender-bathroom/">https://www.snopes.com/fact-check/target-customer-service-transgender-bathroom/</a> | Target      | Target; Transgender; Bathroom; Customer Service Account |
| 105 | Timberland is owned by the Ku Klux Klan, a situation lamented in a poem by Maya Angelou.                                                                                                                                | <a href="https://www.snopes.com/fact-check/shiver-me-timberlands/">https://www.snopes.com/fact-check/shiver-me-timberlands/</a>                                               | Timberland  | Timberland; Ku Klux Klan; KKK                           |
| 106 | A Tootsie Pop wrapper with a picture of an                                                                                                                                                                              | <a href="https://www.snopes.com/fact-check/">https://www.snopes.com/fact-check/</a>                                                                                           | Totsie Roll | Totsie Roll; Tootsie Pop Wrapper;                       |



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|     | Indian shooting an arrow at a star on it can be redeemed with Tootsie Roll Industries for a free bag of candy.   | check/shooting-star/                                                                                                    |                  | Indian; Free Candy                                 |
| 107 | Troop clothing is owned by the KKK, and the name stands for "To Rule Over Oppressed People."                     | <a href="https://www.snopes.com/fact-check/troop-loops/">https://www.snopes.com/fact-check/troop-loops/</a>             | Troop            | Troop; Ku Klux Klan; To Rule Over Oppressed People |
| 108 | Tropical Fantasy contains a special ingredient that will render black men sterile.                               | <a href="https://www.snopes.com/fact-check/tropical-storm/">https://www.snopes.com/fact-check/tropical-storm/</a>       | Tropical Fantasy | Tropical Fantasy; Render Black Men Sterile         |
| 109 | Soles of Vans shoes are adorned with the Star of David so that wearers will "stomp on Jews."                     | <a href="https://www.snopes.com/fact-check/star-quality/">https://www.snopes.com/fact-check/star-quality/</a>           | Vans             | Vans; Star; Jews                                   |
| 110 | The U.S. government prohibits high mileage Volkswagen Passat automobiles from being purchased by U.S. customers. | <a href="https://www.snopes.com/fact-check/volkswagen-passat/">https://www.snopes.com/fact-check/volkswagen-passat/</a> | Volkswagen       | Volkswagen; Prohibit/ Not Allowed; Passat          |
| 111 | Sam's Club and Walmart donated \$10,000 to the Darren Wilson defense fund.                                       | <a href="https://www.snopes.com/fact-check/sams-flub/">https://www.snopes.com/fact-check/sams-flub/</a>                 | Walmart          | Walmart; \$10,000; Darren Wilson                   |
| 112 | Walmart has been adding a "phantom charge" of \$10 to shoppers' bills for 10 years.                              | <a href="https://www.snopes.com/fact-check/walmart-jajket/">https://www.snopes.com/fact-check/walmart-jajket/</a>       | Walmart          | Walmart; Phantom Charge                            |

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| 113 | Walmart has discontinued the sale of all merchandise that is Bible-related or labeled as “made in America” to avoid offending people. | <a href="https://www.snopes.com/fact-check/walmart-made-america-bible/">https://www.snopes.com/fact-check/walmart-made-america-bible/</a>                     | Walmart            | Walmart; Bible Related                        |
| 114 | Under corporate policy, you can get free food at any Wendy’s restaurant by claiming that you’re lost and have no money.               | <a href="https://www.snopes.com/fact-check/business-wendys-free-food/">https://www.snopes.com/fact-check/business-wendys-free-food/</a>                       | Wendy’s            | Wendy’s; Free Food; Lost; No Money            |
| 115 | Monsanto has purchased or merged with the Whole Foods Market grocery chain.                                                           | <a href="https://www.snopes.com/fact-check/monsanto-buys-whole-foods/">https://www.snopes.com/fact-check/monsanto-buys-whole-foods/</a>                       | Whole Foods Market | Whole Foods Market; Monsanto; Purchase; Merge |
| 116 | Elon Musk told a room of investors that he planned on buying Facebook so that he could delete the social media platform.              | <a href="https://www.snopes.com/fact-check/elon-musk-buy-facebook/">https://www.snopes.com/fact-check/elon-musk-buy-facebook/</a>                             | Facebook           | Elon Musk; Facebook; delete                   |
| 117 | Starbucks COO Rosalind Brewer once expressed the viewpoint that “white men are the root of all evil.”                                 | <a href="https://www.snopes.com/fact-check/starbucks-white-men-root-of-all-evil/">https://www.snopes.com/fact-check/starbucks-white-men-root-of-all-evil/</a> | Starbucks          | Starbucks; COO; Rosalind Brewer; White Men    |
| 118 | The ‘Come Alive With Pepsi!’ slogan was interpreted in China as ‘Pepsi Brings Your Ancestors Back From the Dead.’                     | <a href="https://www.snopes.com/fact-check/come-alive/">https://www.snopes.com/fact-check/come-alive/</a>                                                     | Pepsi              | Come Alive With Pepsi; China                  |